Hey! Welcome to Strats Stats, where all the stats are done! Feel free to roam the document tabs if you're already familiar with the algorithms being used, or stay for the explanation!

So you already know Strats? Hopefully… Anyway, essentially we're measuring separate variables and their effects on the game of Strats - for our purposes, we're specifically looking at how many cards you win by at the end of the game (or lose by). Ever won before? Even if you haven't, that's ok :) When you win, you count how many cards you have in your deck after taking half of the winnings from the last round of the game (rounding up). If you lost, you lost by that card count (and it's noted as a negative value). These are the Y values on the graphs you're about to see - and yes, there can technically be a draw where 2 players have the EXACT same cards and are counted as a 0.

So what is a perfect game? Because that is one of the graphs… A perfect game means that somehow, both players knew exactly what each other had for each round (wouldn't that be useful??). Because of this, the algorithm just randomly generates 2 decks for each player and runs through each round with a logic control: Play if I have higher than the other player (and of course equal to or below 20). If I have less than the other player, I play my lowest value card. If there's a joker, both players know it and the logic runs again like normal but the cap is instead at 15 obviously. The winner gets half (rounded up) of the winnings like usual. Sure the game is definitely simplified (the biggest part being the fact that people don't "reveal" cards in succession), but even then it's actually still quite a good measure of the game. The X axis for all of the plotted graphs is the original deck disparity, aces count as 1's, jacks - 11, queens - 12, and kings worth 13 (jokers with 0). All of the cards in a player's starting deck are summed for the total, and player 1's total is subtracted by player 2's total for the deck disparity.

Confidence games are a little confusing but follow the same setup and plotting. The only difference is that both players now have no idea what each other has at the start of the game. Only thing is, these players have a way better memory than you. Both players can remember their own deck (after 9 rounds they will have completely memorized their deck) and over time begin figuring out what the other deck's order is. Even though these players have a great memory, they can't see what they can't see: if the other player wins, they have no idea what the other player picked from the winnings. But even then, they know what the other player *might* have when that hand comes around. Both players then take this data each round and calculate almost* every possibility of plays they could make paired with what the other player could make and take the amount of wins they could have / the total plays that could be made as their "win percentage".

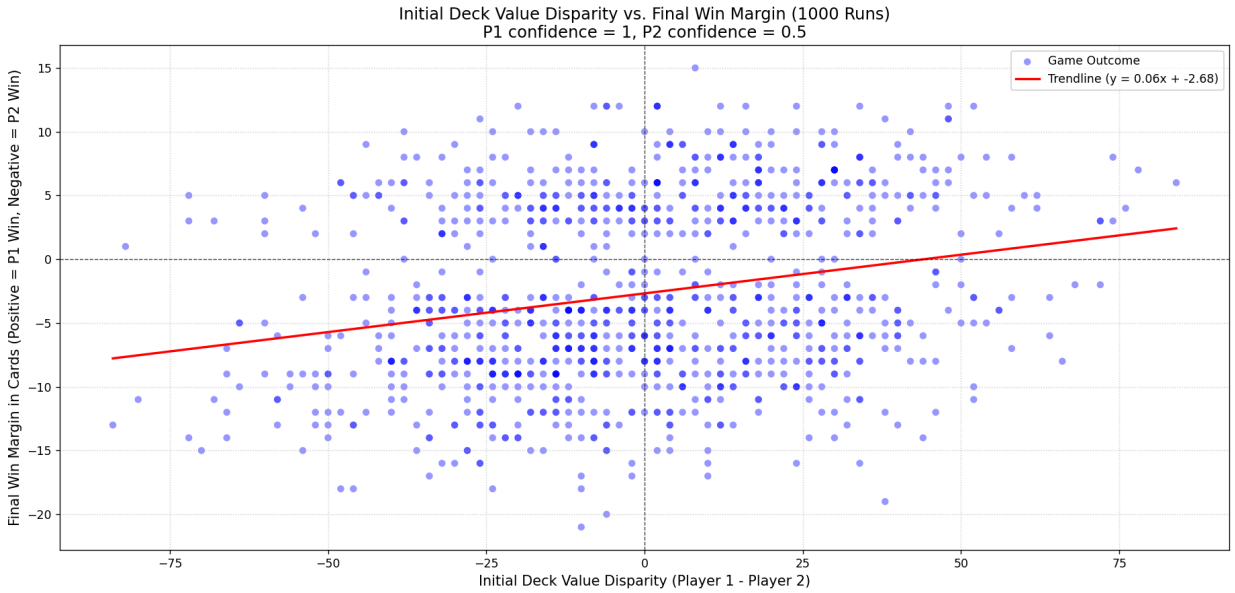
* - Not every single combination of plays can be calculated each time; the first round of a two player game has almost 20 thousand possible plays from a player's perspective - they only know the 3 cards they have right now. Because of this, the algorithm will look at how many possibilities there are, if there are equal to or under 2 thousand possibilities, it will run all of them - otherwise, the algorithm will pick 2 thousand at random and perform a win percentage on these possible hands.

Once the player has a win percentage, whether they make a play or play safe is up to their confidence level. The way this is done in the algorithm is quite simple. A player's confidence level is between 0 and 1. A player with a confidence level of 1 will make a play if the win percentage is above 50 percent. A player with a confidence level of 0 will make a play if the win percentage is above 100 percent… you can kind of predict how that'd go. To find the win percentage value at which players will make a play, just use those 2 coordinates and make a linear regression. Or just use -50x + 100… Also, if the win percentage happens to ever be equal to that minimum value for that player, they will choose at random to make a play or not make a play.

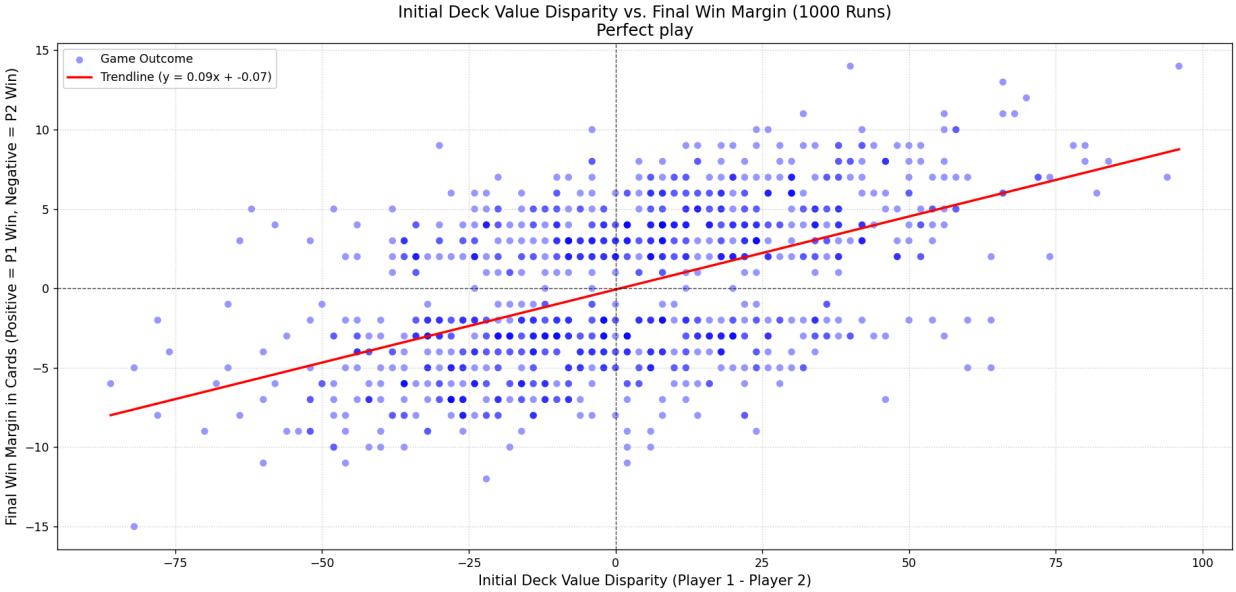
Lastly, 1000 different games (each having a different shuffle) are then played using these bots and the data of each game is plotted.

These confidence bots are essentially what a perfect player would be - capable of knowing everything that there is to be known. But wait… is Strats a solved game then? Kind of… To be fair, we're missing the biggest aspect of Strats: the Strats!! We're human - we give clues, whether true or false, to our opponent and take in clues as to what the opponent has; even further, flipping over cards in succession or choosing to keep cards hidden are also major aspects of the manipulation game that is Strats!

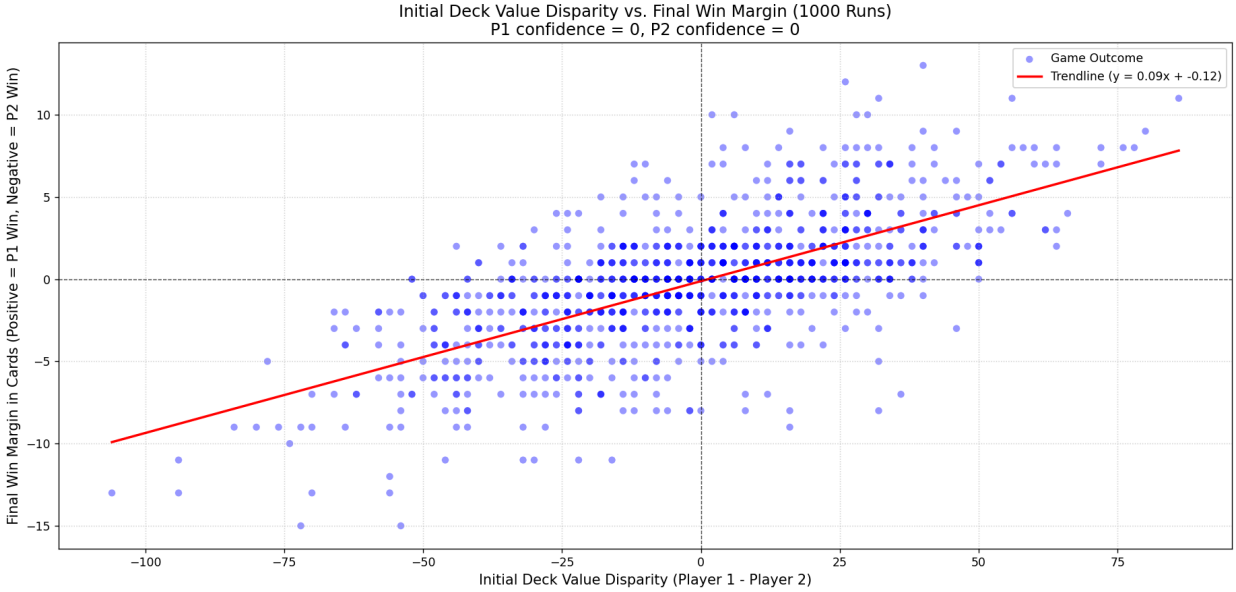
So in some ways, you have statistics with or against you every game you play, but regardless, the outcome of the game is most definitely up to how you play! Just take a look at the point at -80 that still won the game! That'd be like being down 8 face cards and still winning! (And you're an overconfident player in this case!):



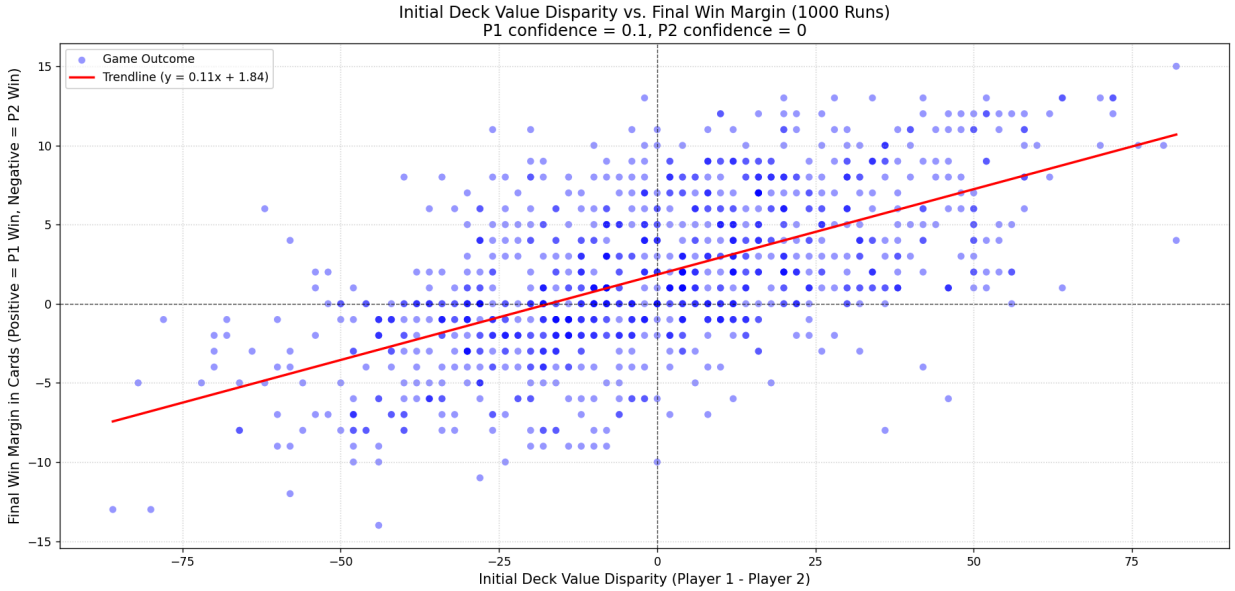
Absolutely perfect game



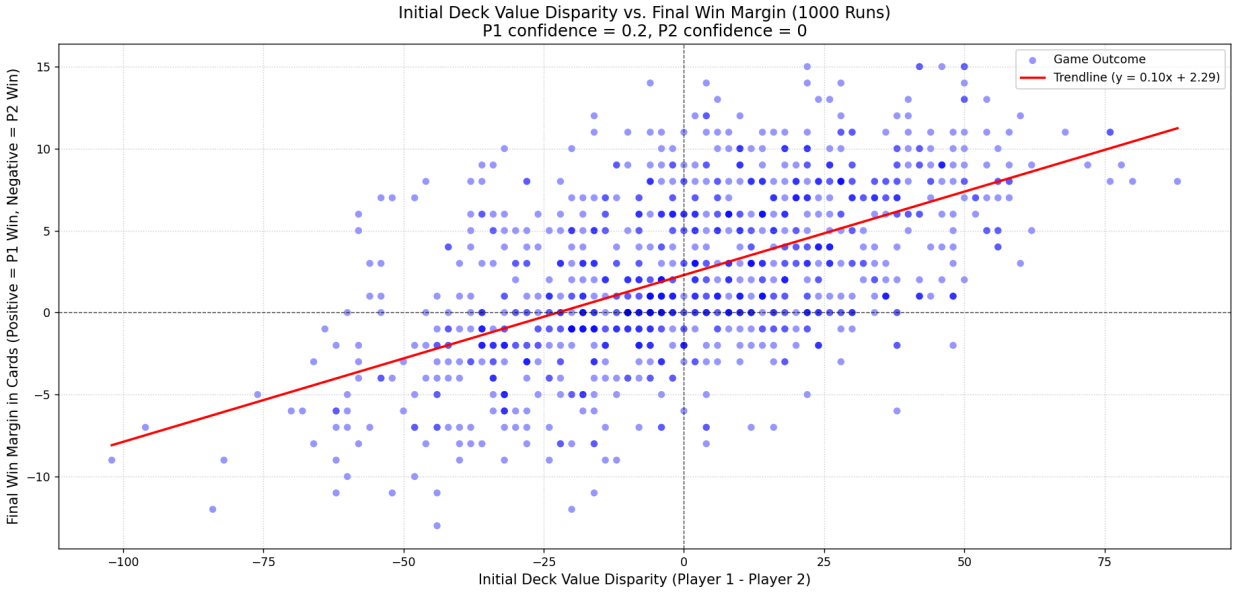
P1 confidence - 0, P2 confidence - 0



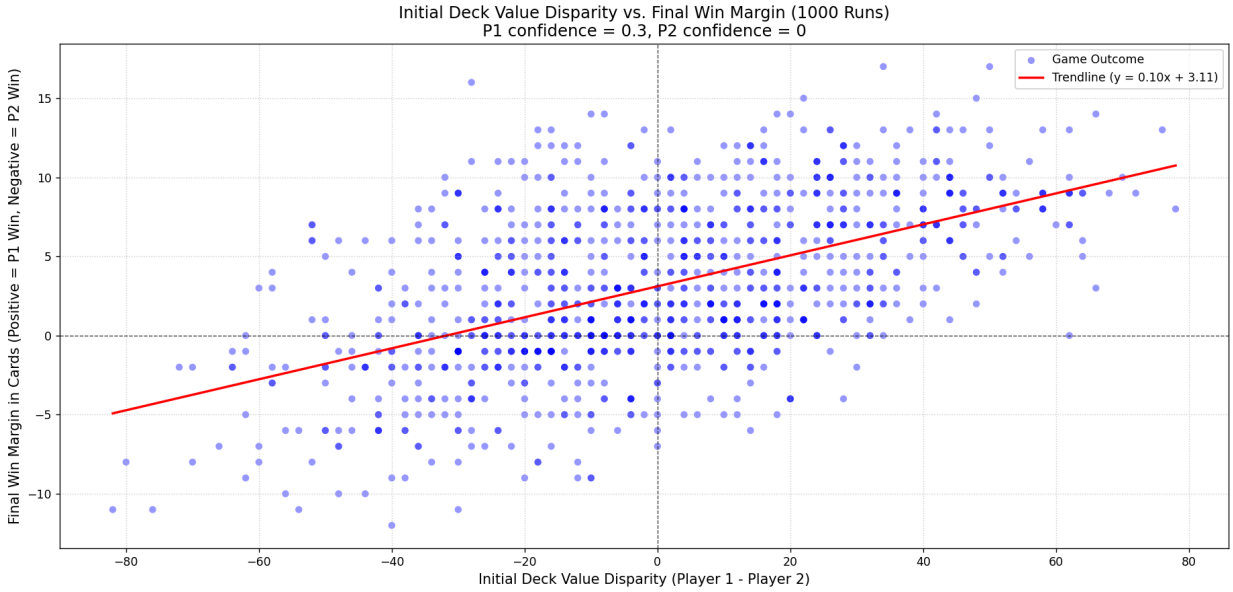
P1 confidence - 0.1, P2 confidence - 0



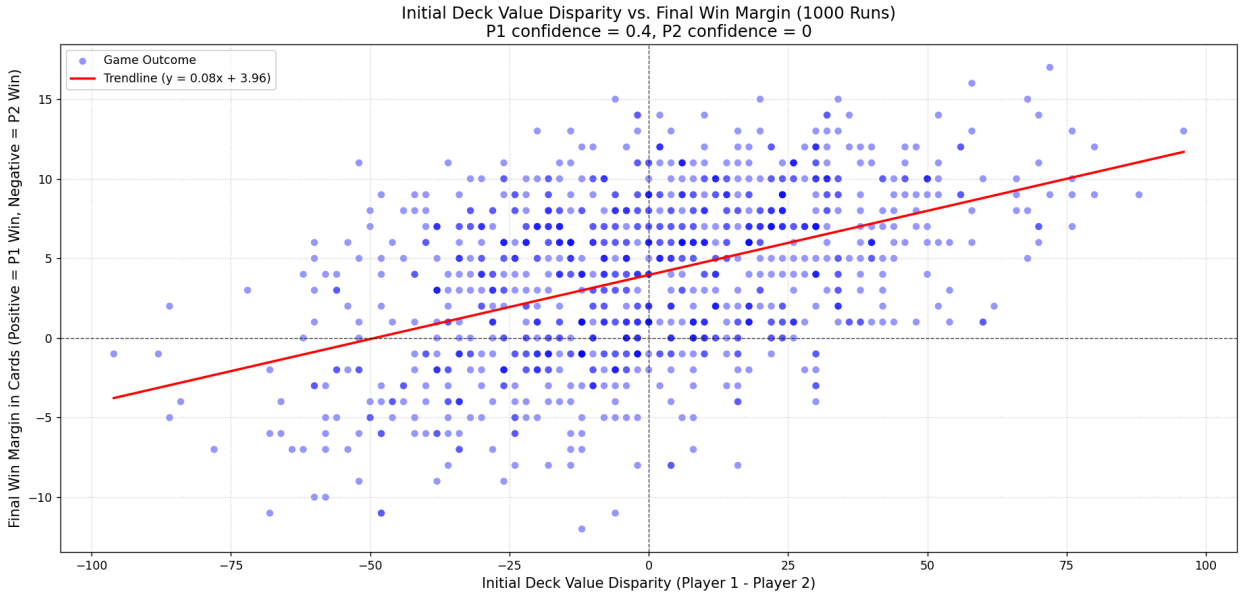
P1 confidence - 0.2, P2 confidence - 0



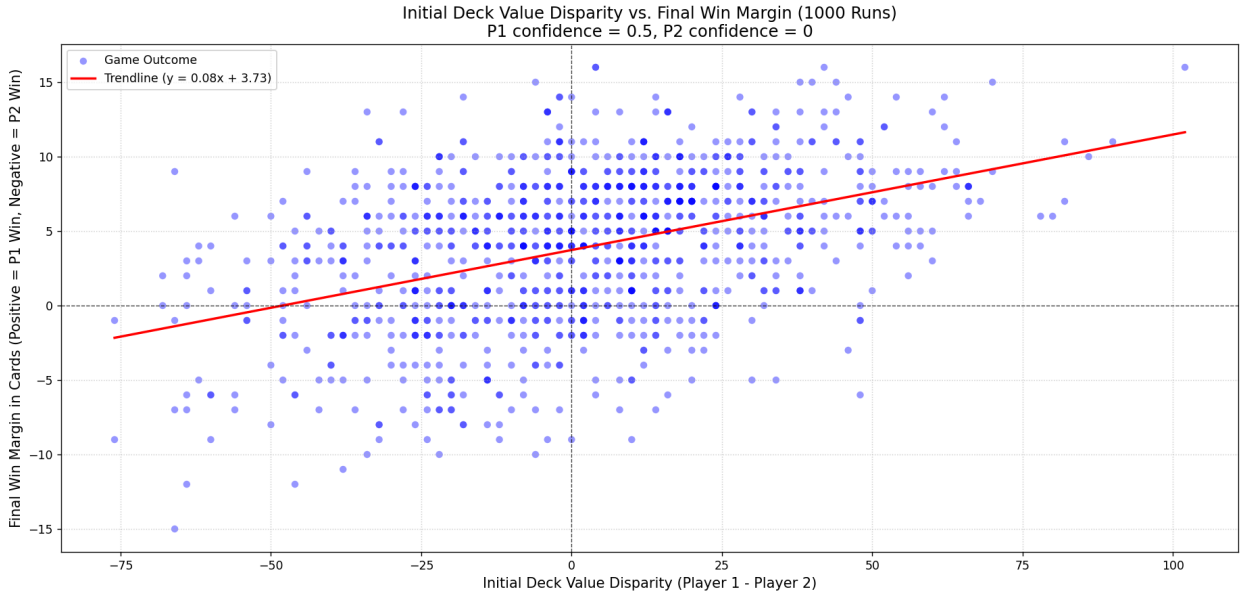
P1 confidence - 0.3, P2 confidence - 0



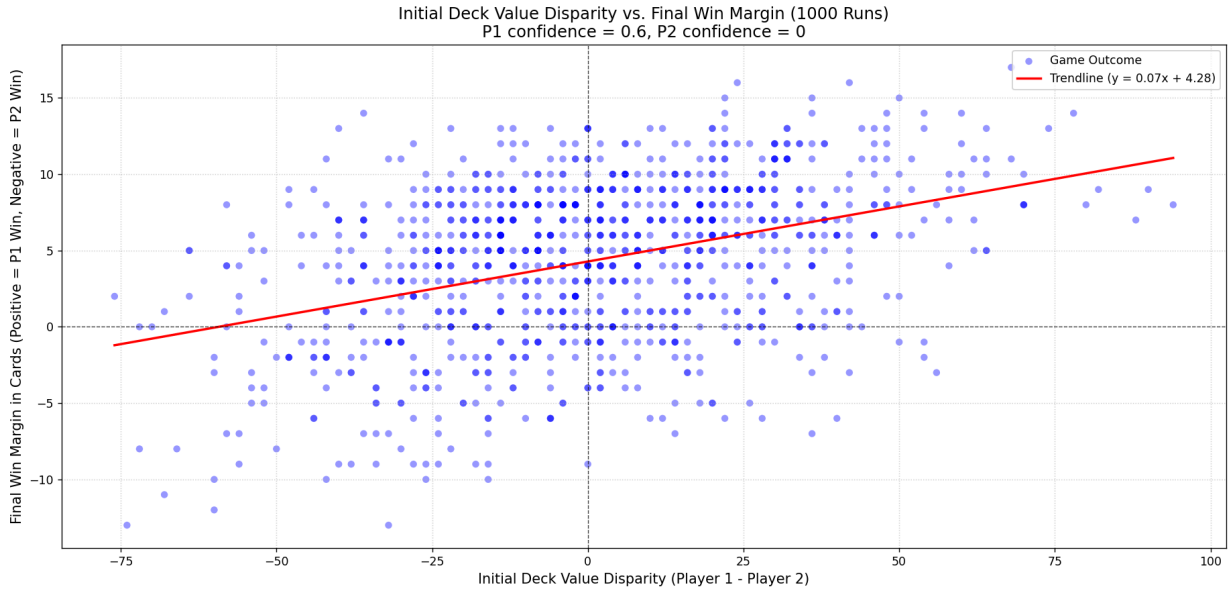
P1 confidence - 0.4, P2 confidence - 0



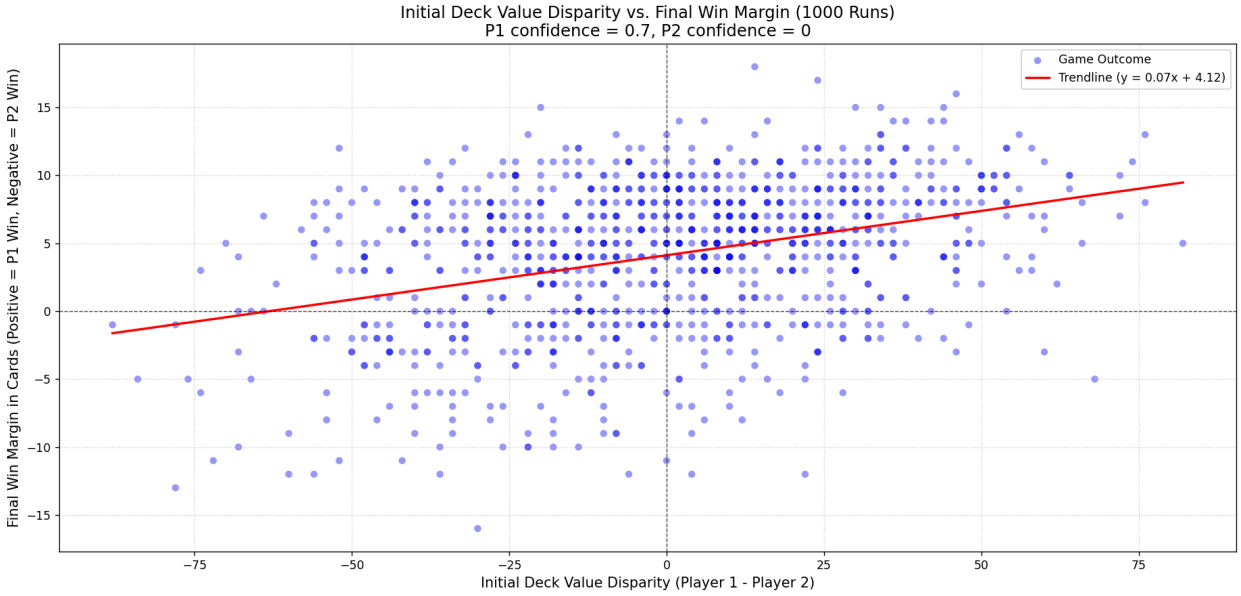
P1 confidence - 0.5, P2 confidence - 0



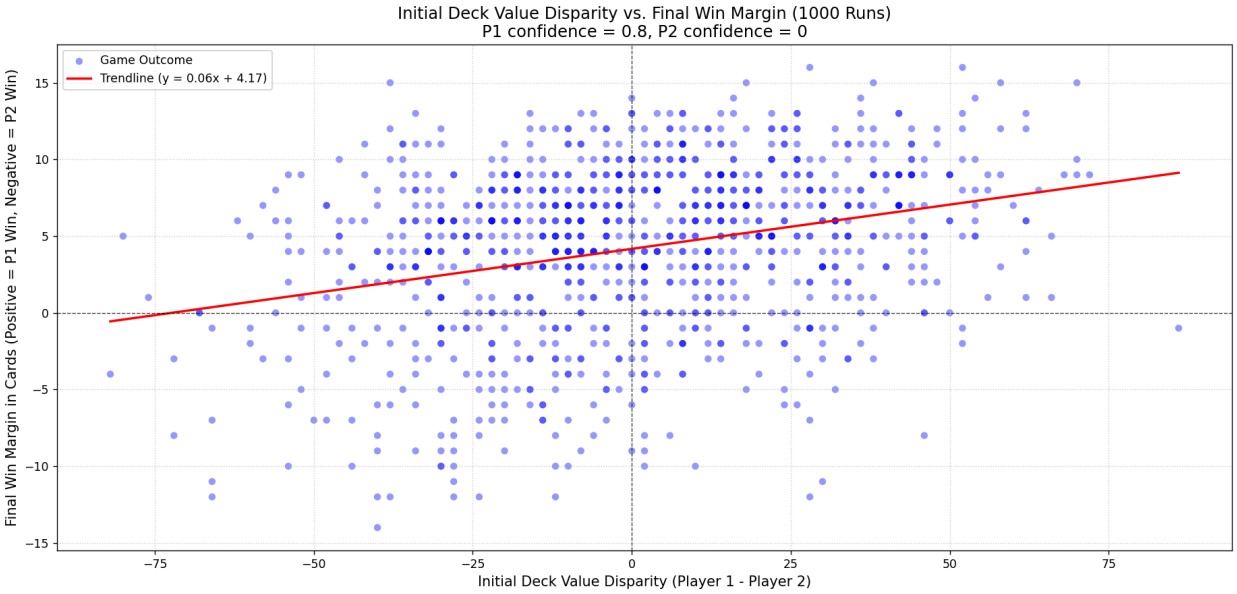
P1 confidence - 0.6, P2 confidence - 0



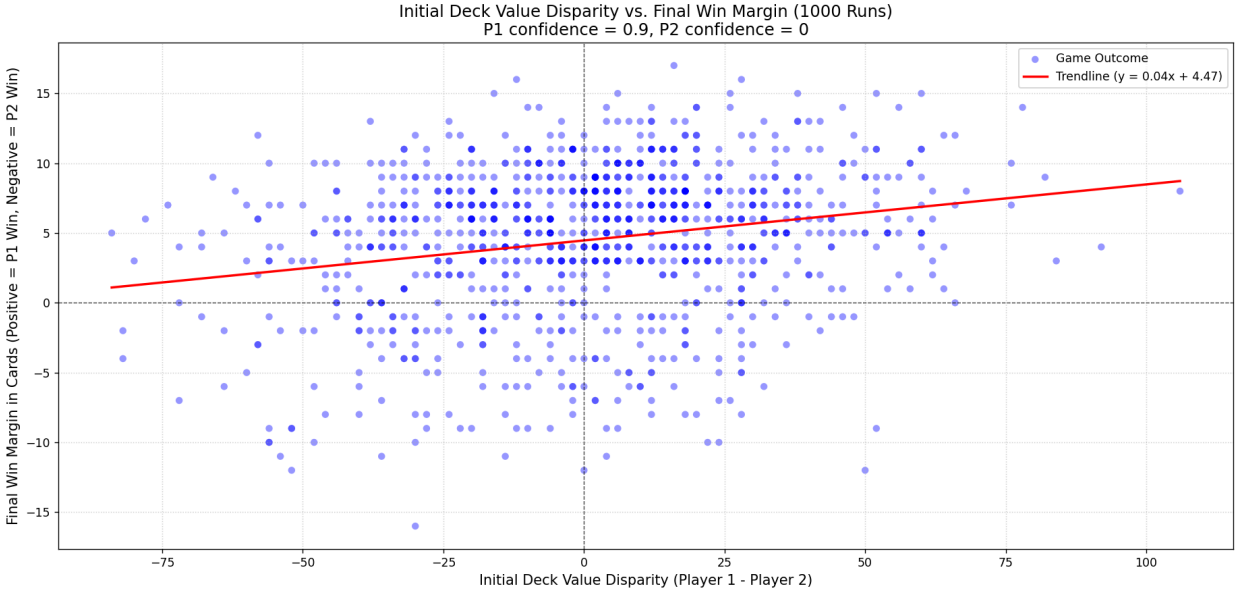
P1 confidence - 0.7, P2 confidence - 0



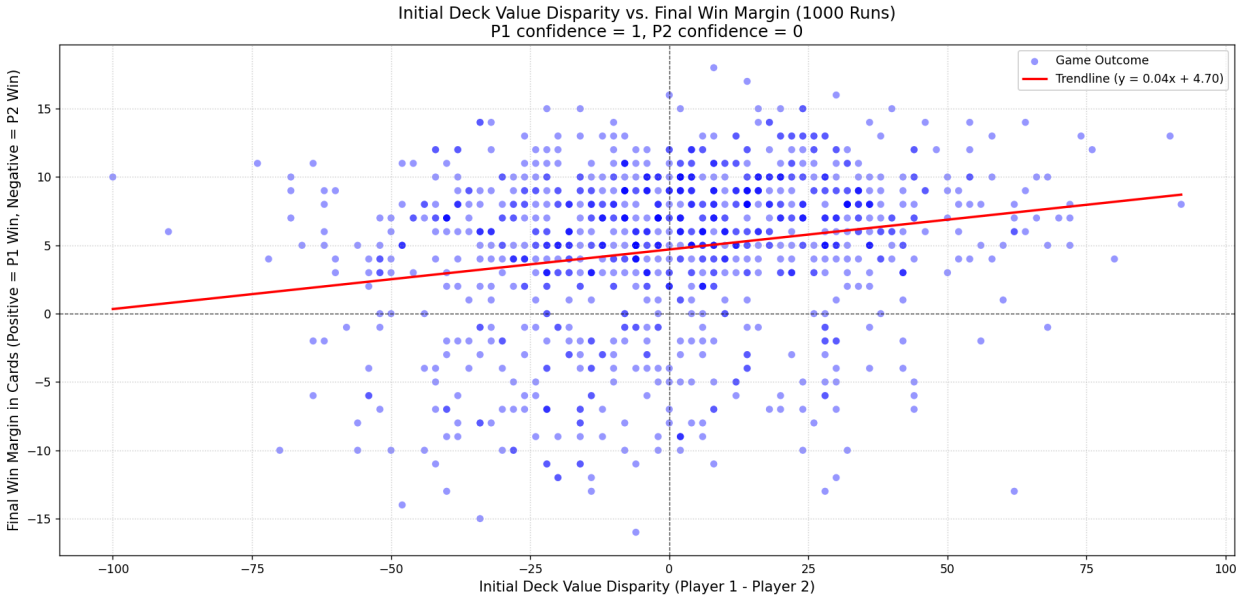
P1 confidence - 0.8, P2 confidence - 0



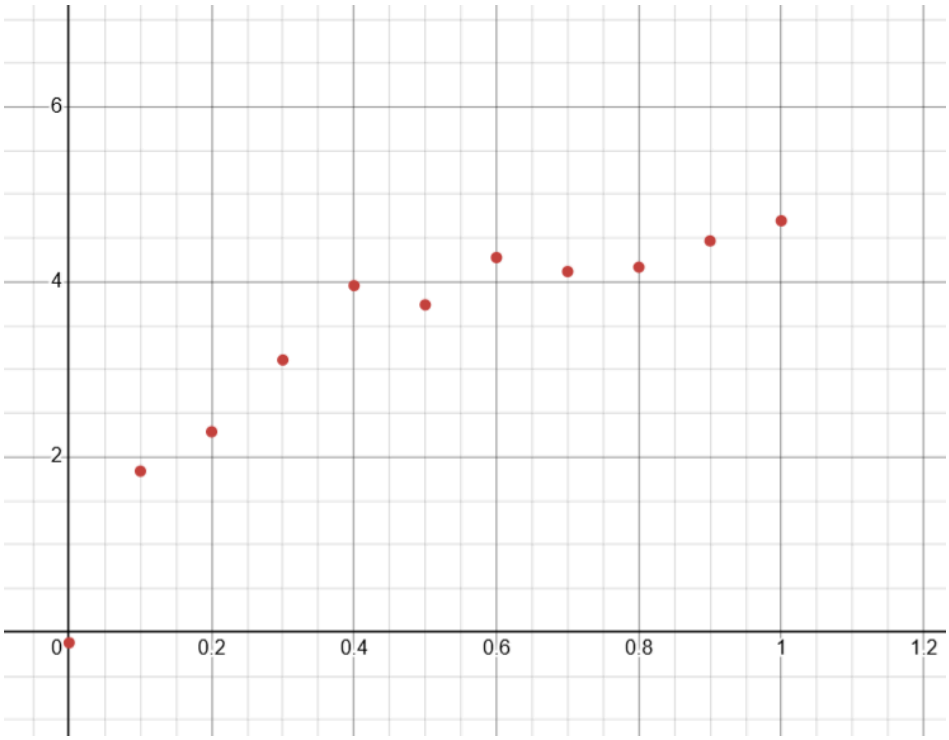
P1 confidence - 0.9, P2 confidence - 0



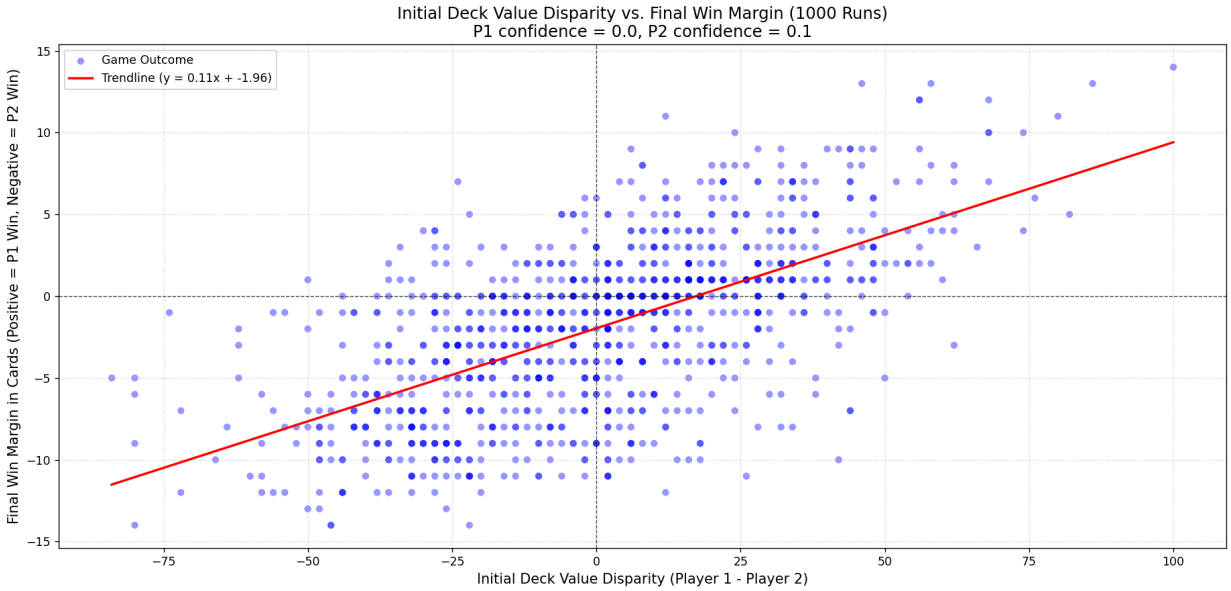
P1 confidence - 1, P2 confidence - 0



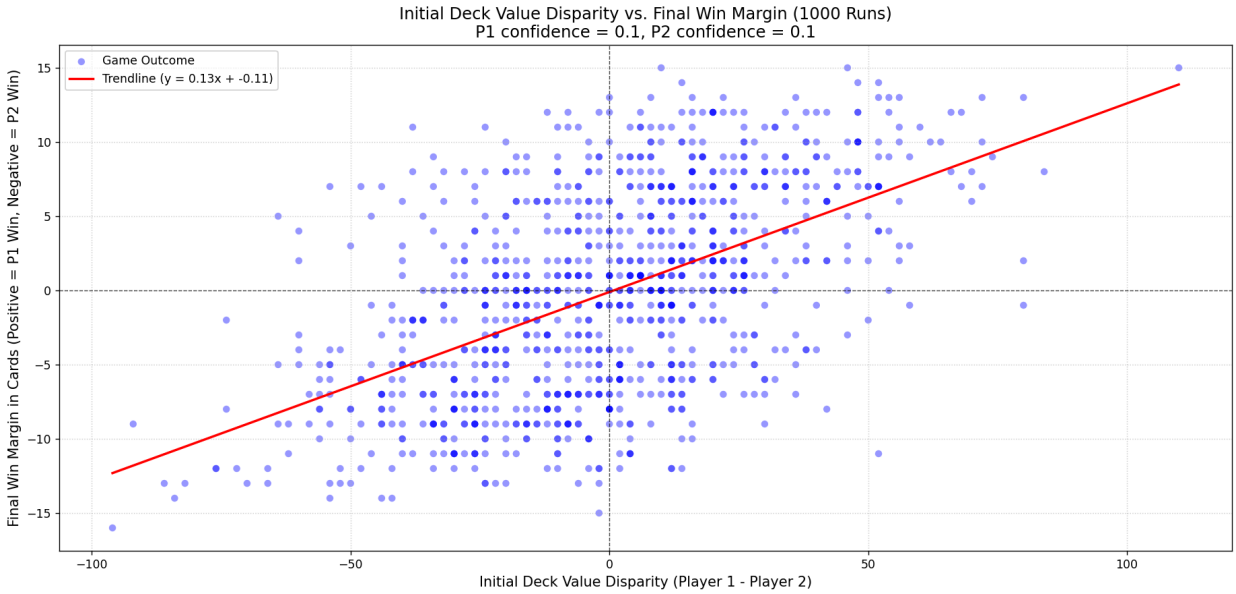
Against someone with a confidence level of 0, these are the amount of cards you will win / lose by with different confidence levels.



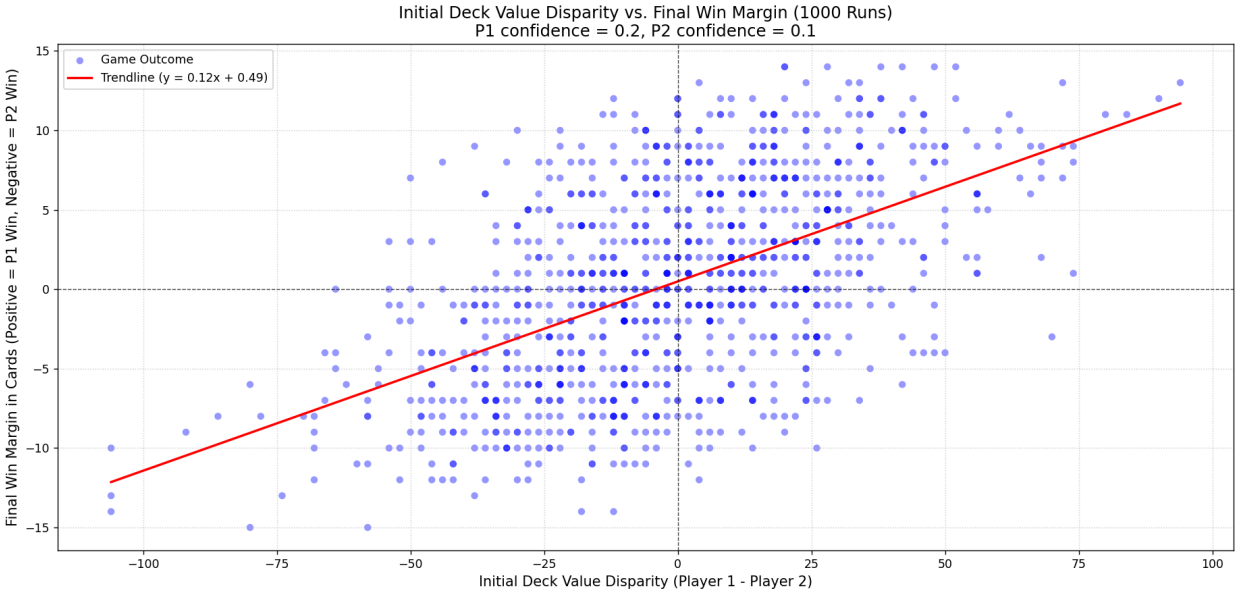
P1 confidence - 0, P2 confidence - 0.1



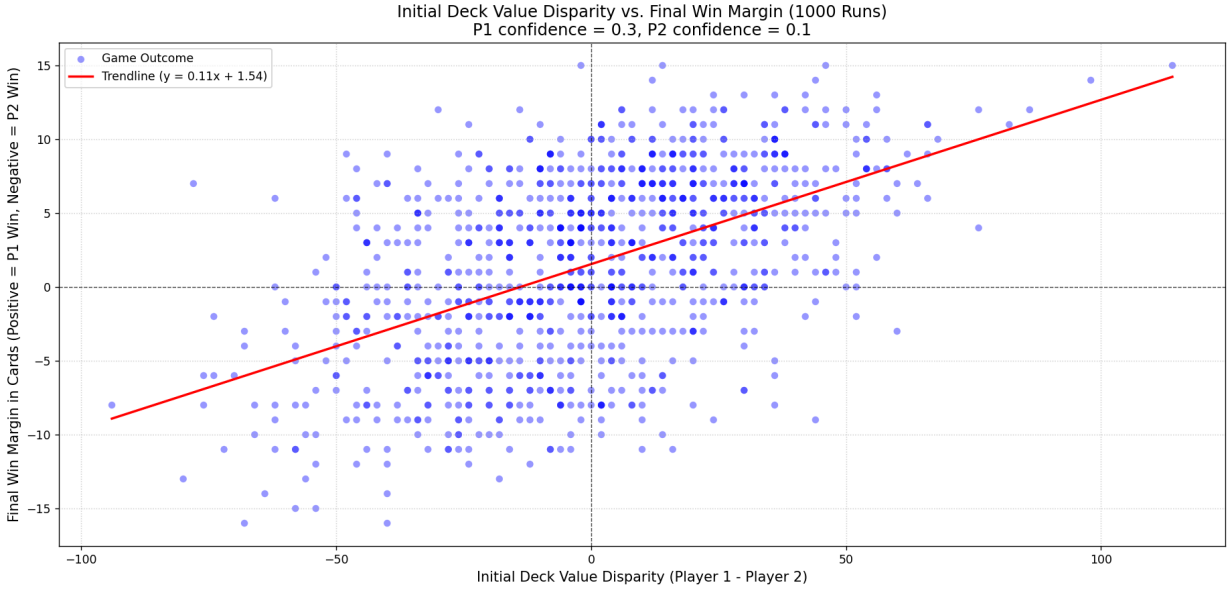
P1 confidence - 0.1, P2 confidence - 0.1



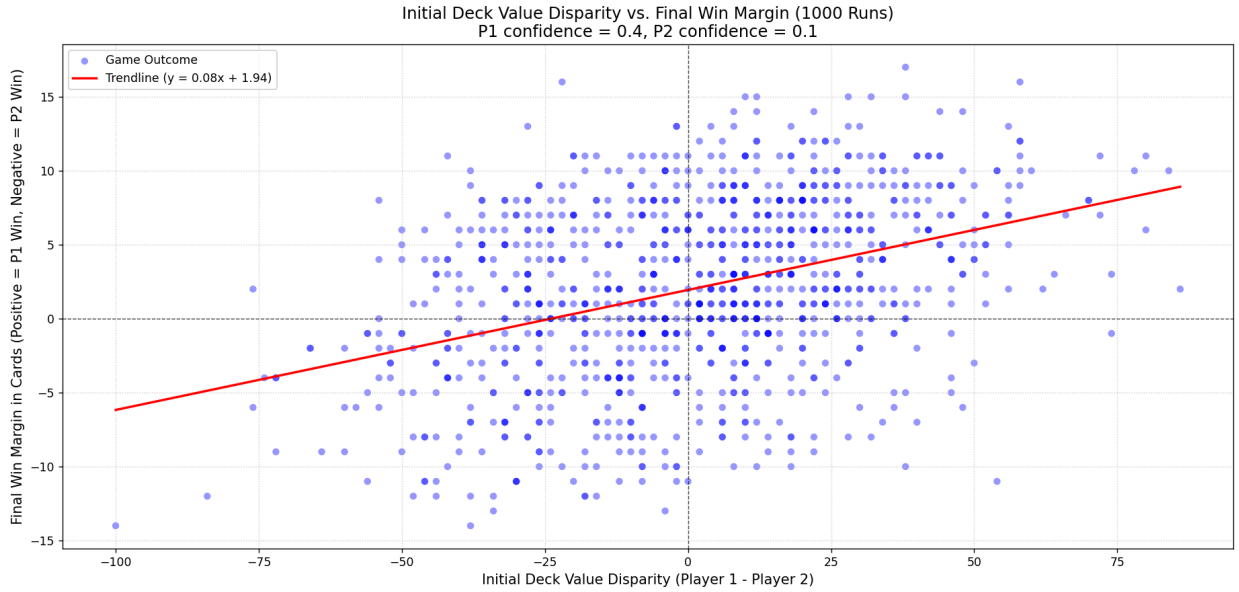
P1 confidence - 0.2, P2 confidence - 0.1



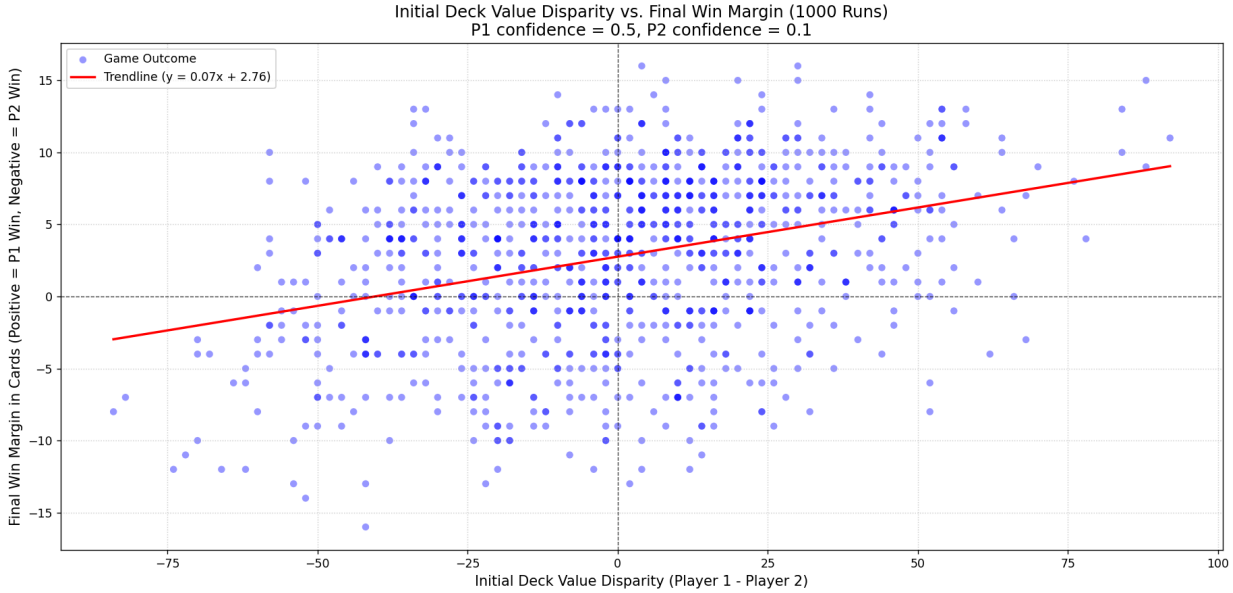
P1 confidence - 0.3, P2 confidence - 0.1



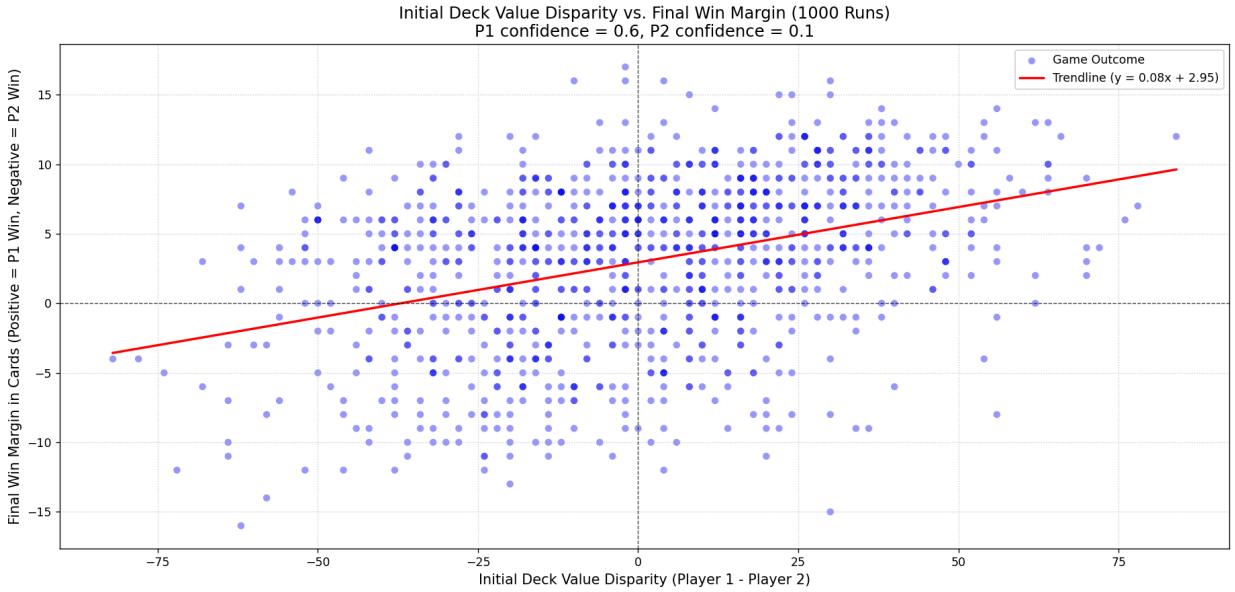
P1 confidence - 0.4, P2 confidence - 0.1



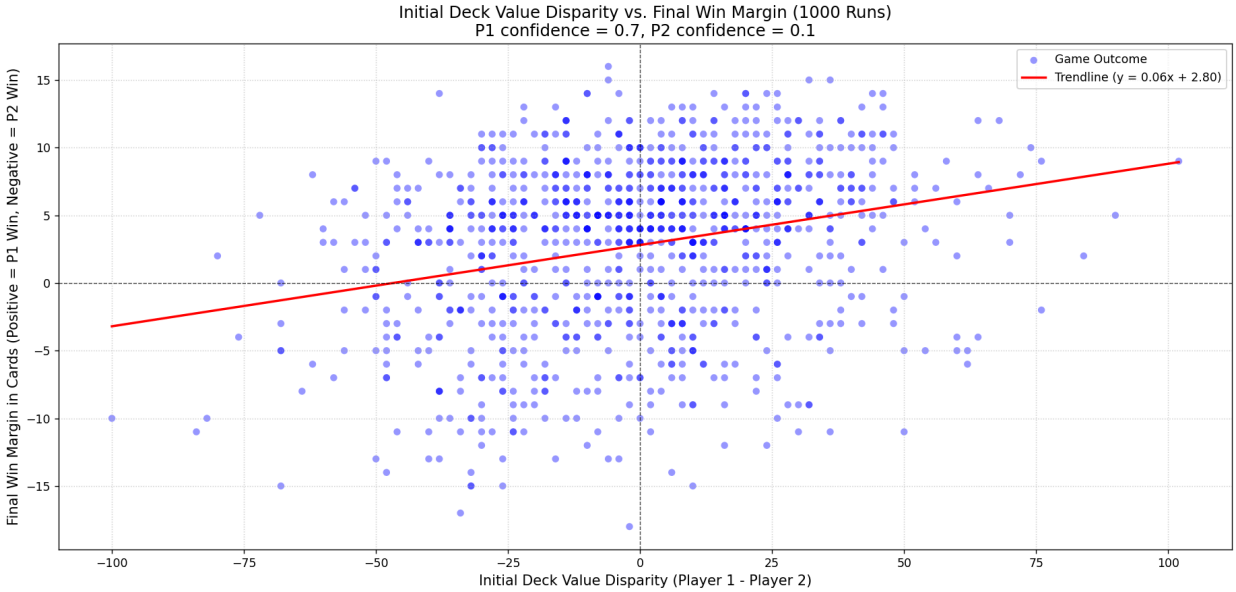
P1 confidence - 0.5, P2 confidence - 0.1



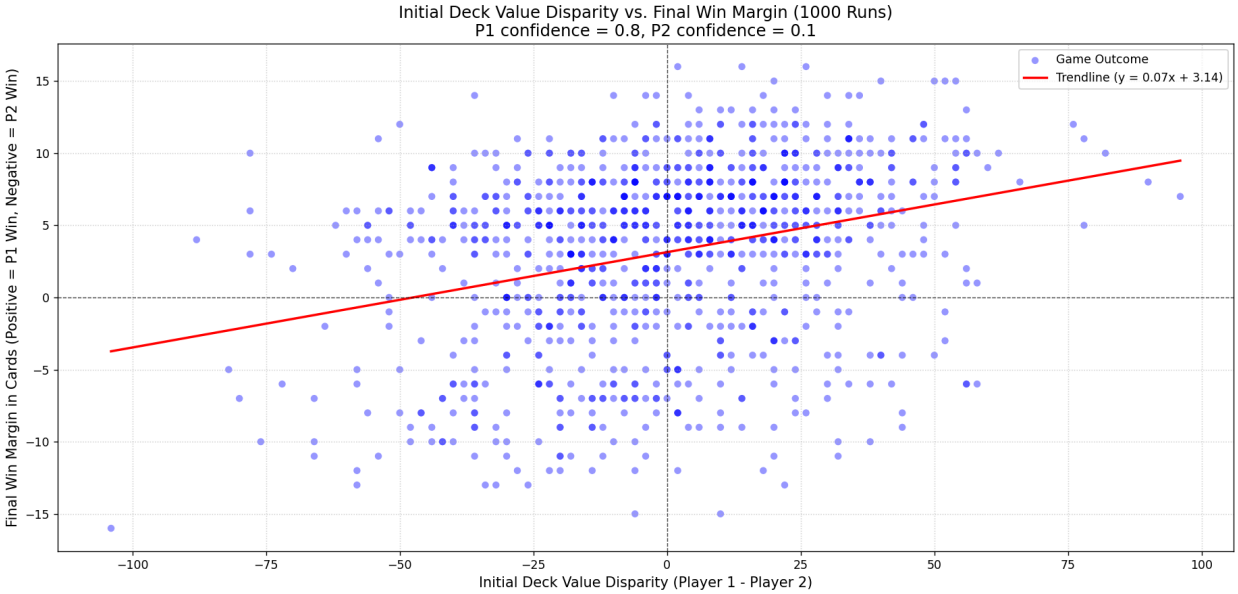
P1 confidence - 0.6, P2 confidence - 0.1



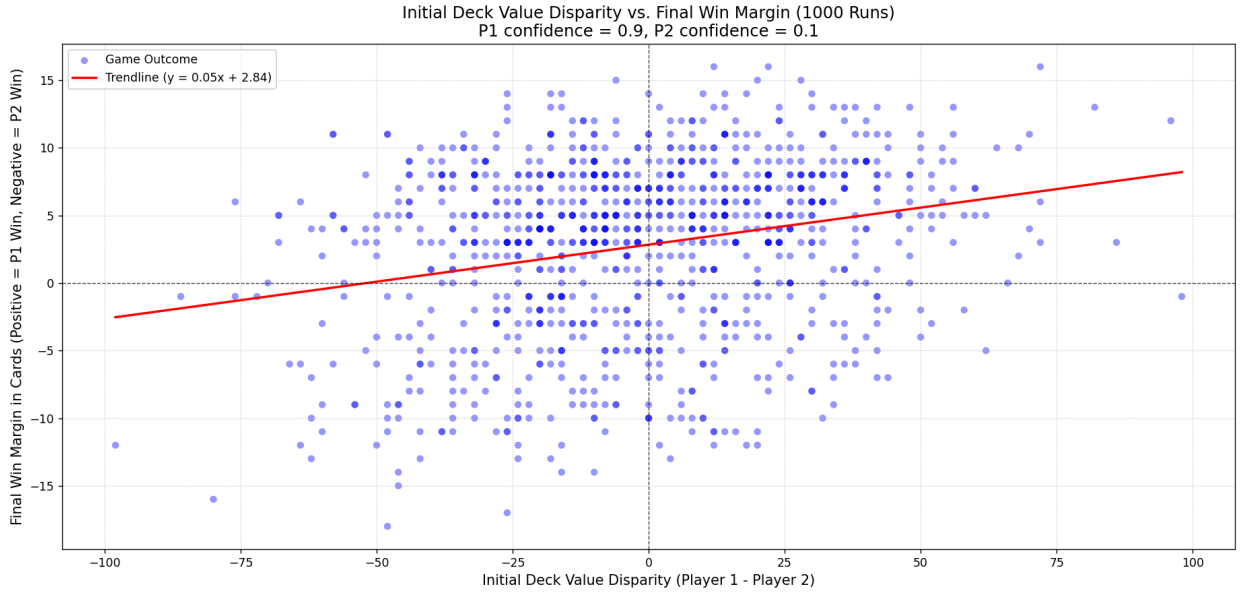
P1 confidence - 0.7, P2 confidence - 0.1



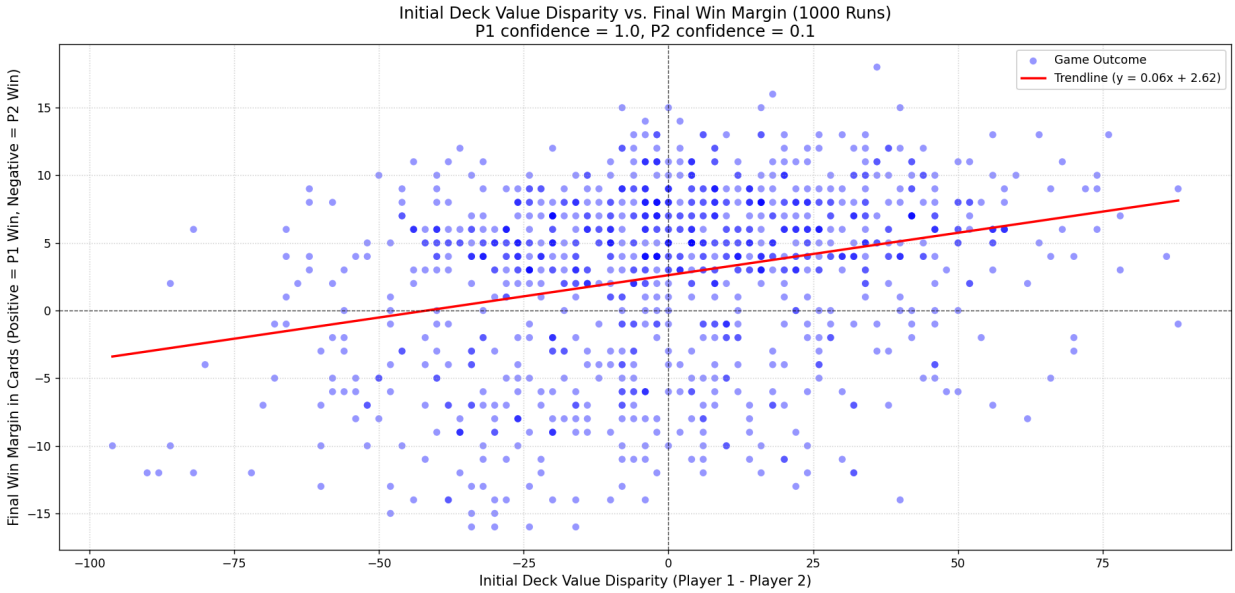
P1 confidence - 0.8, P2 confidence - 0.1



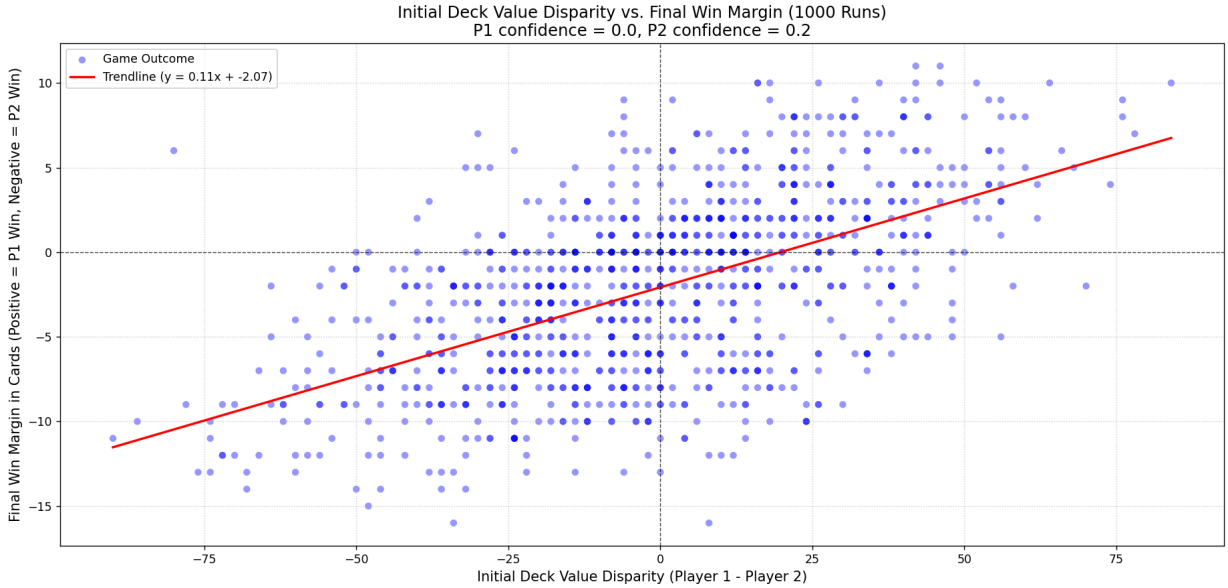
P1 confidence - 0.9, P2 confidence - 0.1



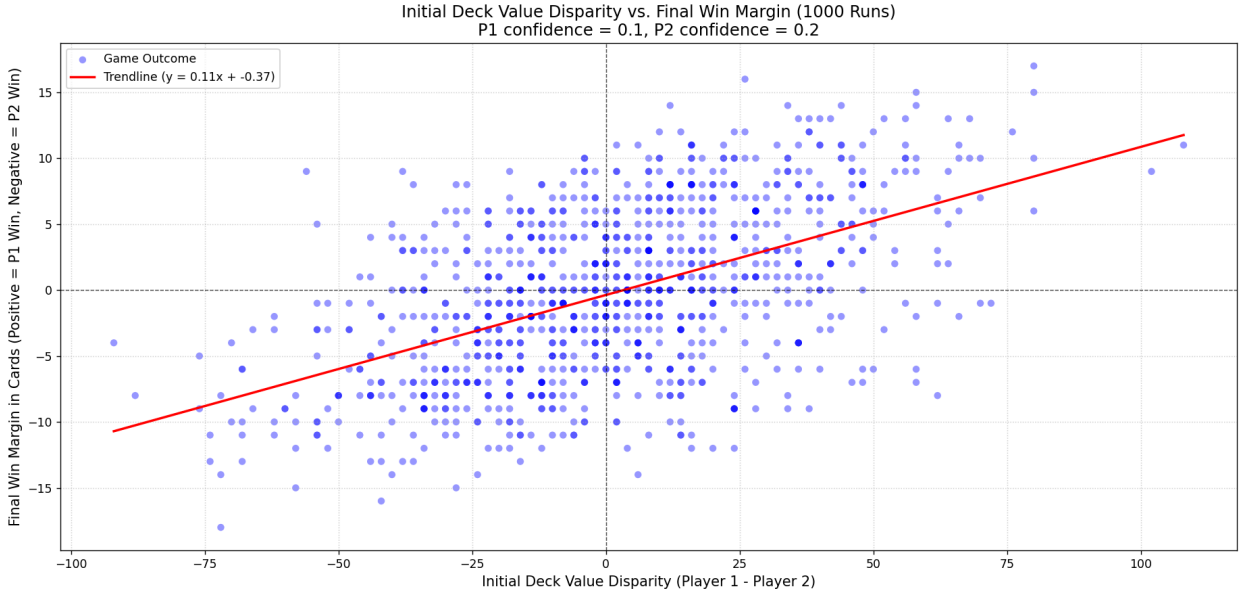
P1 confidence - 1, P2 confidence - 0.1


Initial Deck Value Disparity vs. Final Win Margin (1000 Runs)
P1 confidence = 1.0, P2 confidence = 0.1
Game Outcome
Trendline (y = 0.06x + 2.62)
Final Win Margin in Cards (Positive = P1 Win, Negative = P2 Win)
Initial Deck Value Disparity (Player 1 - Player 2)
15
10
5
0
−5
−10
−15
−100
−75
−50
−25
0
25
50
75

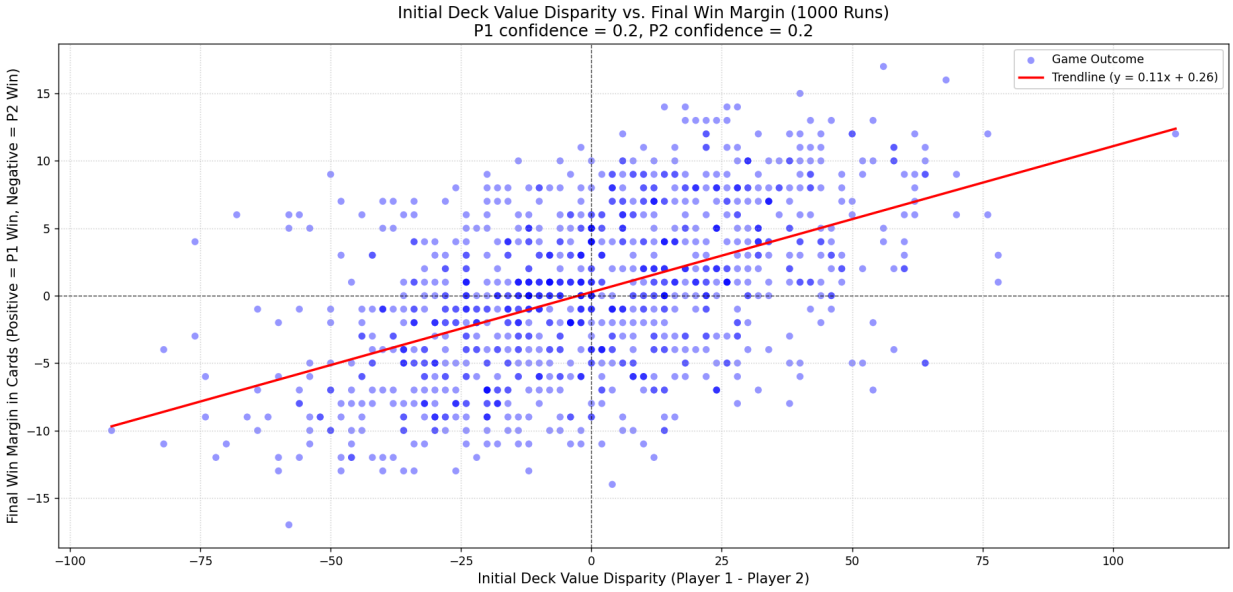
P1 confidence - 0, P2 confidence - 0.2



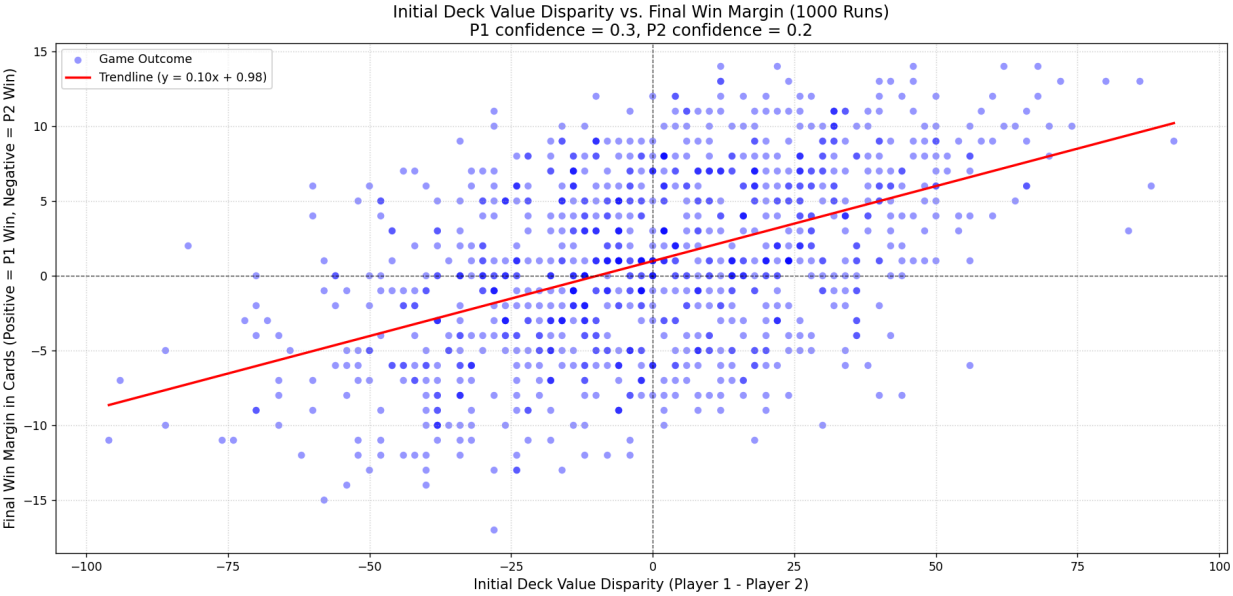
P1 confidence - 0.1, P2 confidence - 0.2



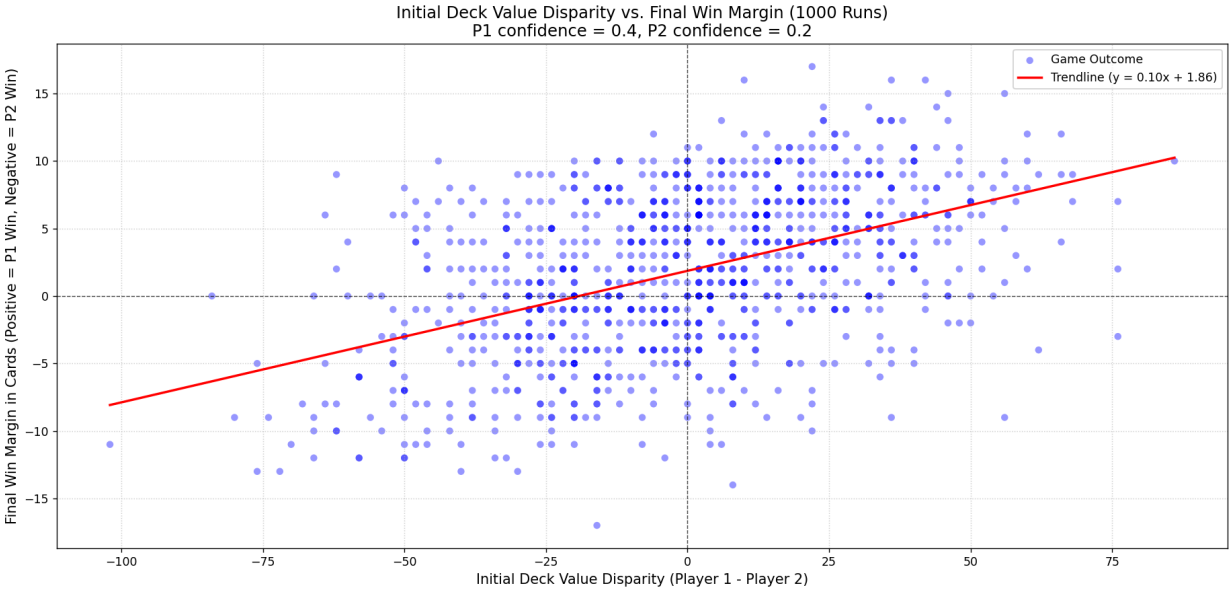
P1 confidence - 0.2, P2 confidence - 0.2



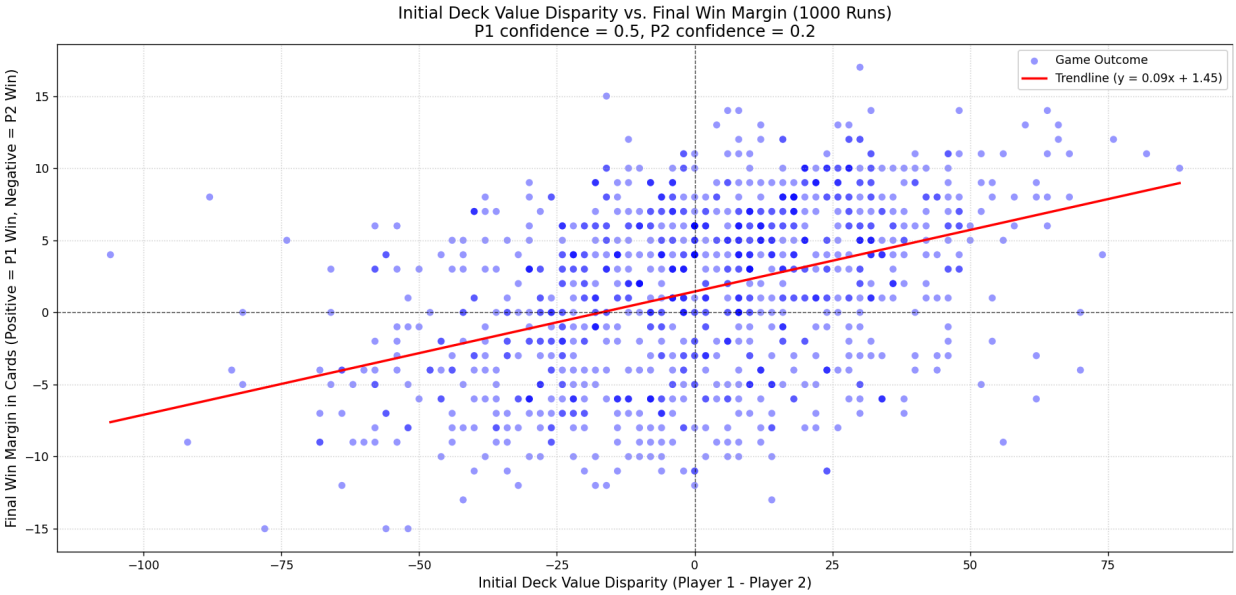
P1 confidence - 0.3, P2 confidence - 0.2



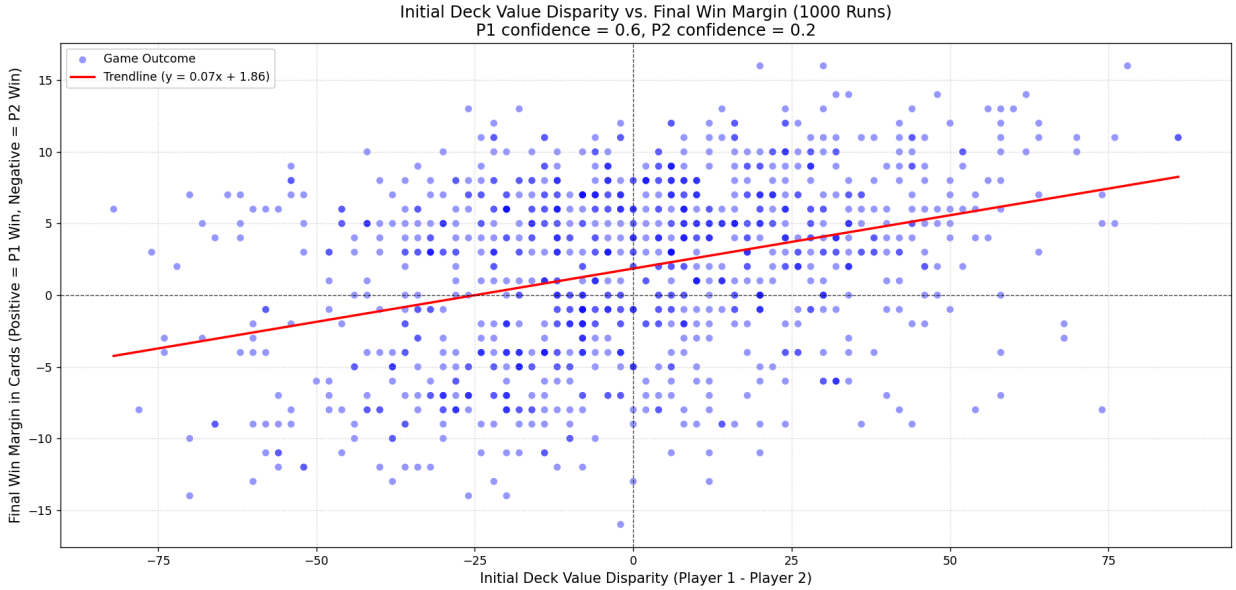
P1 confidence - 0.4, P2 confidence - 0.2



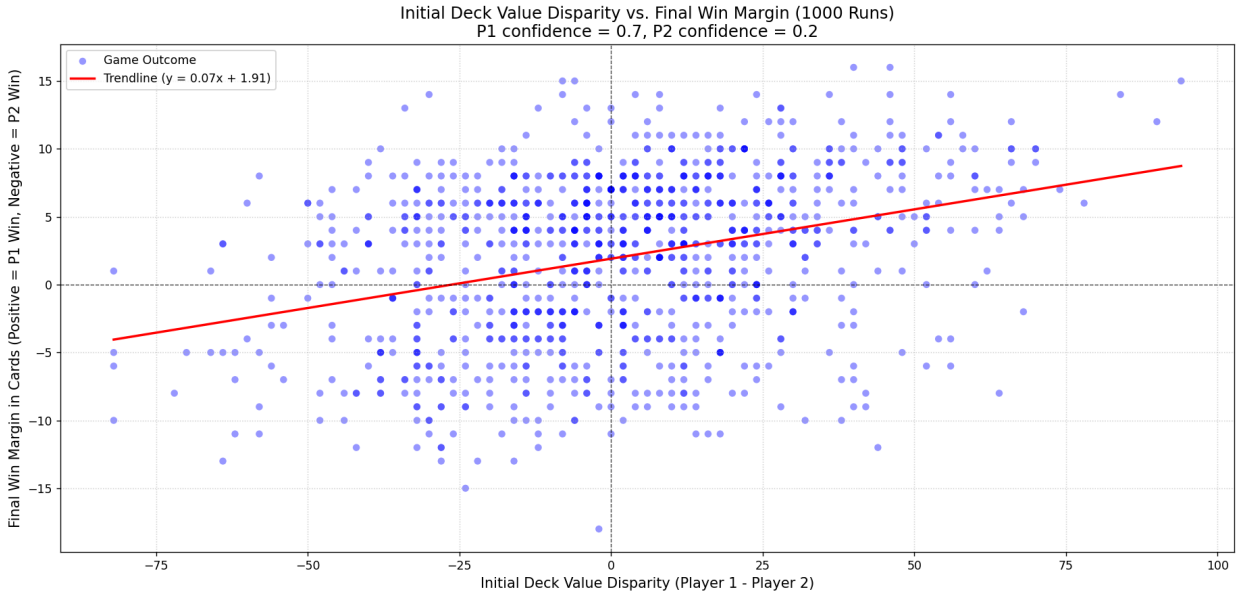
P1 confidence - 0.5, P2 confidence - 0.2



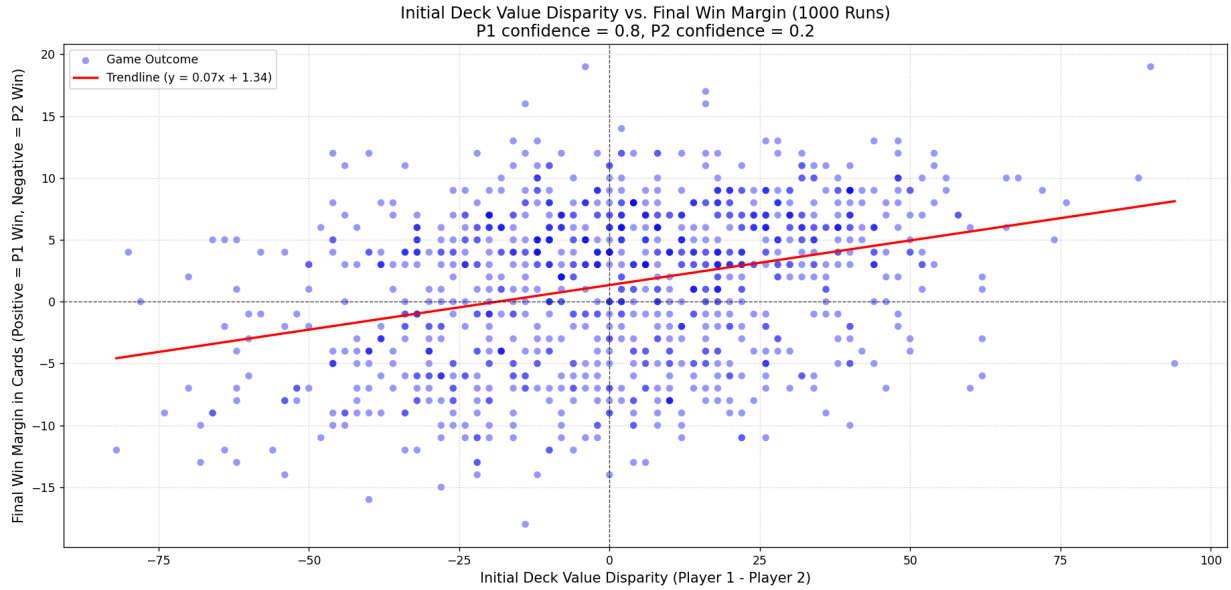
P1 confidence - 0.6, P2 confidence - 0.2



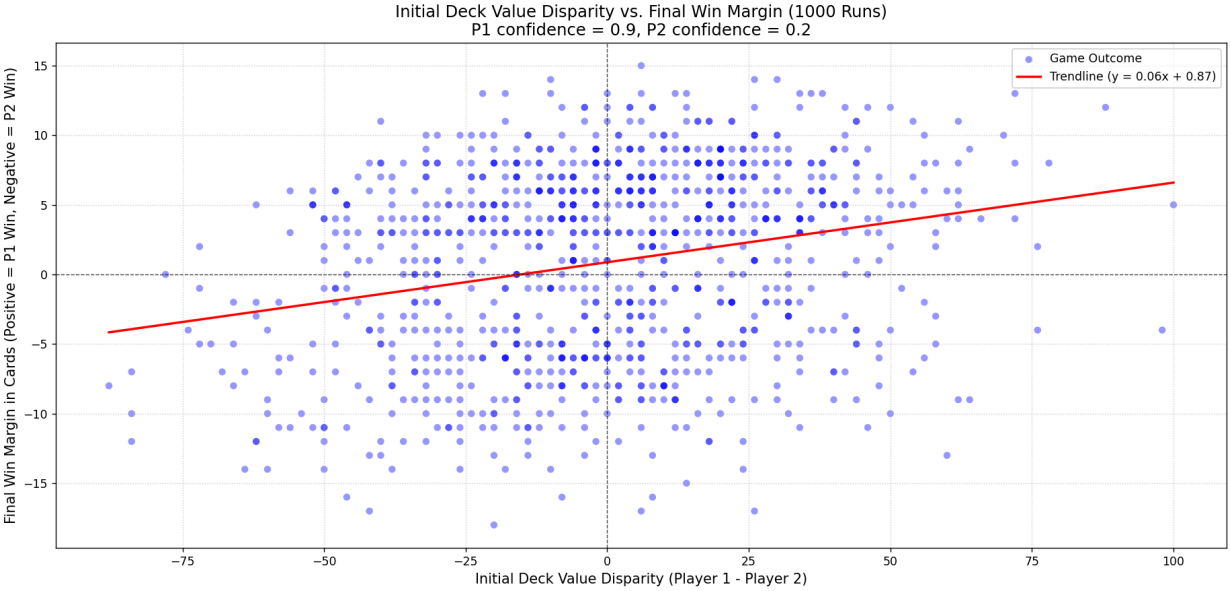
P1 confidence - 0.7, P2 confidence - 0.2



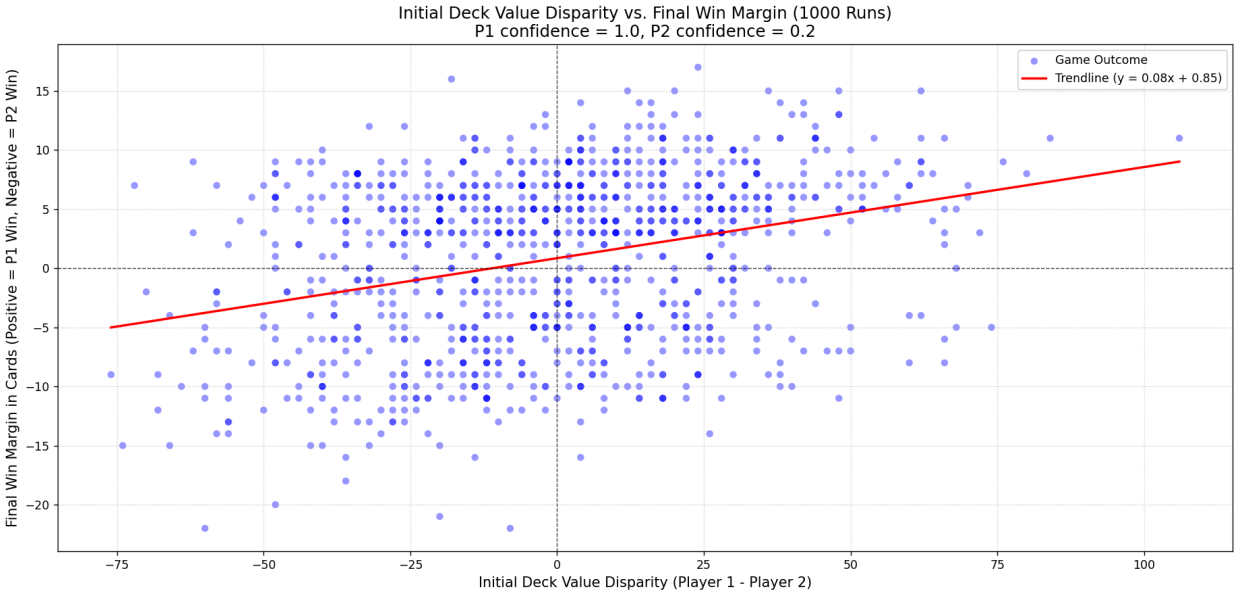
P1 confidence - 0.8, P2 confidence - 0.2



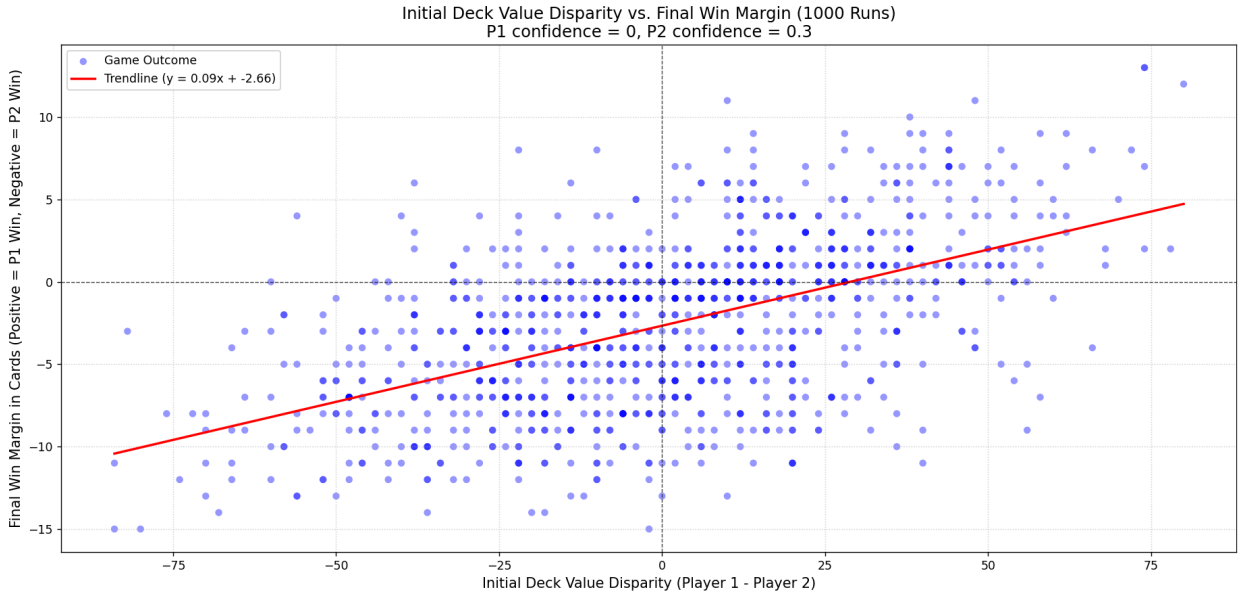
P1 confidence - 0.9, P2 confidence - 0.2



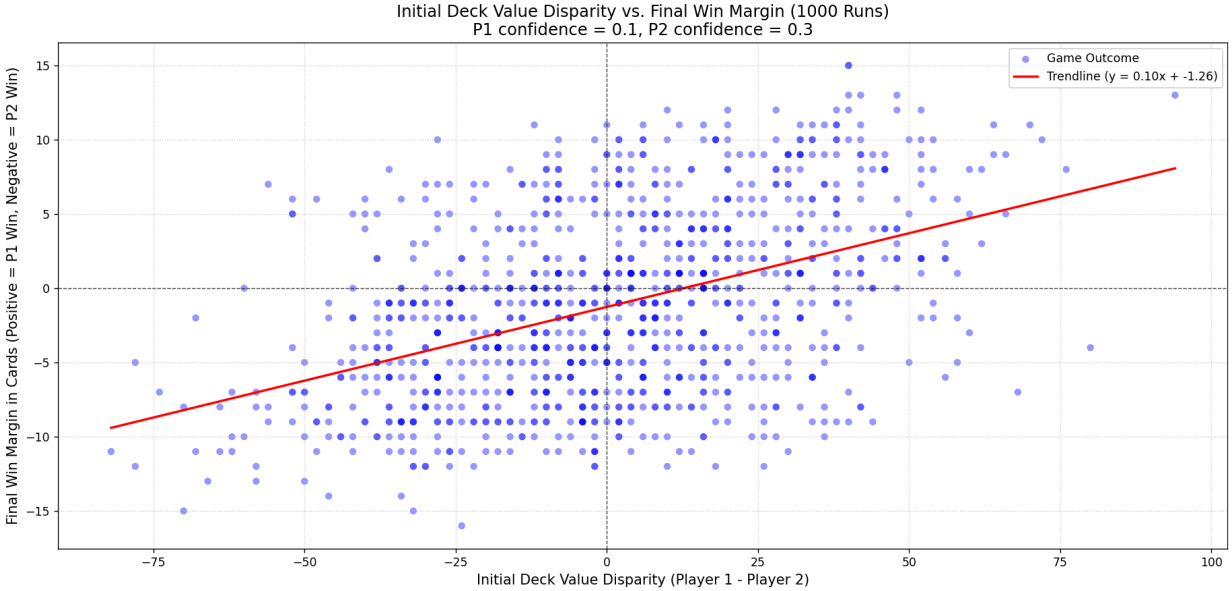
P1 confidence - 1, P2 confidence - 0.2


Initial Deck Value Disparity vs. Final Win Margin (1000 Runs)
P1 confidence = 1.0, P2 confidence = 0.2
Game Outcome
Trendline (y = 0.08x + 0.85)
Final Win Margin in Cards (Positive = P1 Win, Negative = P2 Win)
Initial Deck Value Disparity (Player 1 - Player 2)
15
10
5
0
−5
−10
−15
−20
−75
−50
−25
0
25
50
75
100

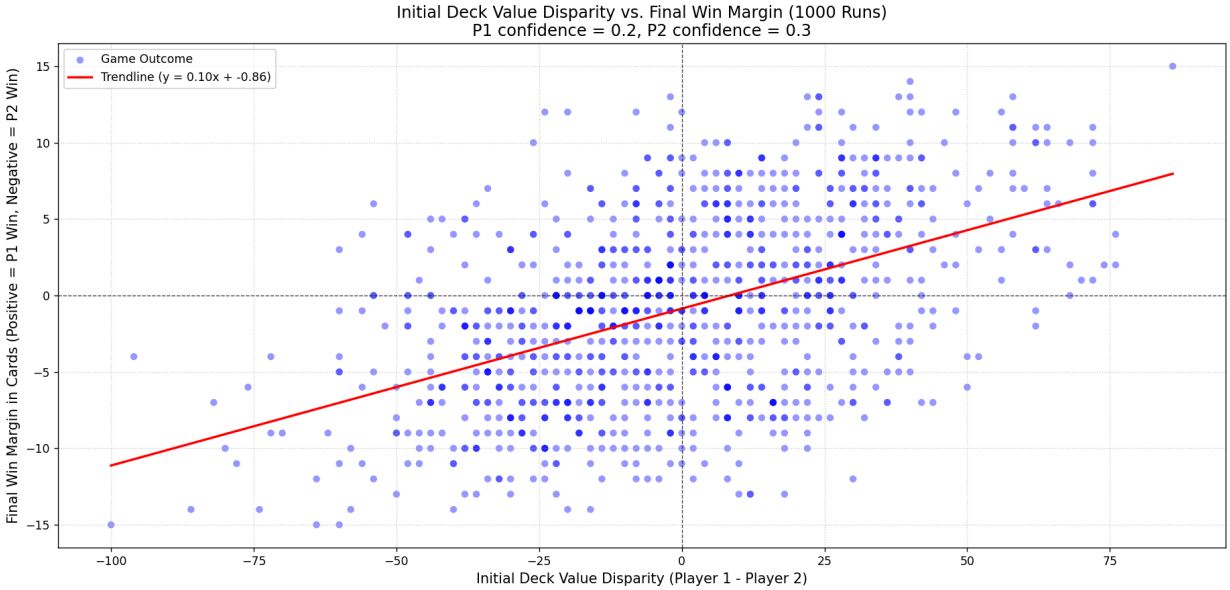
P1 confidence - 0, P2 confidence - 0.3



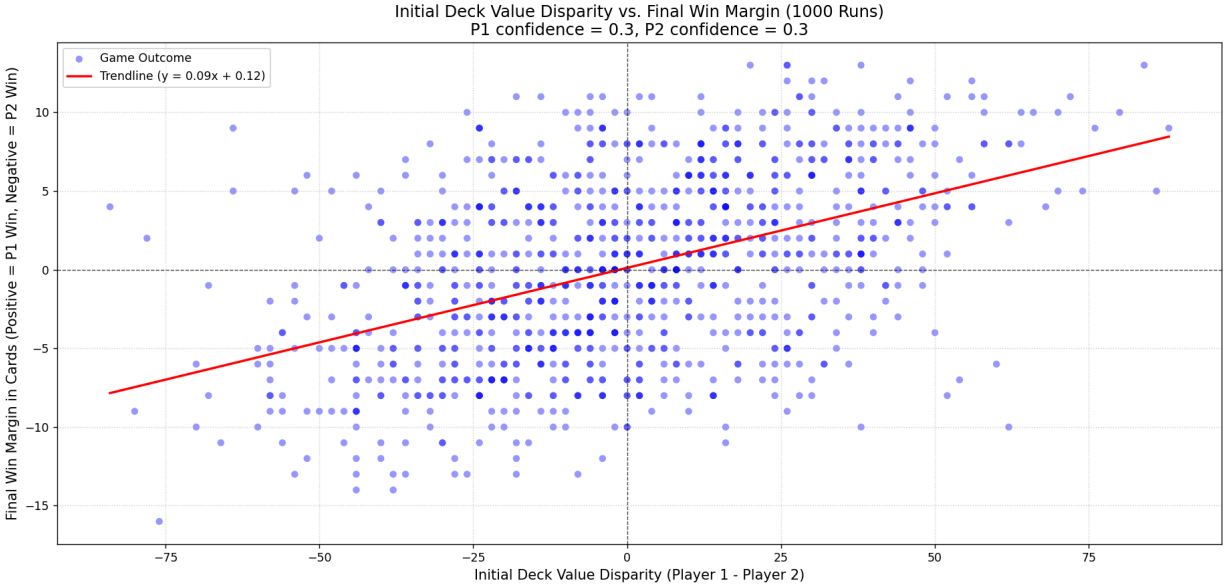
P1 confidence - 0.1, P2 confidence - 0.3



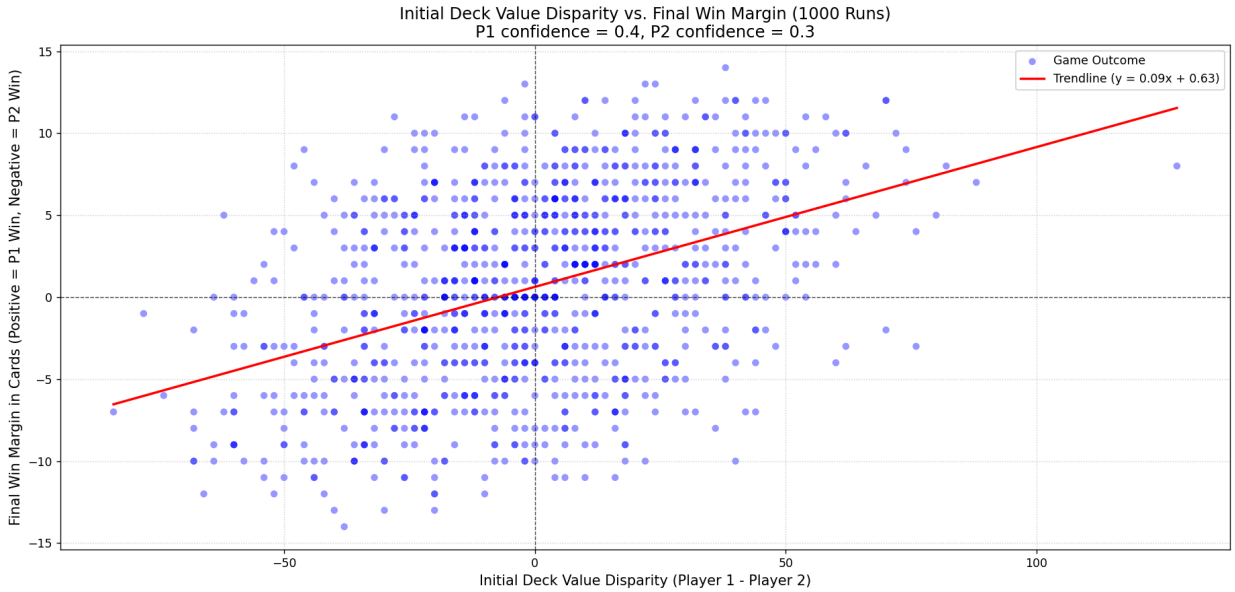
P1 confidence - 0.2, P2 confidence - 0.3



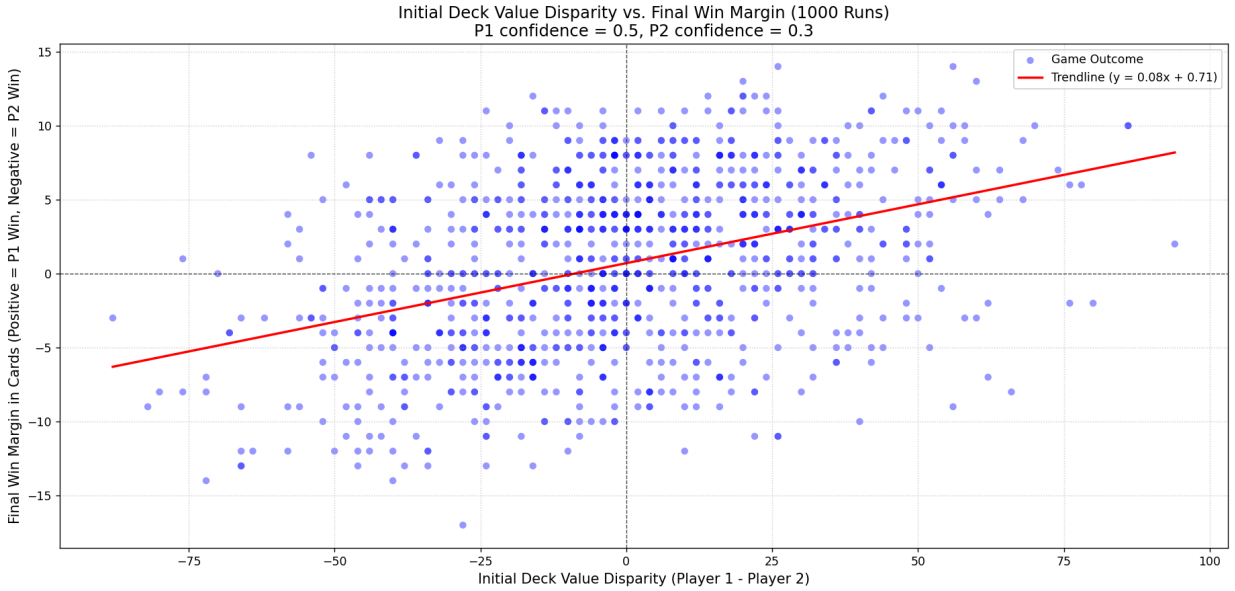
P1 confidence - 0.3, P2 confidence - 0.3



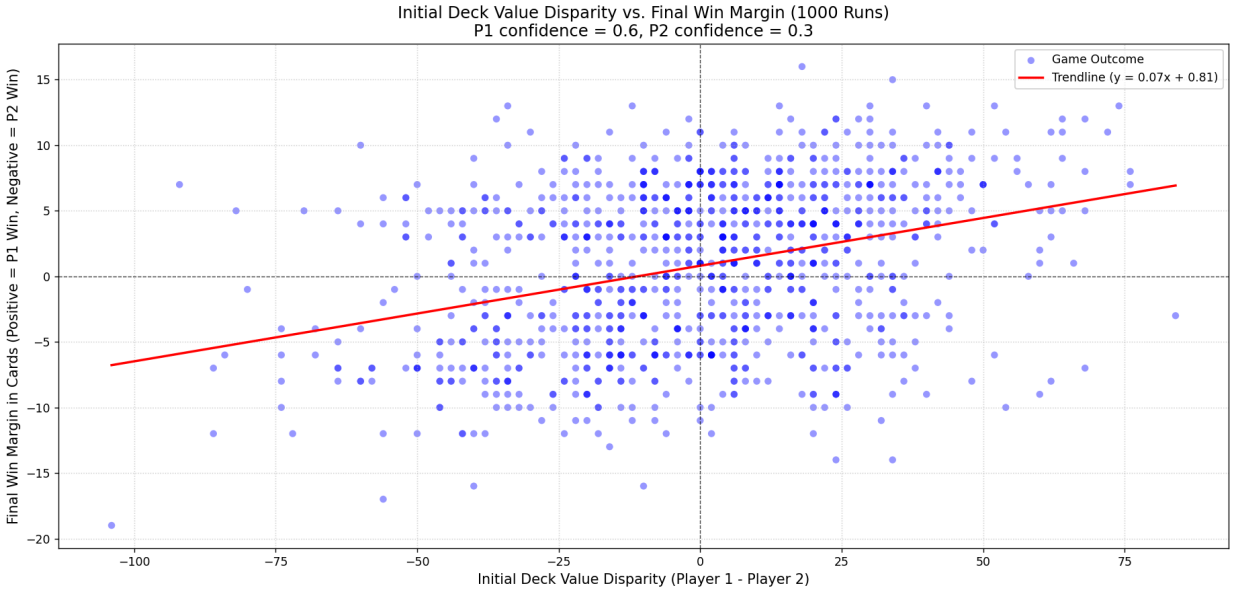
P1 confidence - 0.4, P2 confidence - 0.3



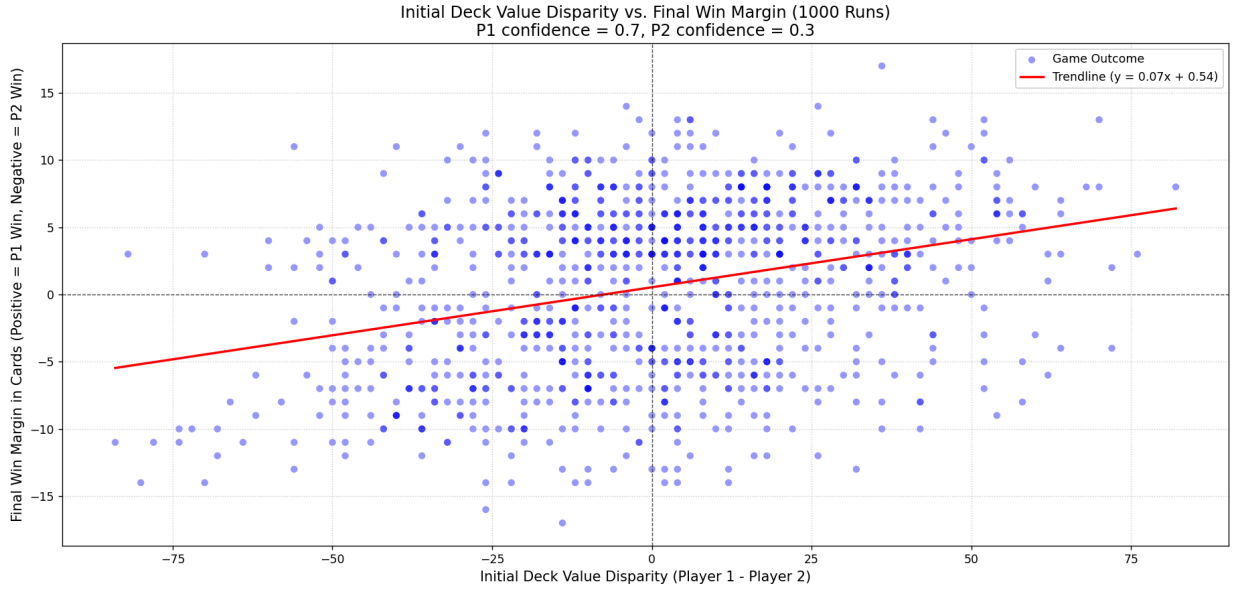
P1 confidence - 0.5, P2 confidence - 0.3



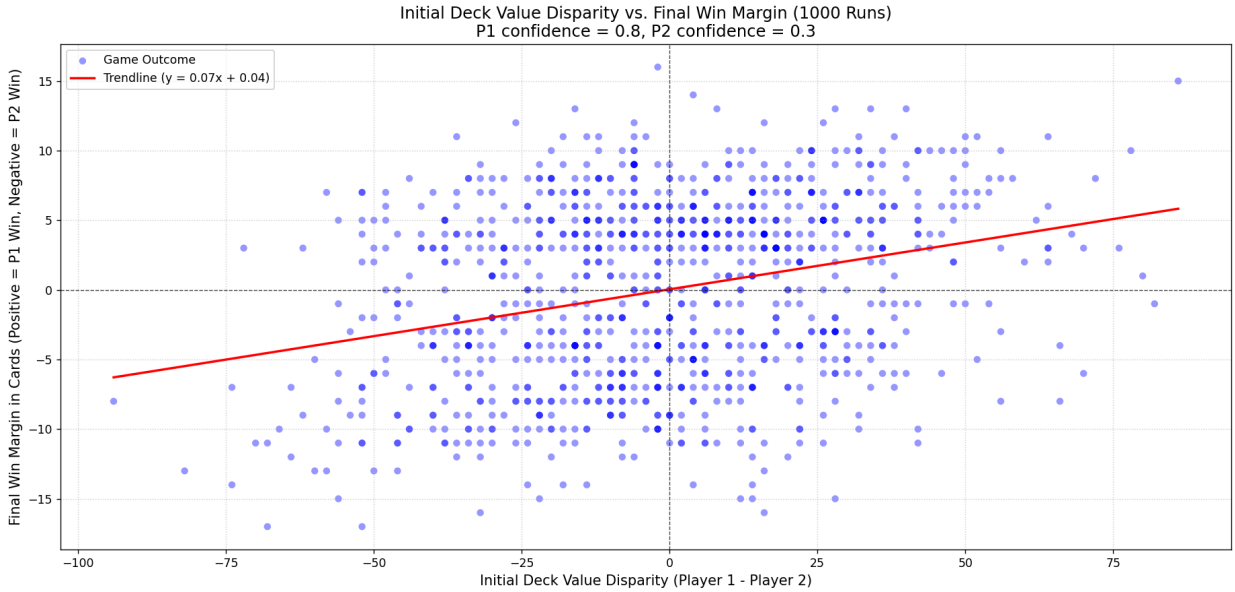
P1 confidence - 0.6, P2 confidence - 0.3



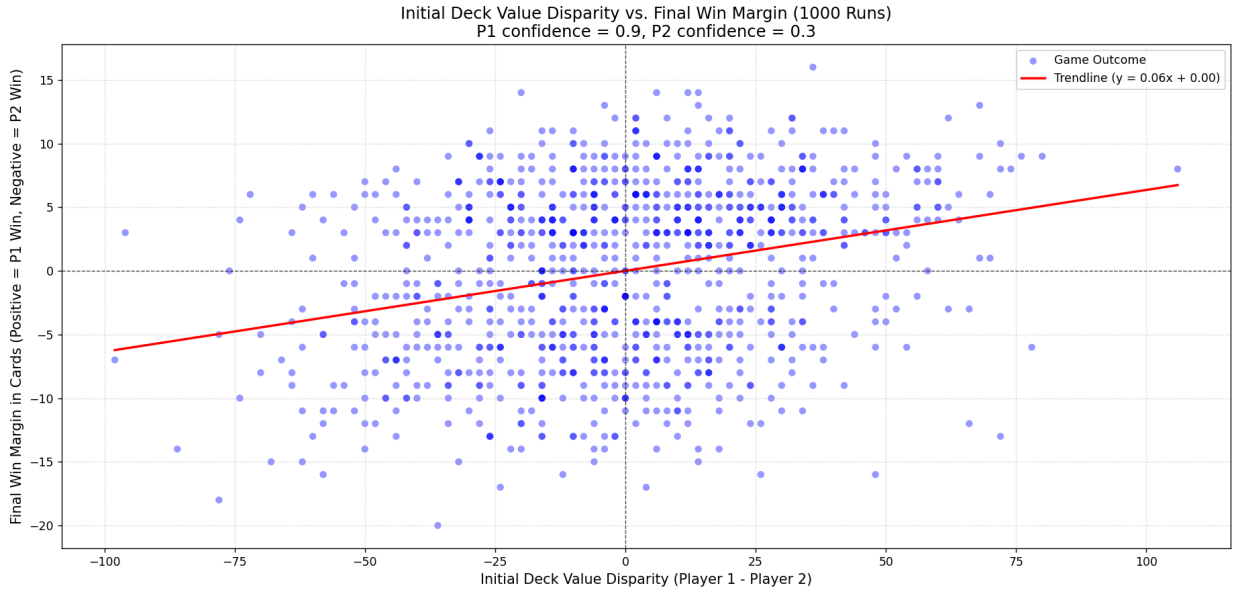
P1 confidence - 0.7, P2 confidence - 0.3



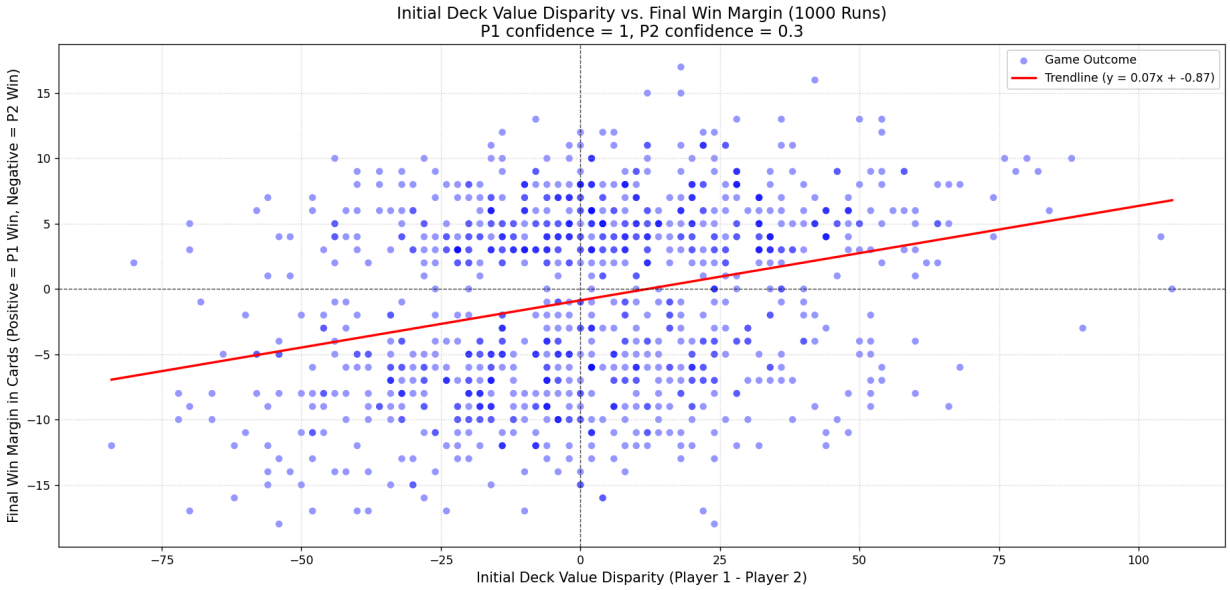
P1 confidence - 0.8, P2 confidence - 0.3



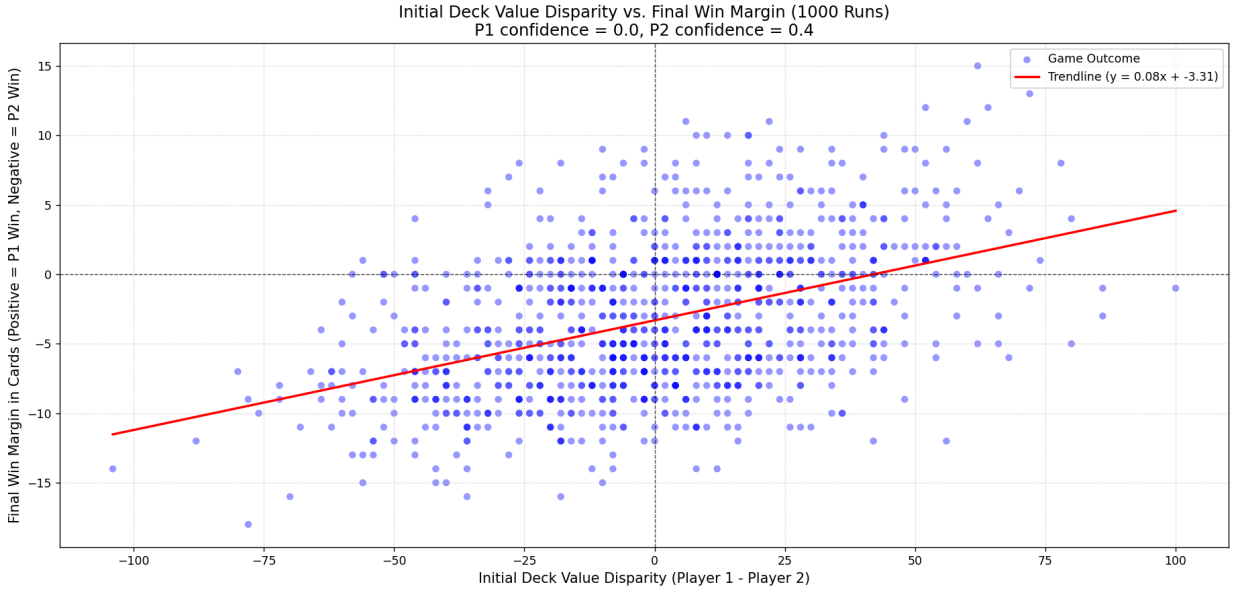
P1 confidence - 0.9, P2 confidence - 0.3



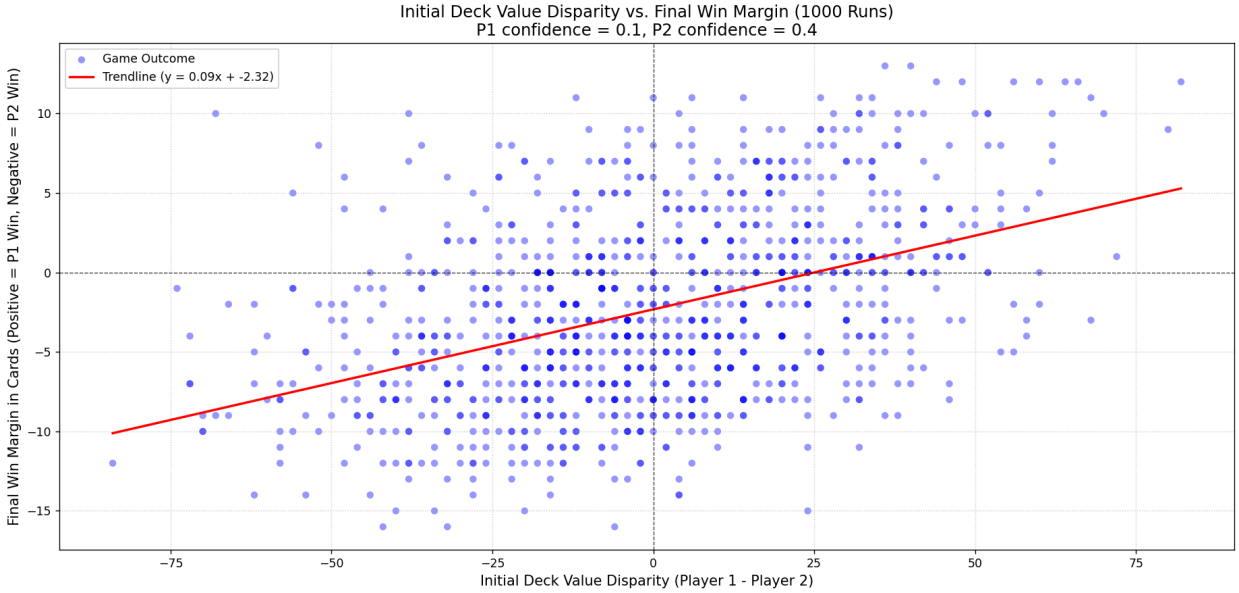
P1 confidence - 1, P2 confidence - 0.3


Initial Deck Value Disparity vs. Final Win Margin (1000 Runs)
P1 confidence = 1, P2 confidence = 0.3
Game Outcome
Trendline (y = 0.07x + -0.87)
Final Win Margin in Cards (Positive = P1 Win, Negative = P2 Win)
Initial Deck Value Disparity (Player 1 - Player 2)
15
10
5
0
−5
−10
−15
−75
−50
−25
0
25
50
75
100

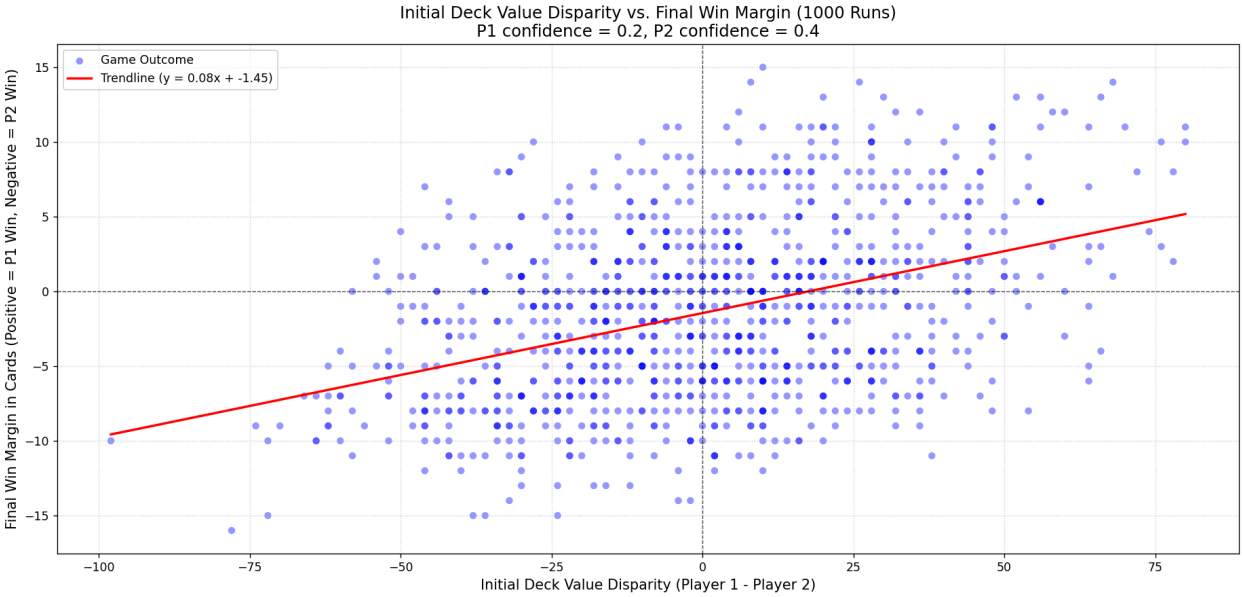
P1 confidence - 0, P2 confidence - 0.4



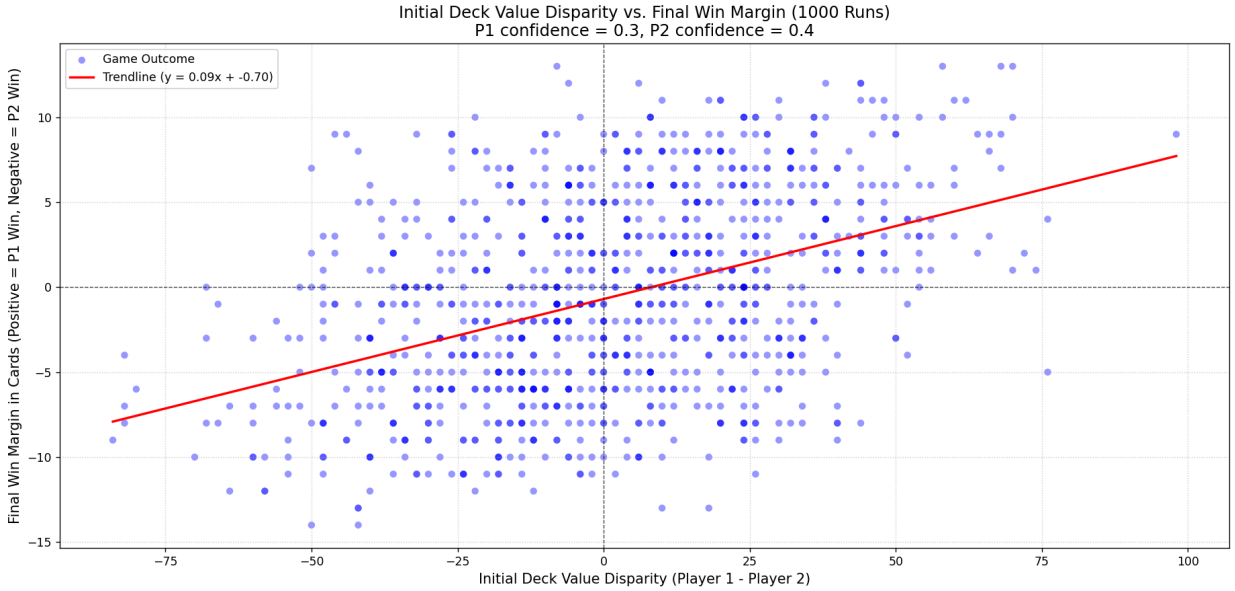
P1 confidence - 0.1, P2 confidence - 0.4



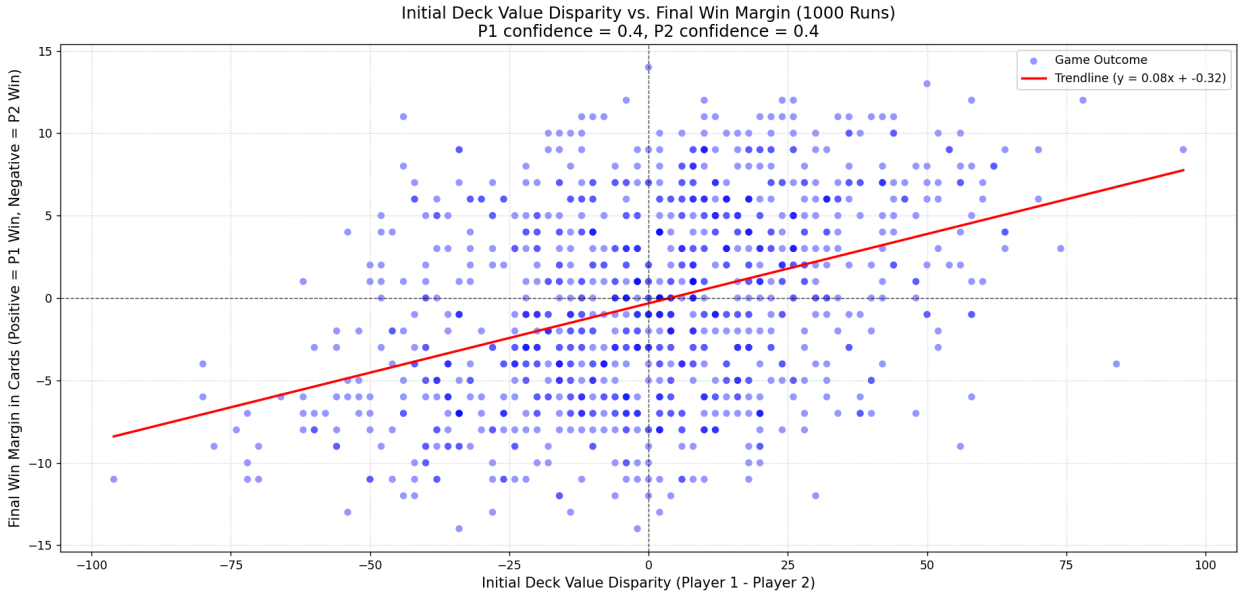
P1 confidence - 0.2, P2 confidence - 0.4



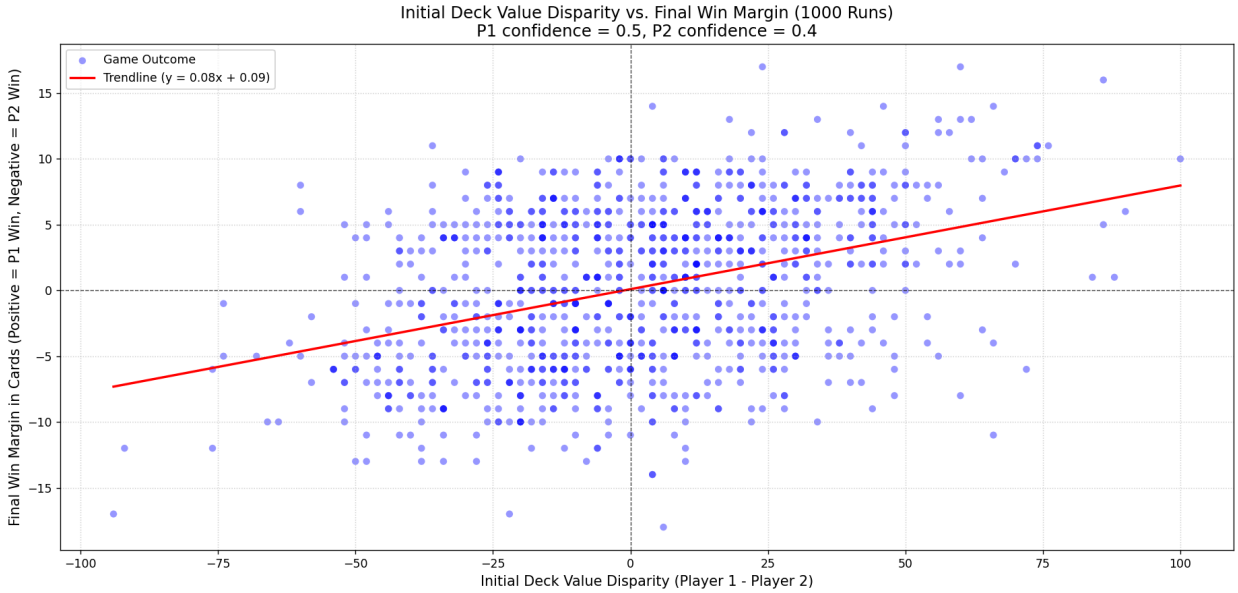
P1 confidence - 0.3, P2 confidence - 0.4



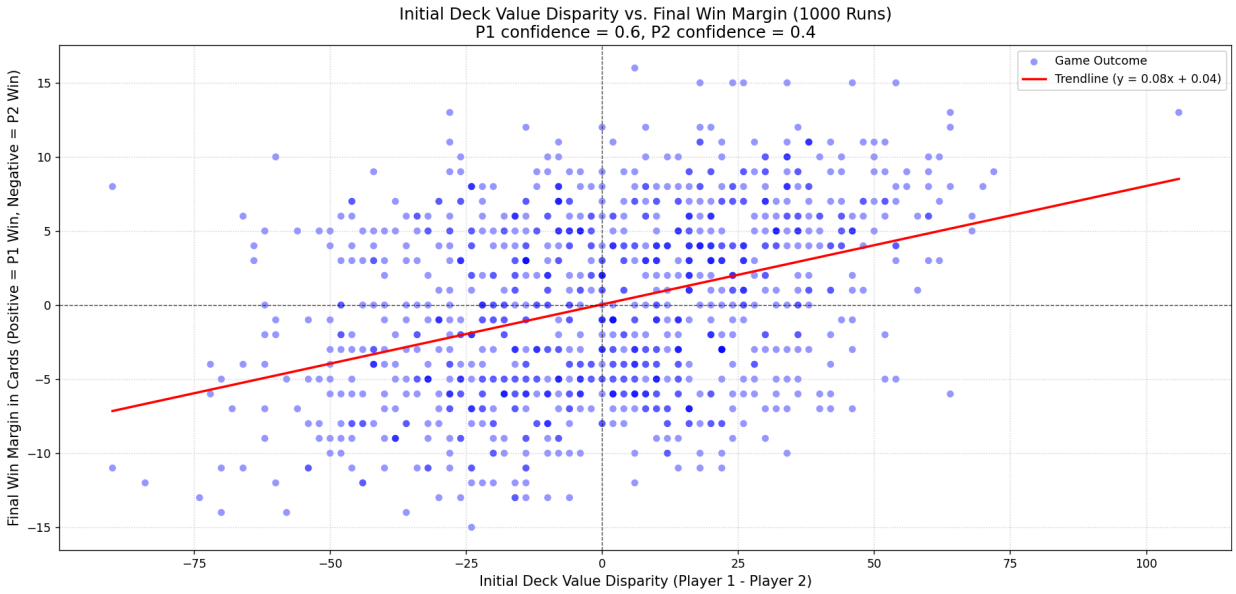
P1 confidence - 0.4, P2 confidence - 0.4



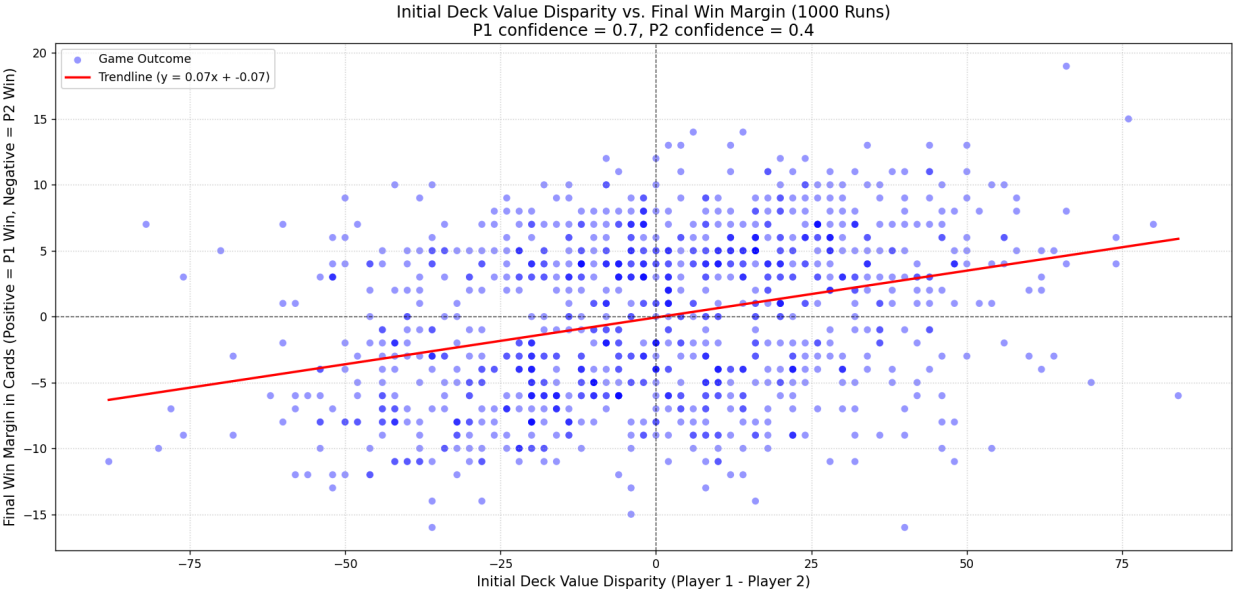
P1 confidence - 0.5, P2 confidence - 0.4



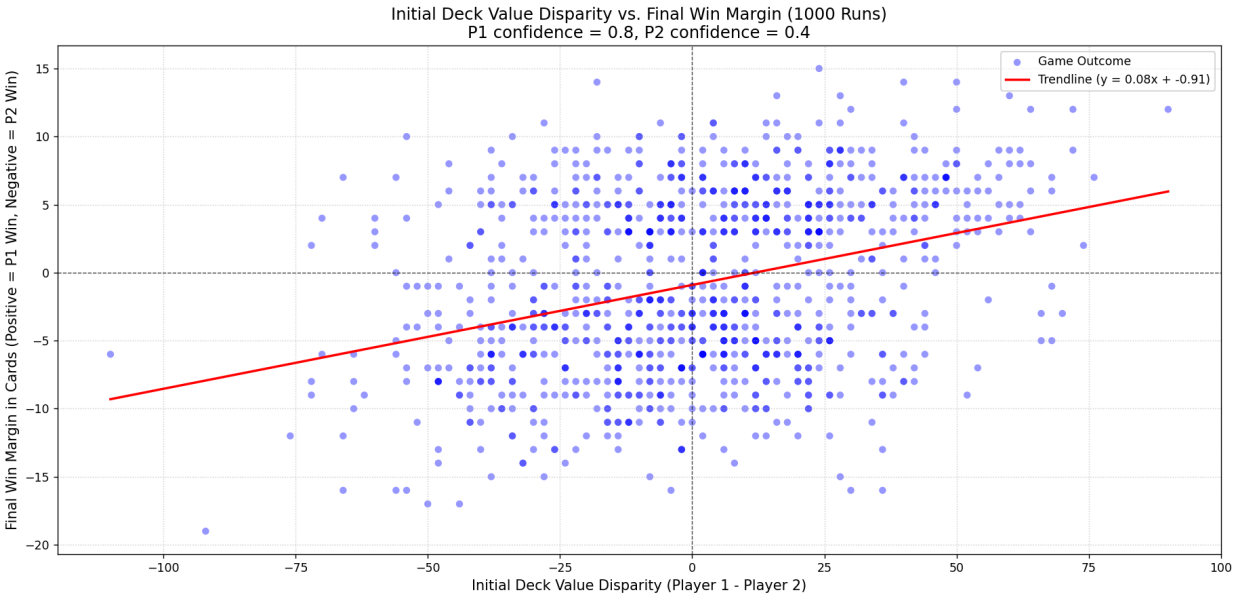
P1 confidence - 0.6, P2 confidence - 0.4



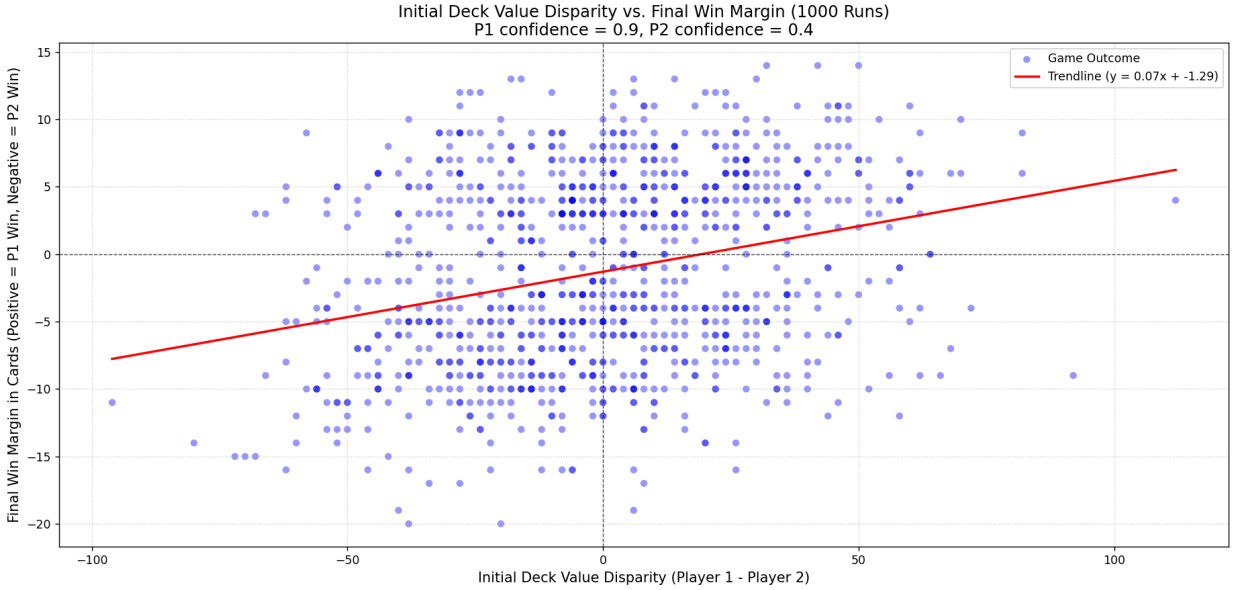
P1 confidence - 0.7, P2 confidence - 0.4



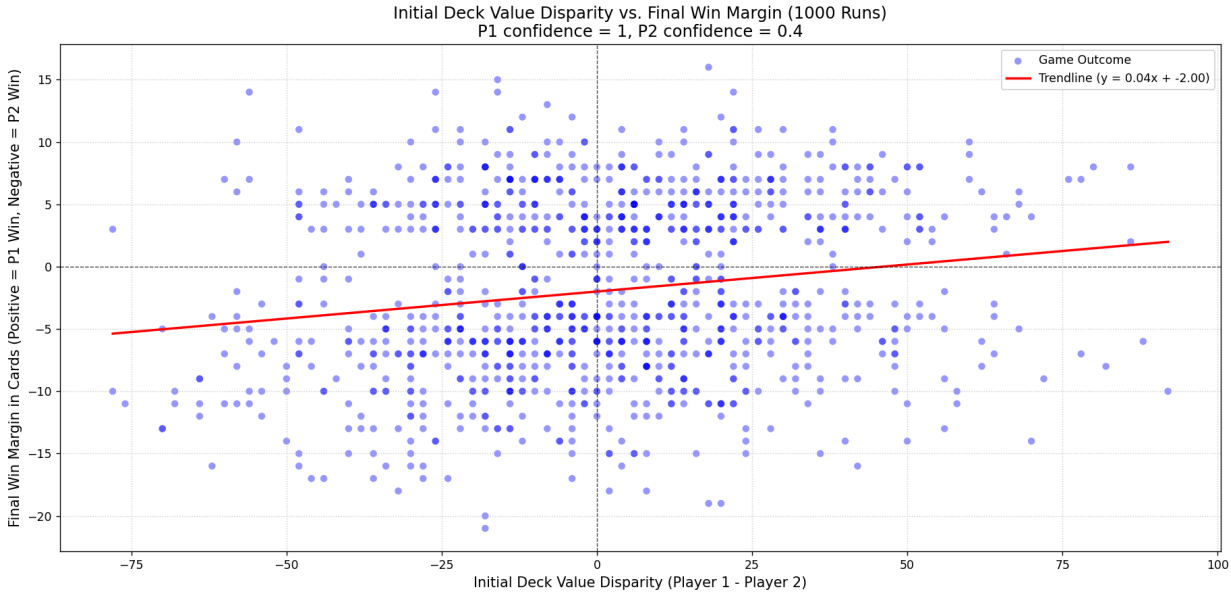
P1 confidence - 0.8, P2 confidence - 0.4



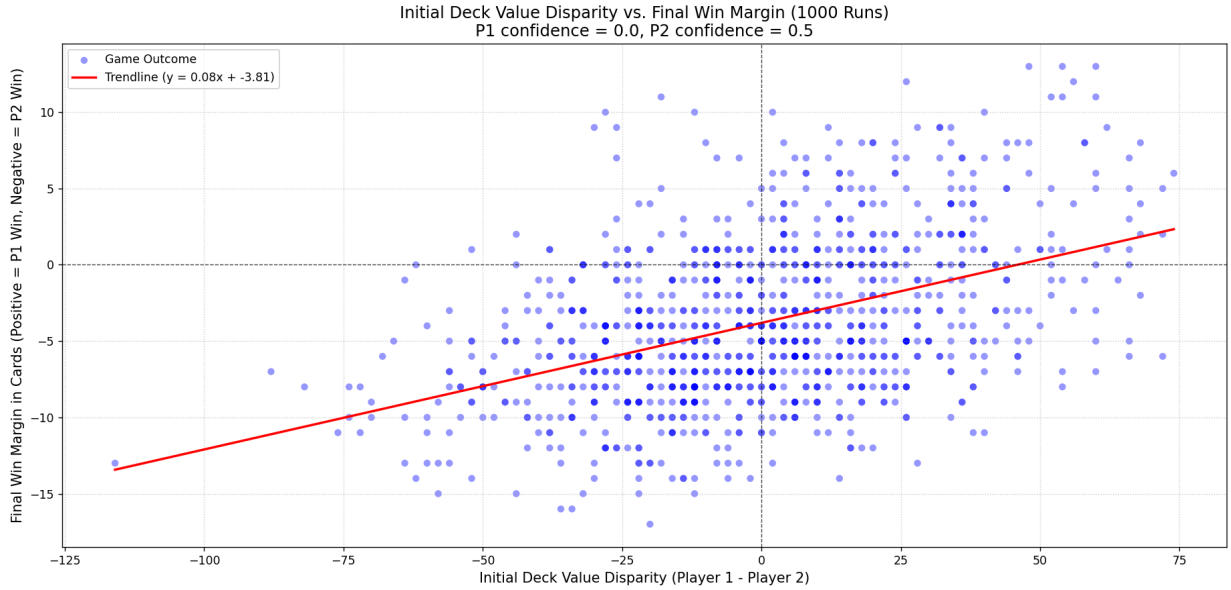
P1 confidence - 0.9, P2 confidence - 0.4



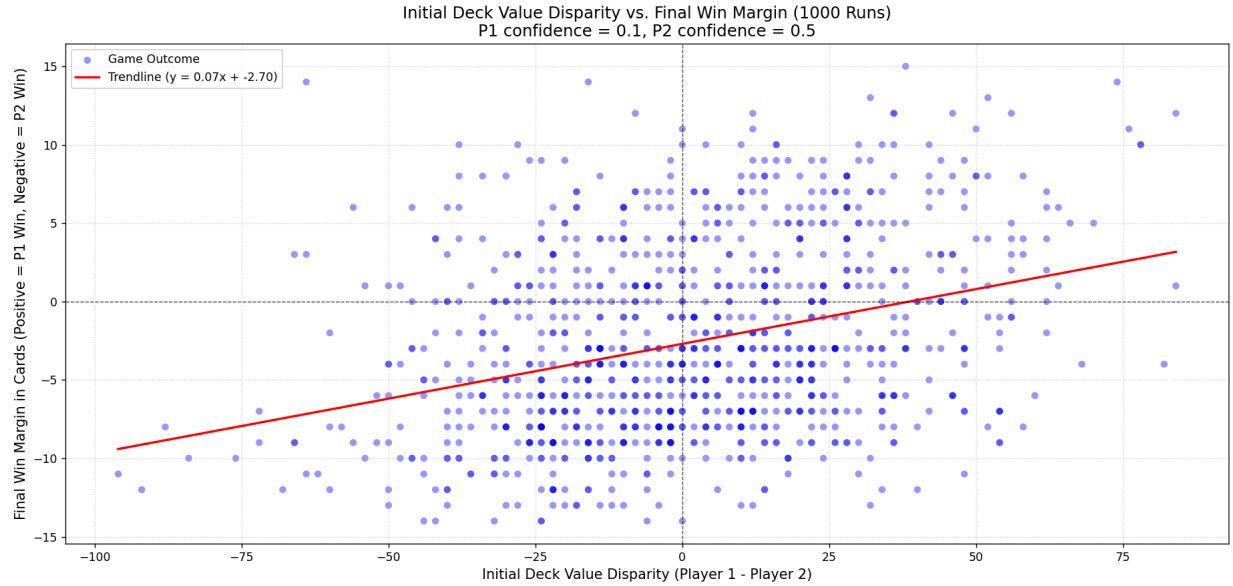
P1 confidence - 1, P2 confidence - 0.4


Initial Deck Value Disparity vs. Final Win Margin (1000 Runs)
P1 confidence = 1, P2 confidence = 0.4
Game Outcome
Trendline (y = 0.04x + -2.00)
Final Win Margin in Cards (Positive = P1 Win, Negative = P2 Win)
Initial Deck Value Disparity (Player 1 - Player 2)
15
10
5
0
−5
−10
−15
−20
−75
−50
−25
0
25
50
75
100

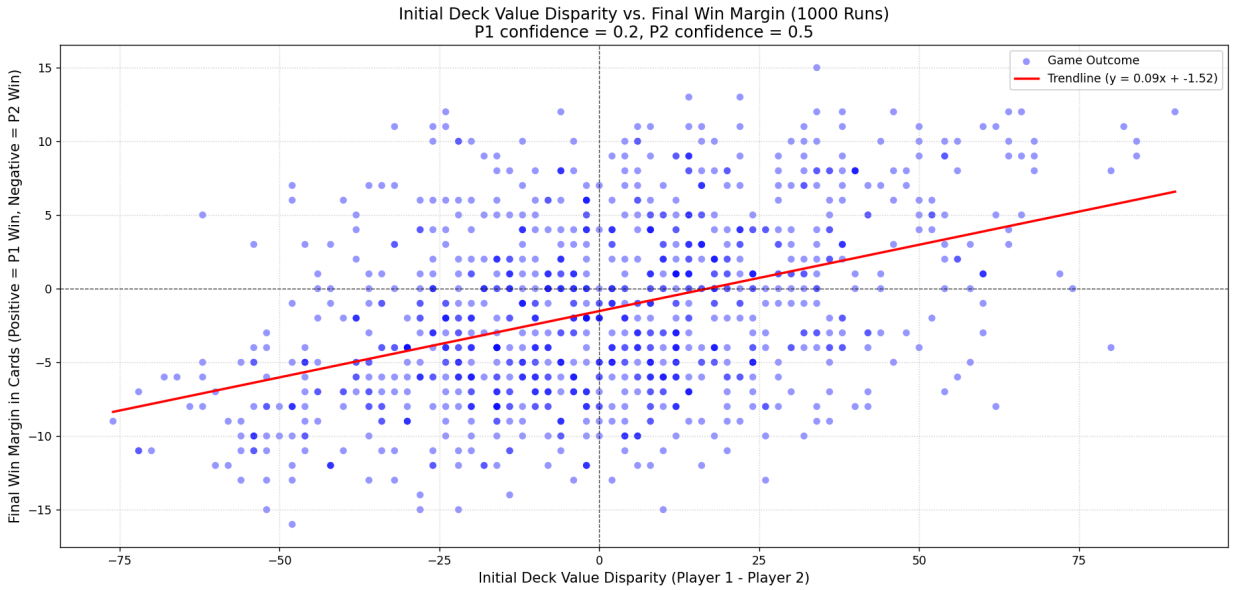
P1 confidence - 0, P2 confidence - 0.5



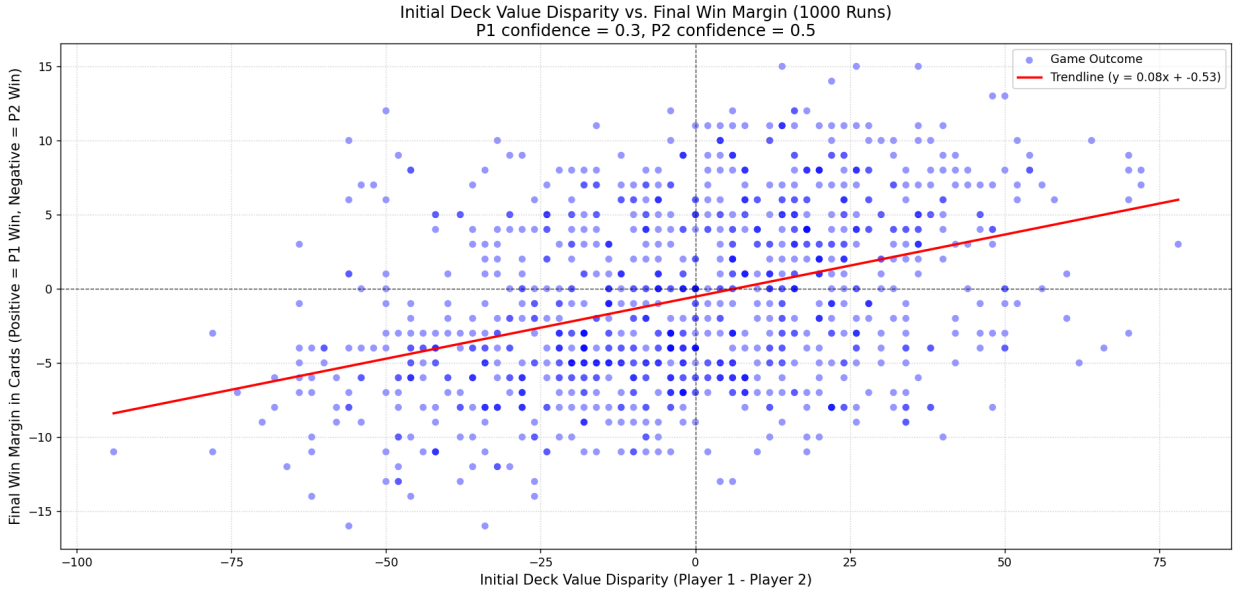
P1 confidence - 0.1, P2 confidence - 0.5



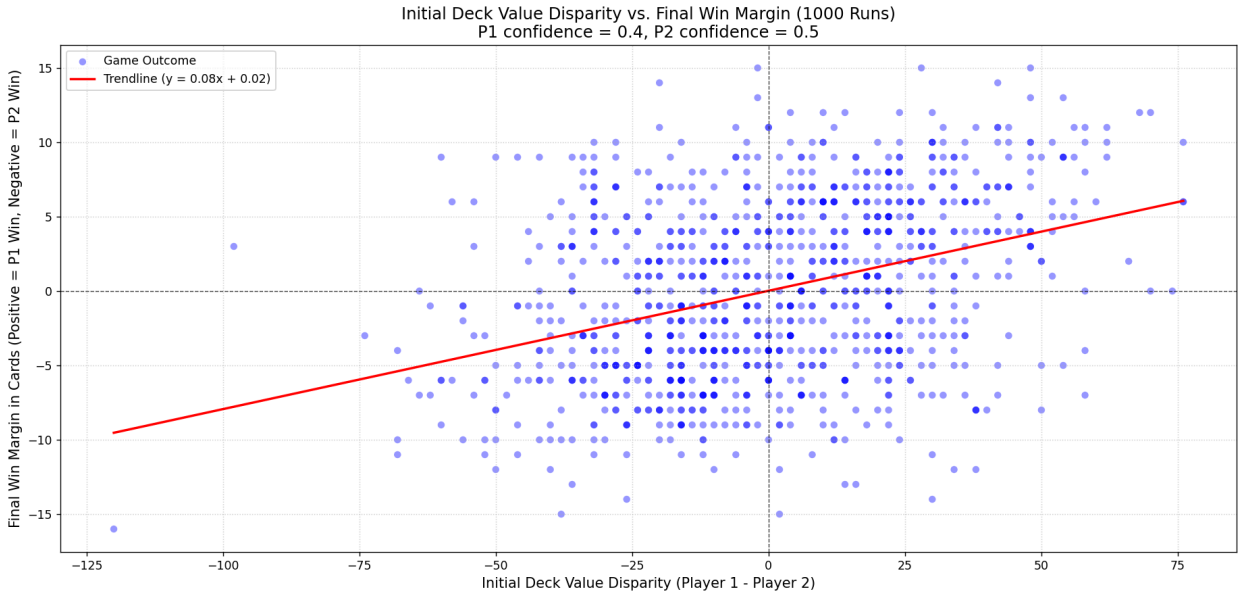
P1 confidence - 0.2, P2 confidence - 0.5



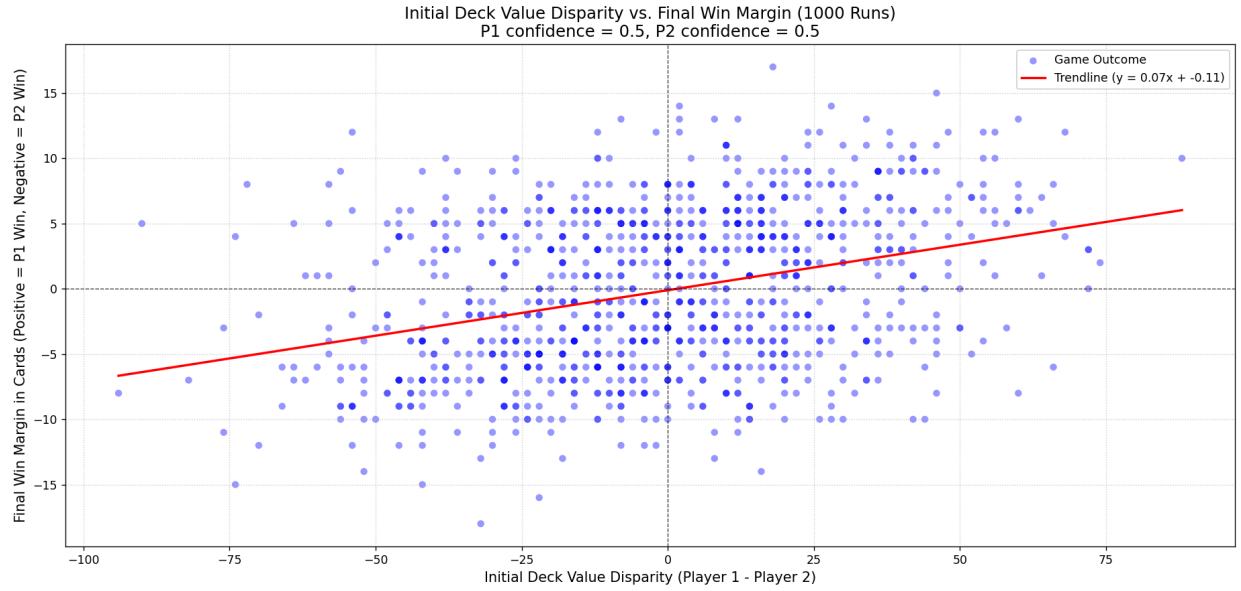
P1 confidence - 0.3, P2 confidence - 0.5



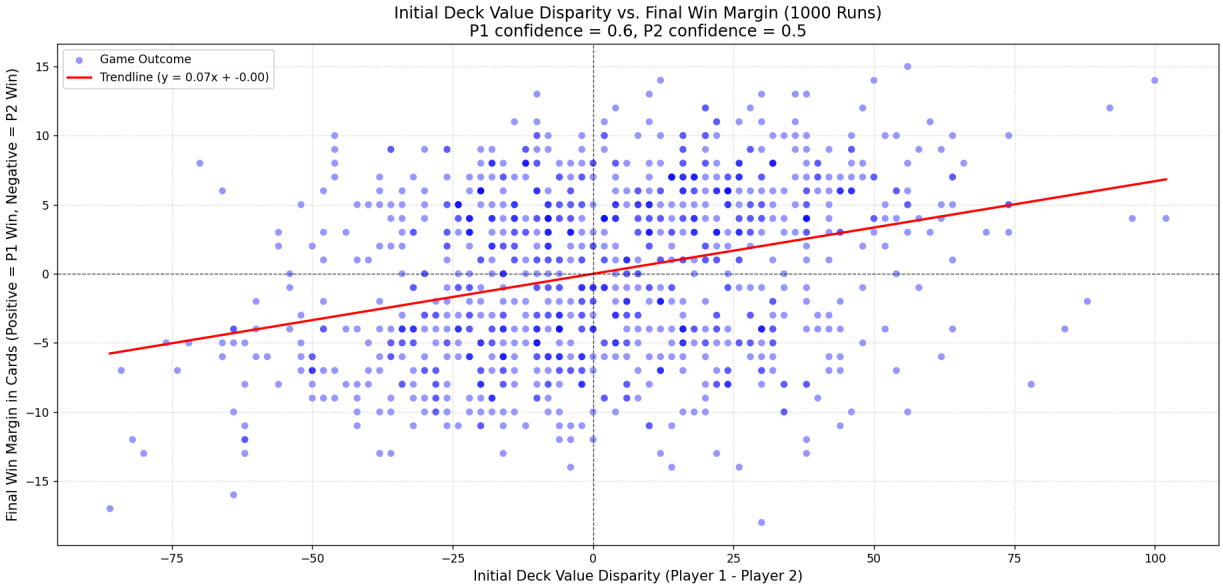
P1 confidence - 0.4, P2 confidence - 0.5



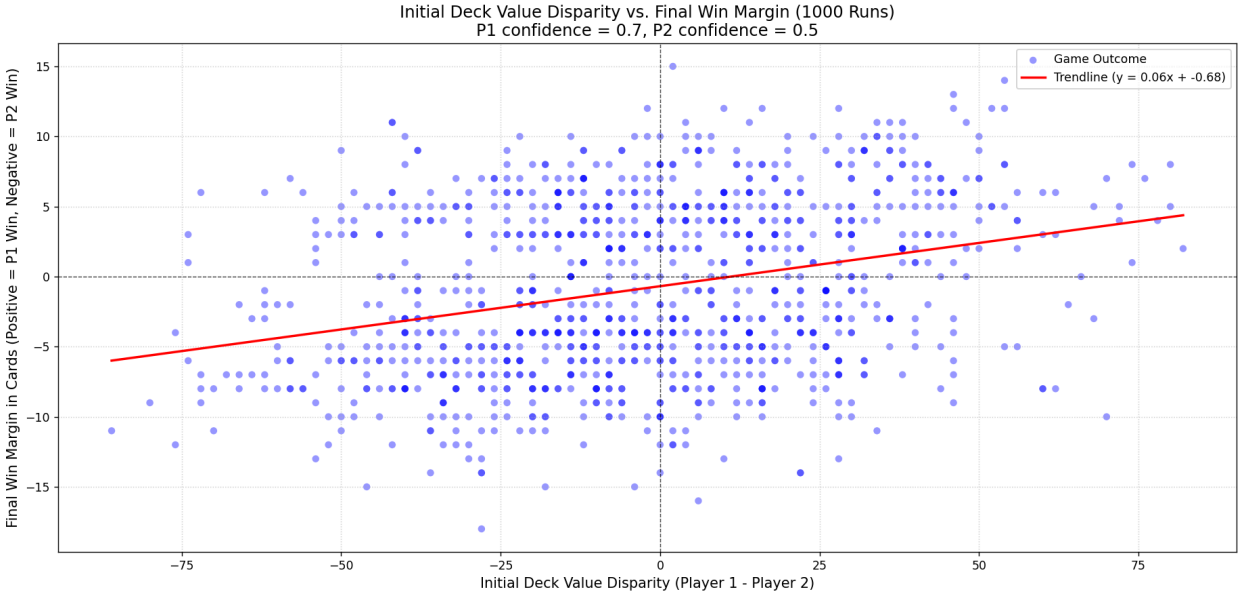
P1 confidence - 0.5, P2 confidence - 0.5



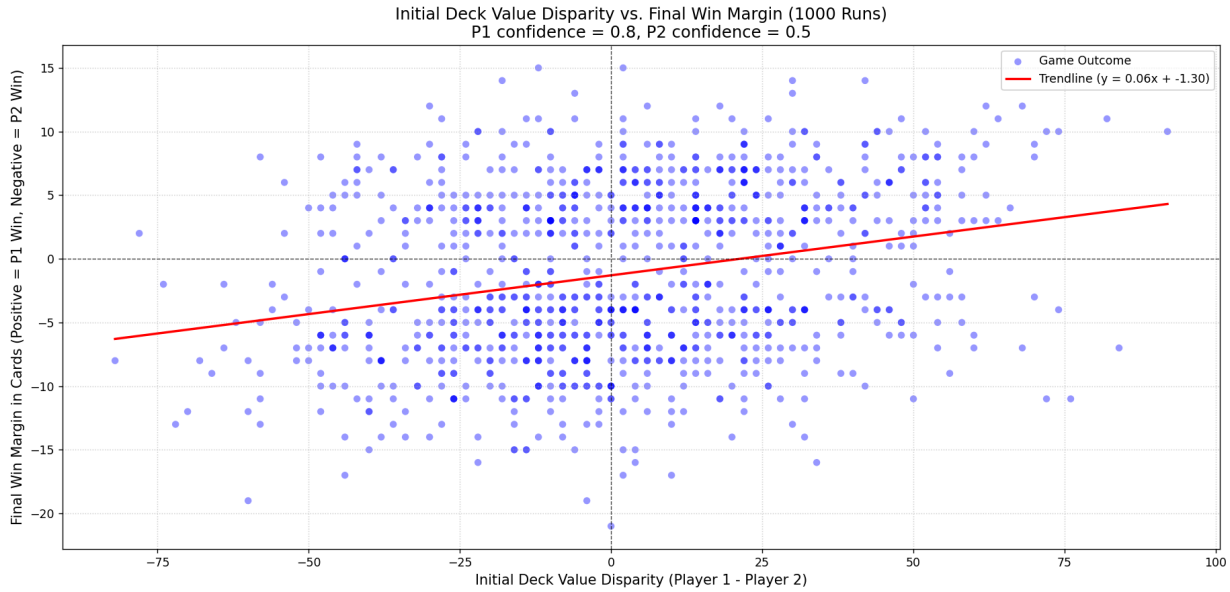
P1 confidence - 0.6, P2 confidence - 0.5



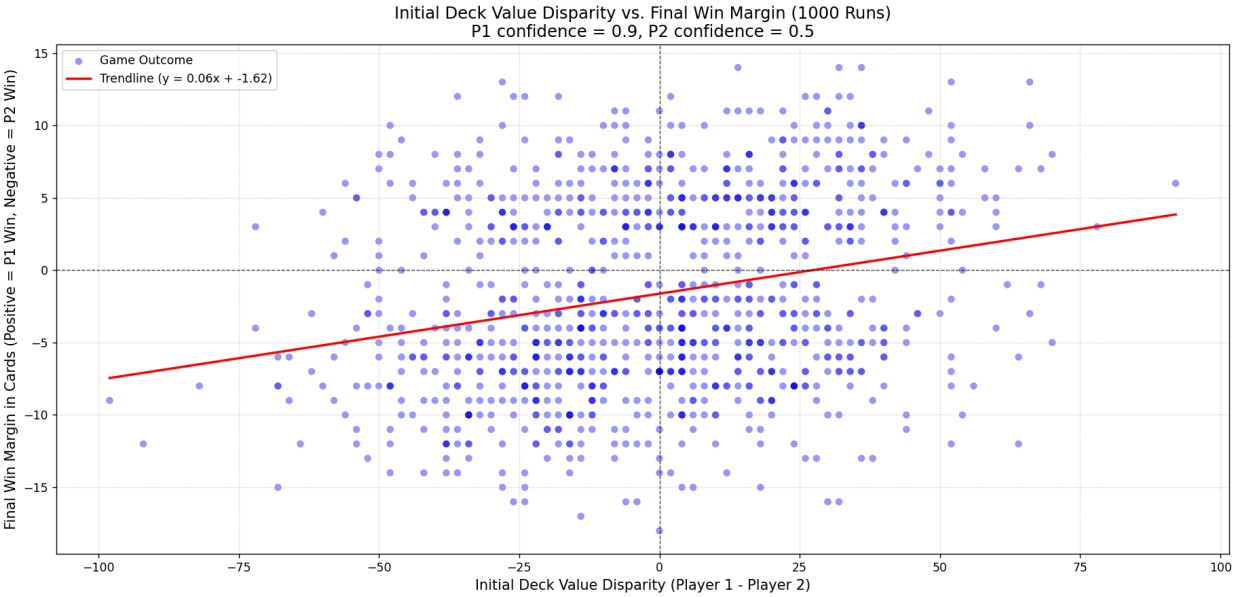
P1 confidence - 0.7, P2 confidence - 0.5



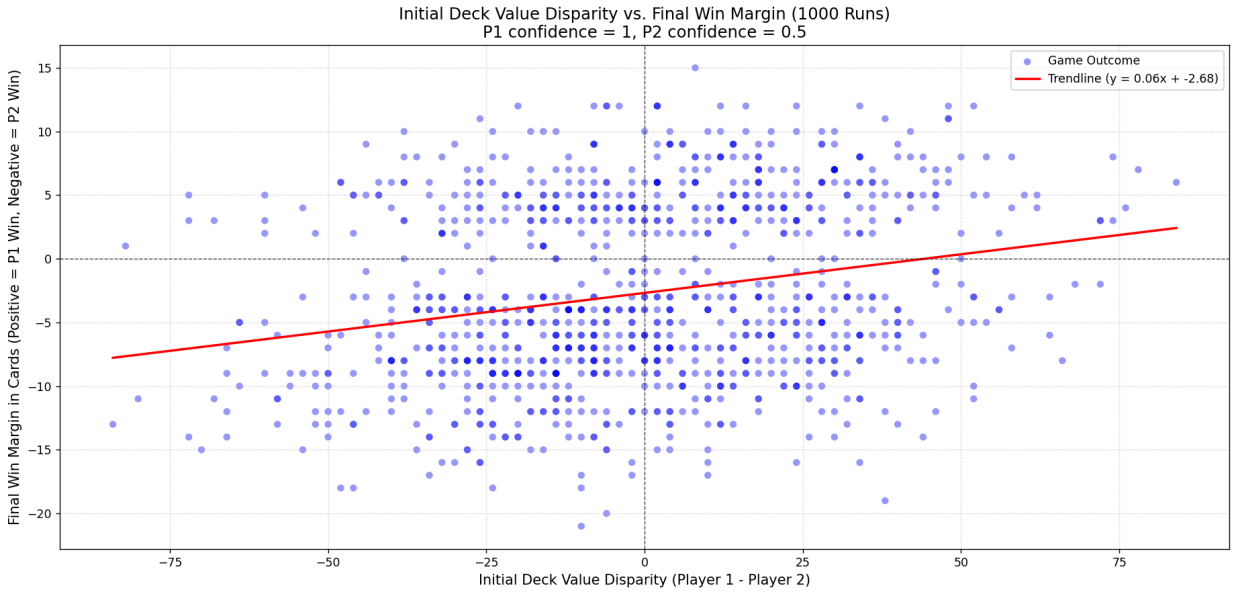
P1 confidence - 0.8, P2 confidence - 0.5



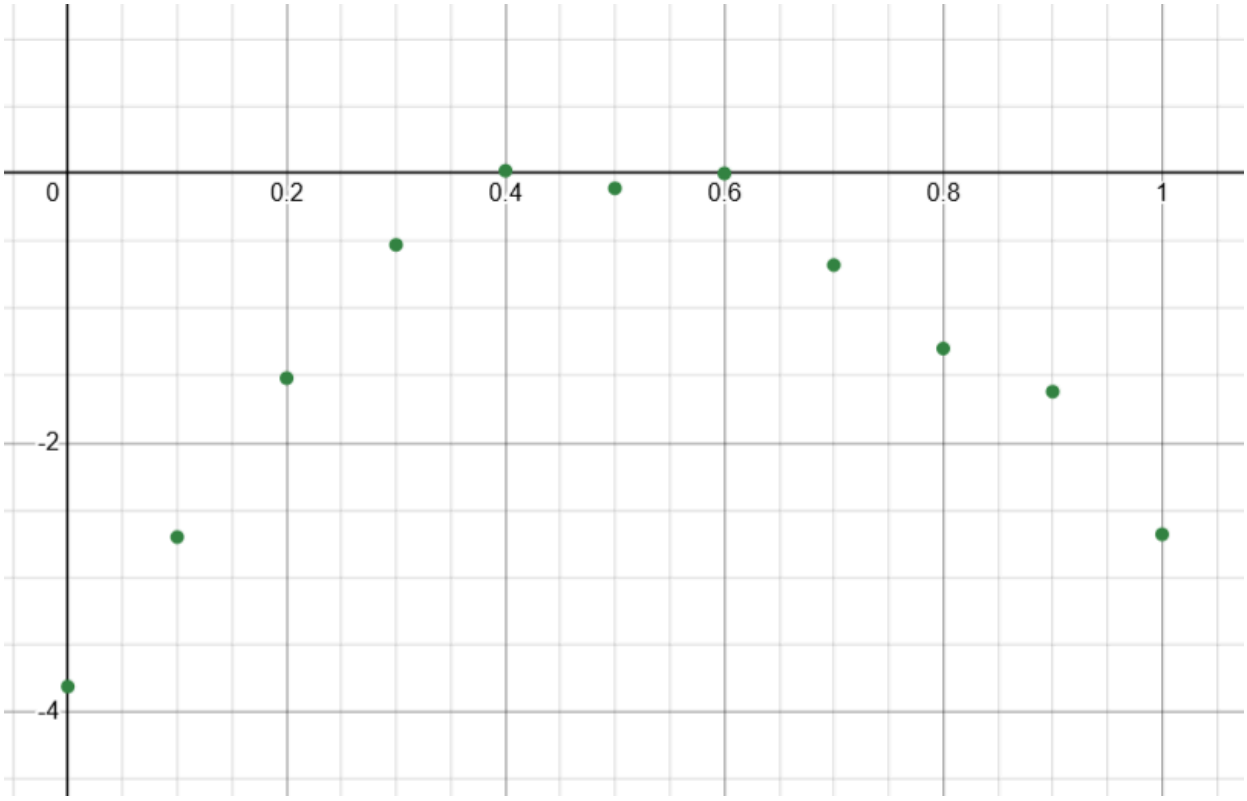
P1 confidence - 0.9, P2 confidence - 0.5



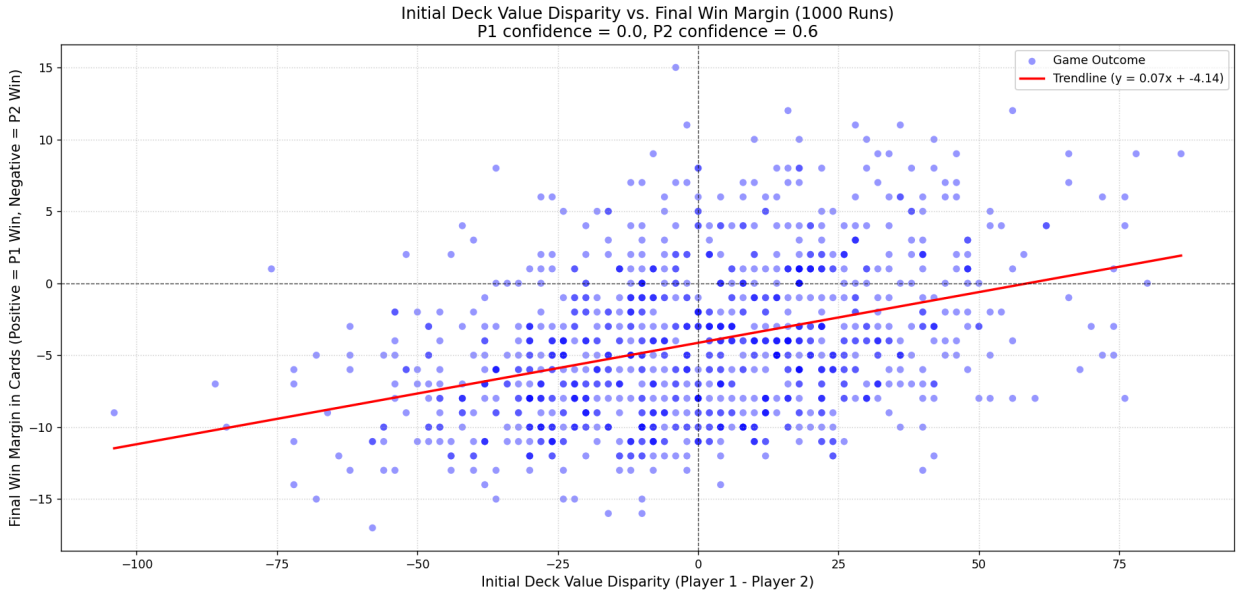
P1 confidence - 1, P2 confidence - 0.5



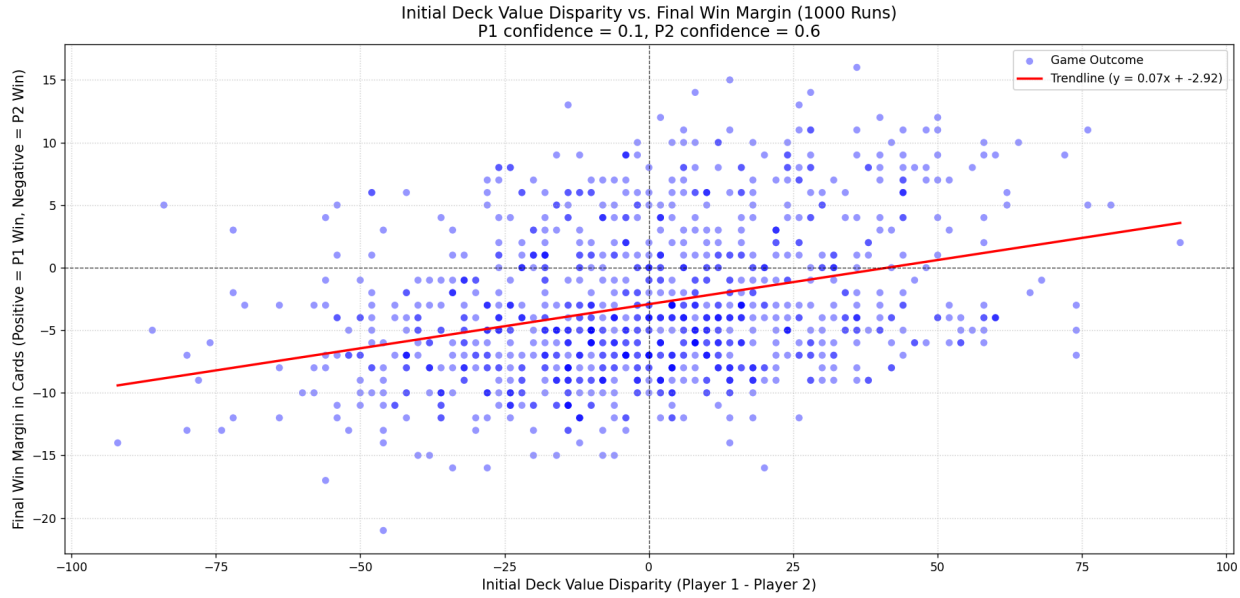
Against someone with a confidence level of 0.5, these are the amount of cards you will win / lose by with different confidence levels.



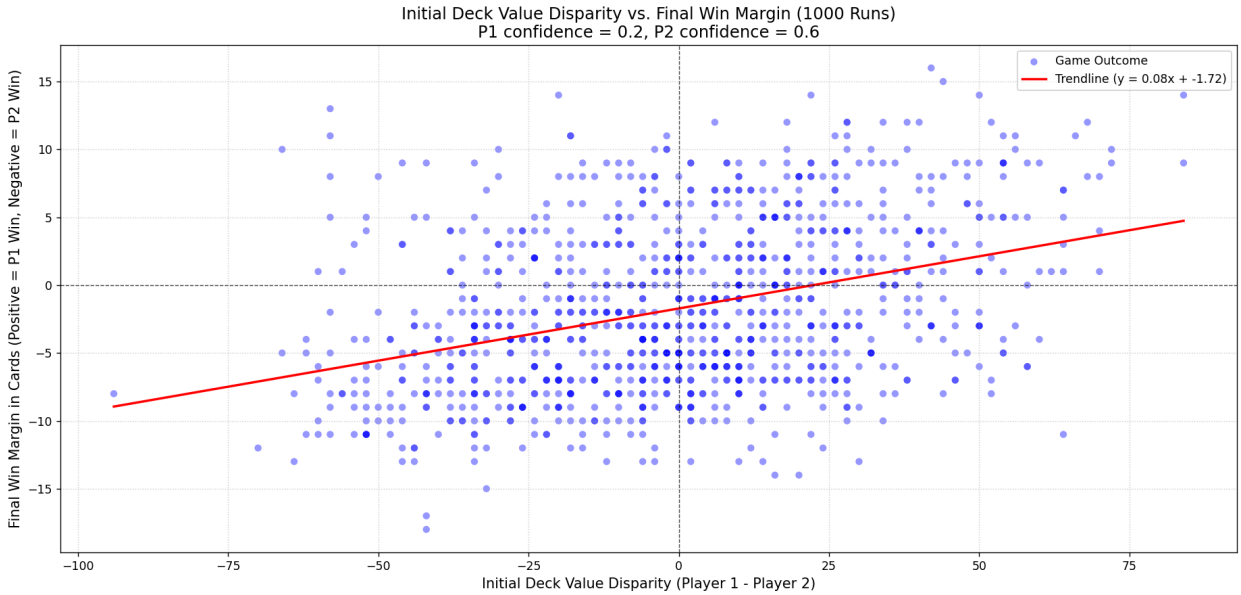
P1 confidence - 0, P2 confidence - 0.6



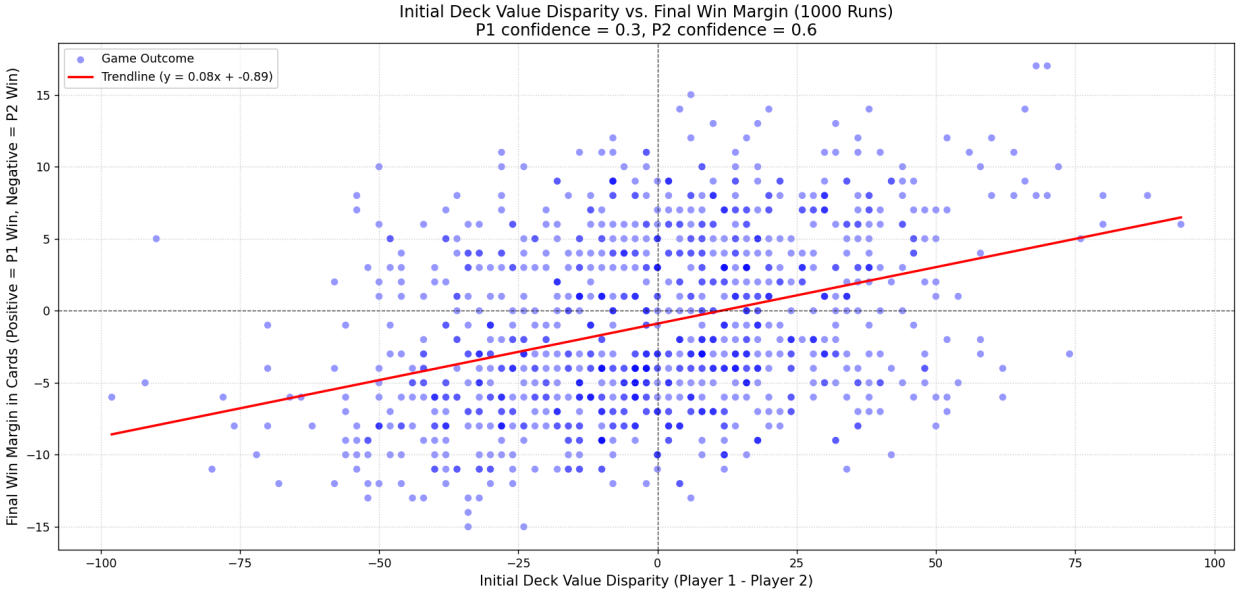
P1 confidence - 0.1, P2 confidence - 0.6



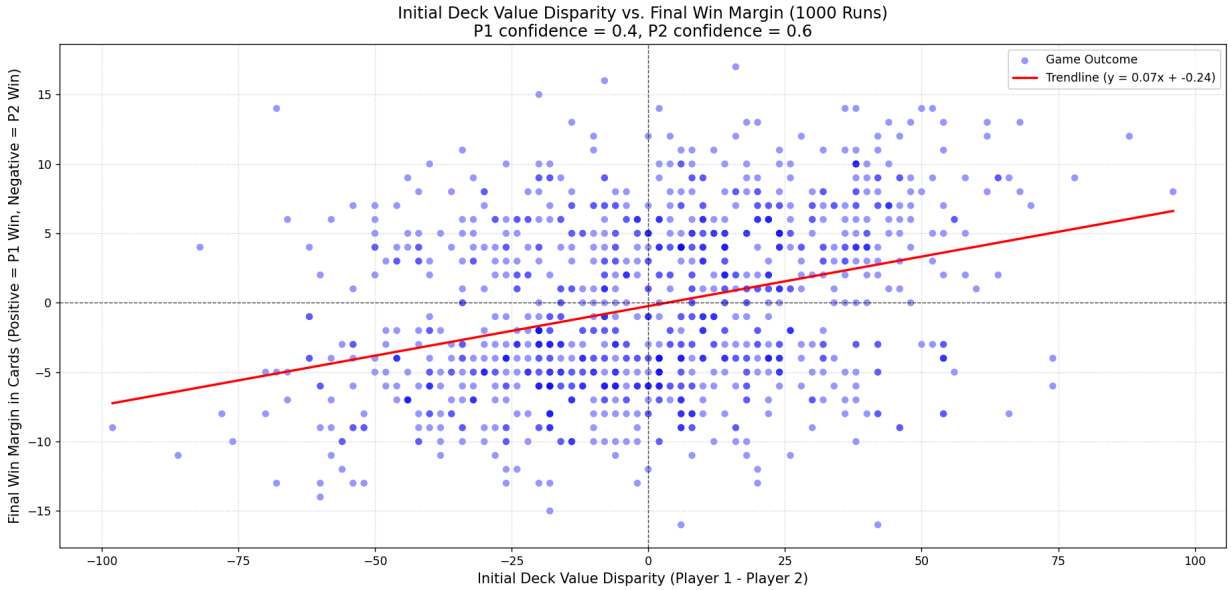
P1 confidence - 0.2, P2 confidence - 0.6



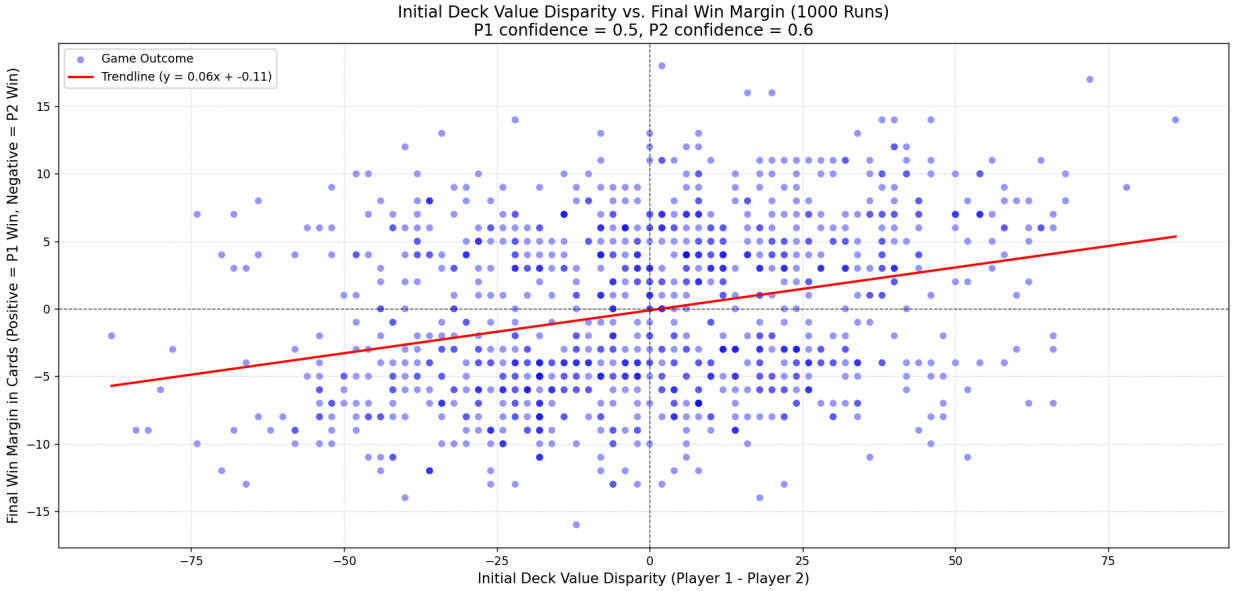
P1 confidence - 0.3, P2 confidence - 0.6



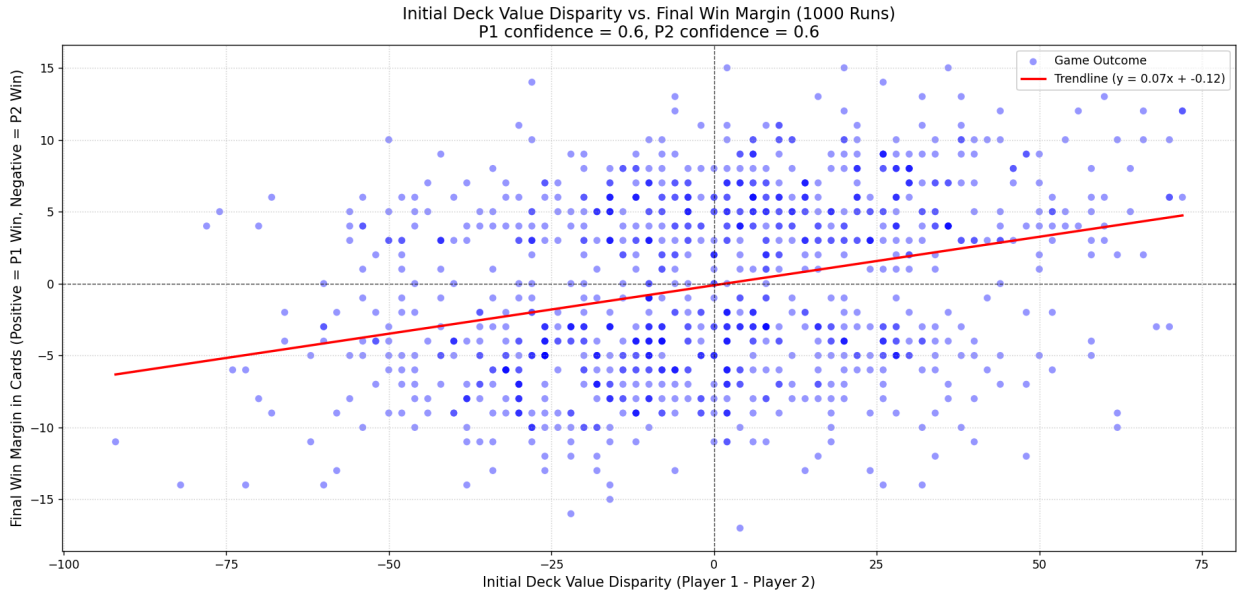
P1 confidence - 0.4, P2 confidence - 0.6



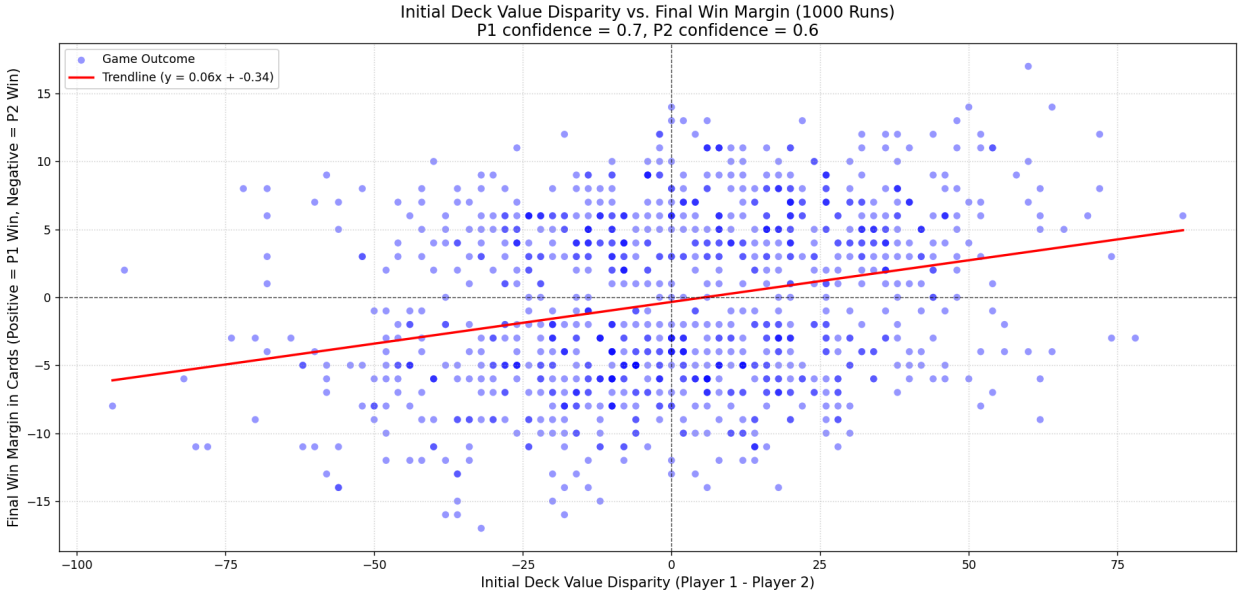
P1 confidence - 0.5, P2 confidence - 0.6



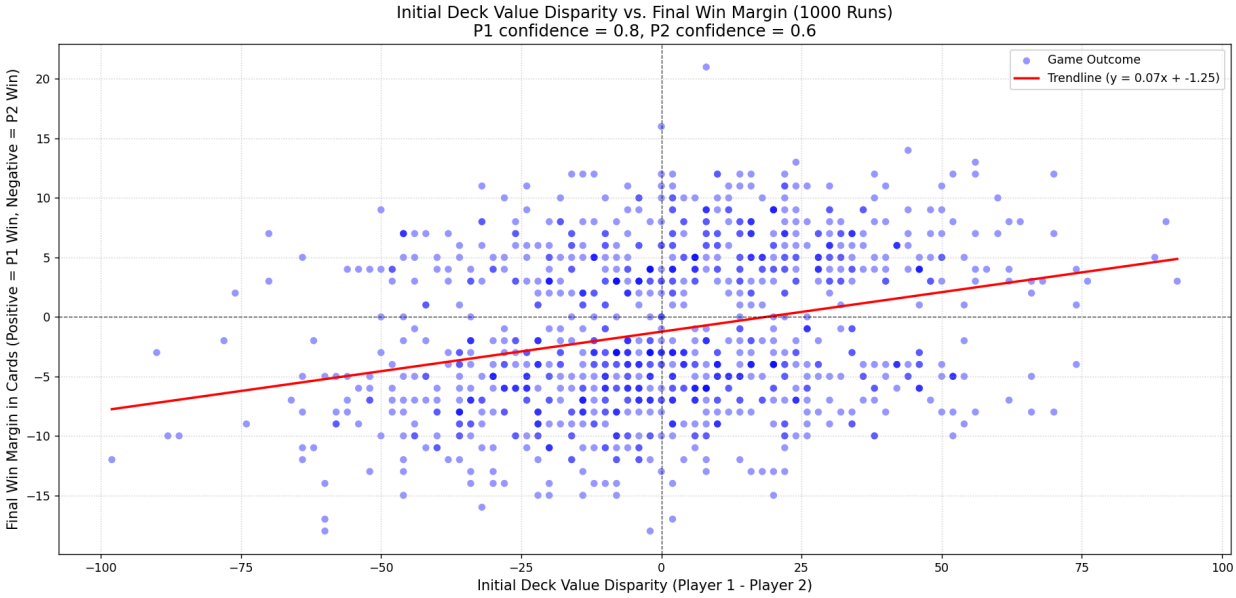
P1 confidence - 0.6, P2 confidence - 0.6



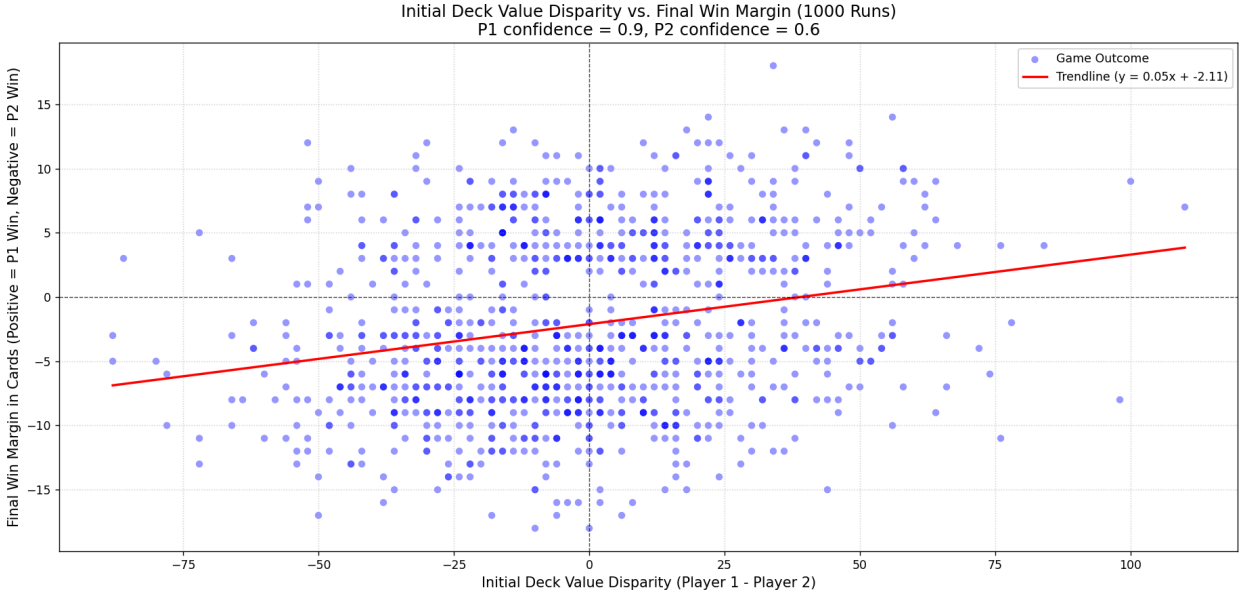
P1 confidence - 0.7, P2 confidence - 0.6



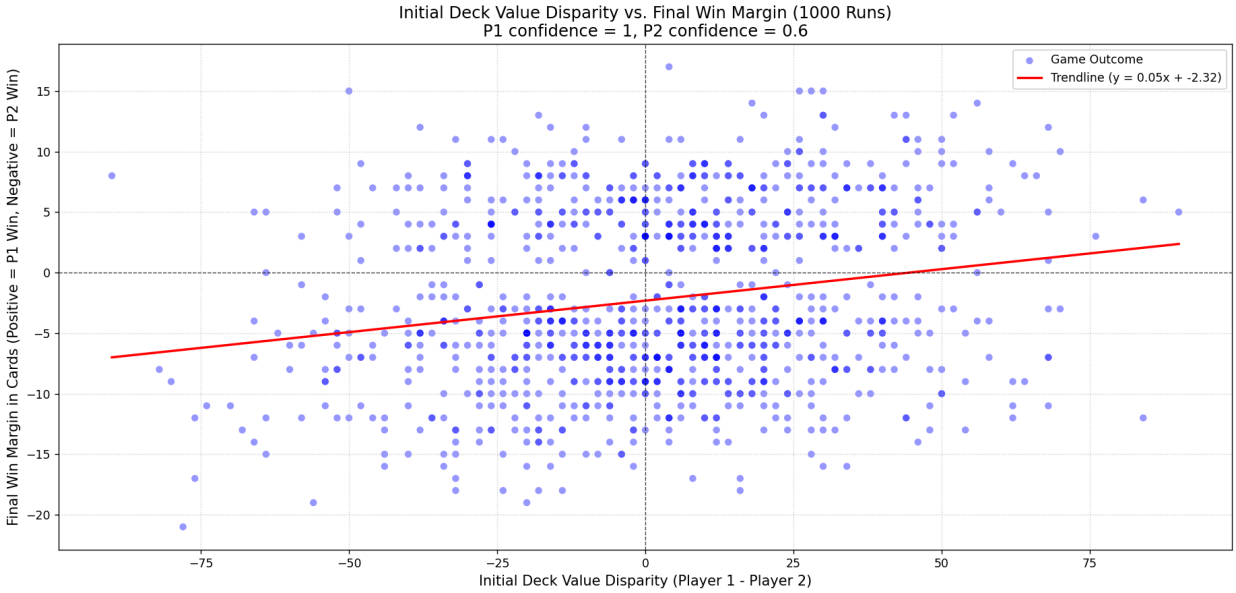
P1 confidence - 0.8, P2 confidence - 0.6



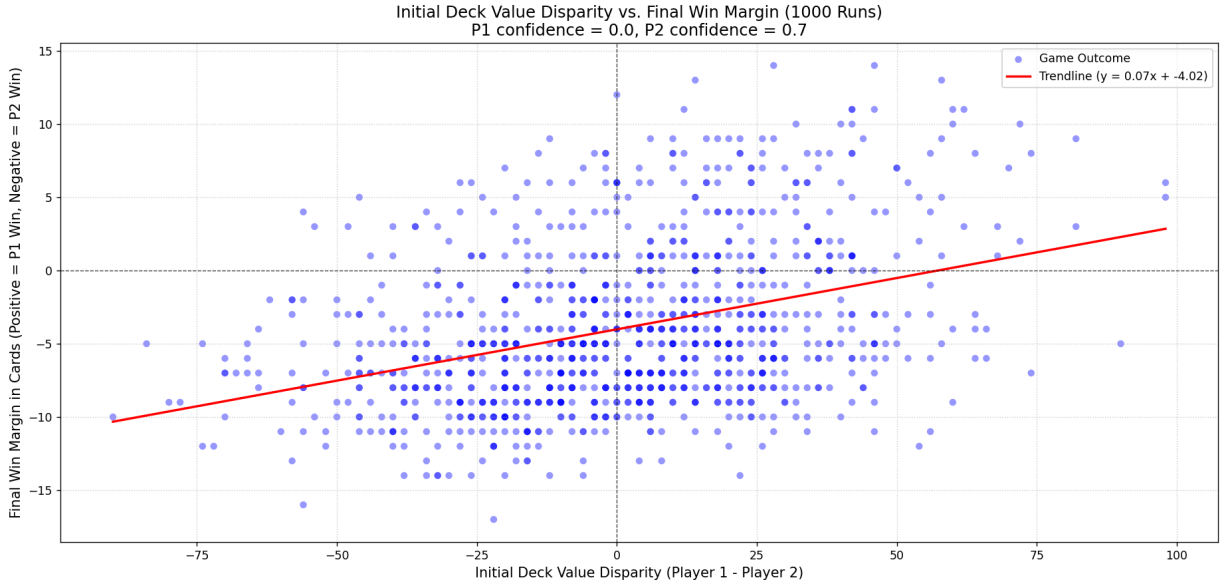
P1 confidence - 0.9, P2 confidence - 0.6



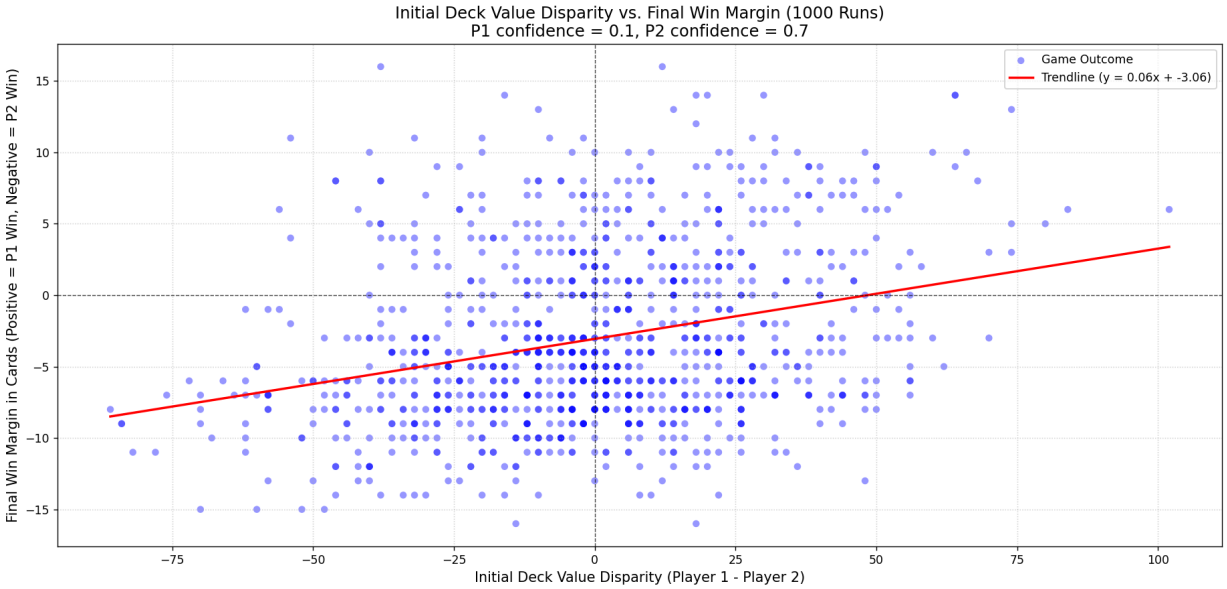
P1 confidence - 1, P2 confidence - 0.6


Initial Deck Value Disparity vs. Final Win Margin (1000 Runs)
P1 confidence = 1, P2 confidence = 0.6
Game Outcome
Trendline (y = 0.05x + -2.32)
Final Win Margin in Cards (Positive = P1 Win, Negative = P2 Win)
Initial Deck Value Disparity (Player 1 - Player 2)
15
10
5
0
−5
−10
−15
−20
−75
−50
−25
0
25
50
75

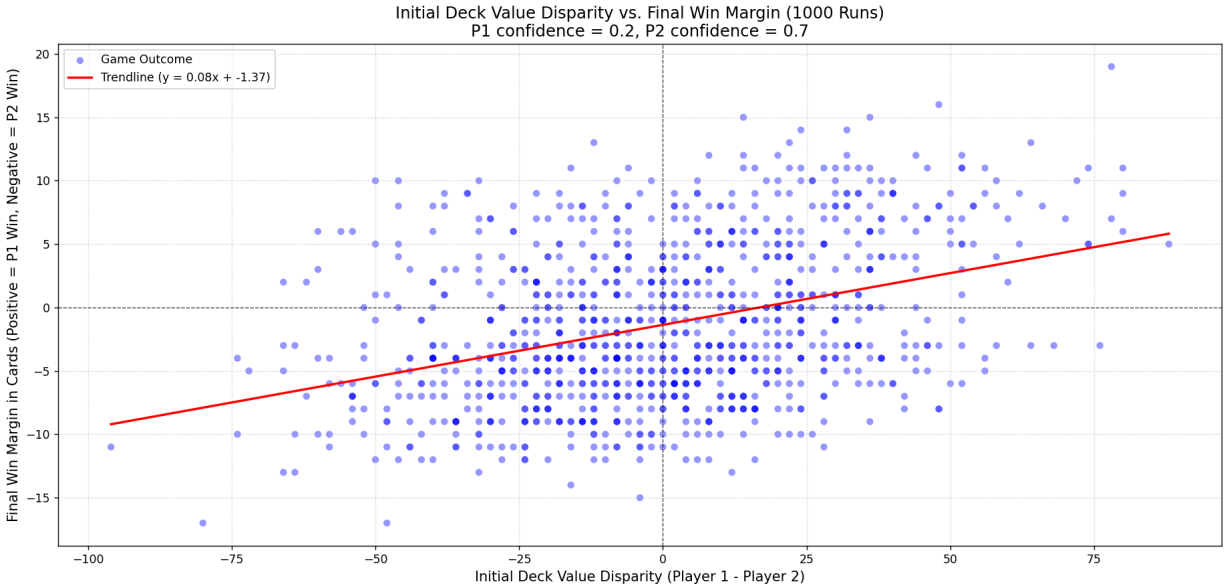
P1 confidence - 0, P2 confidence - 0.7



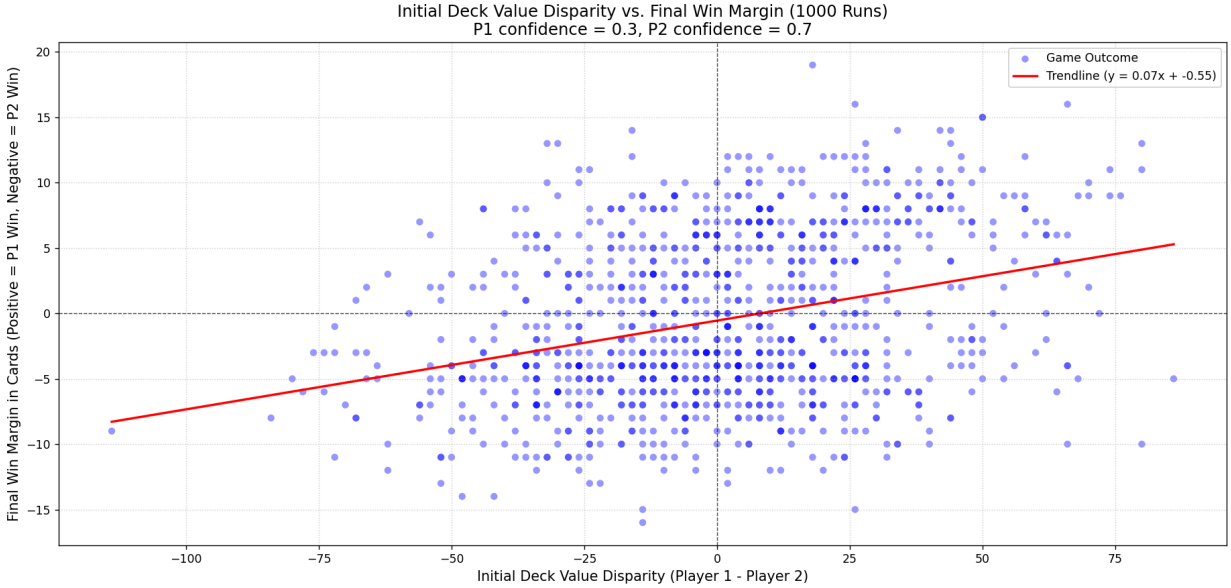
P1 confidence - 0.1, P2 confidence - 0.7



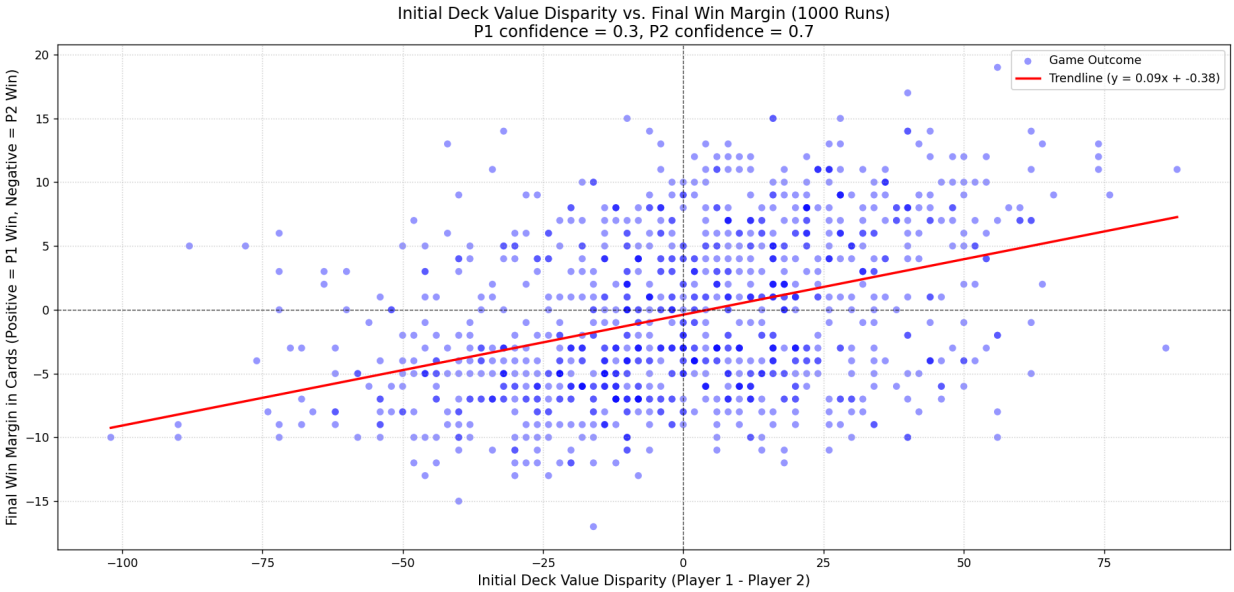
P1 confidence - 0.2, P2 confidence - 0.7



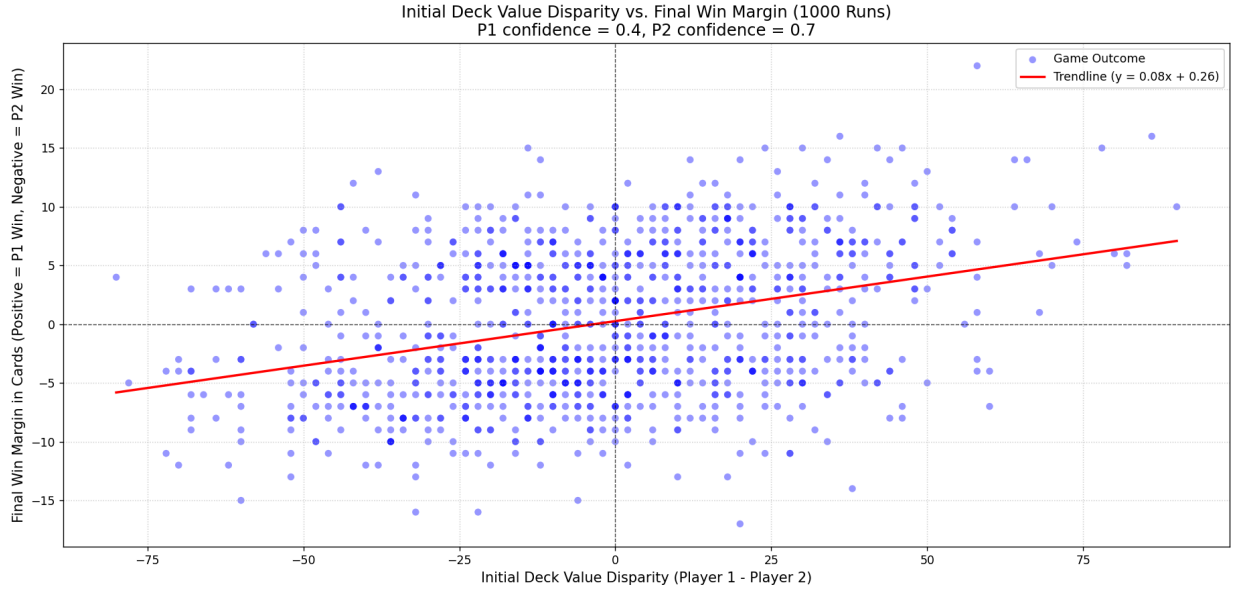
P1 confidence - 0.3, P2 confidence - 0.7



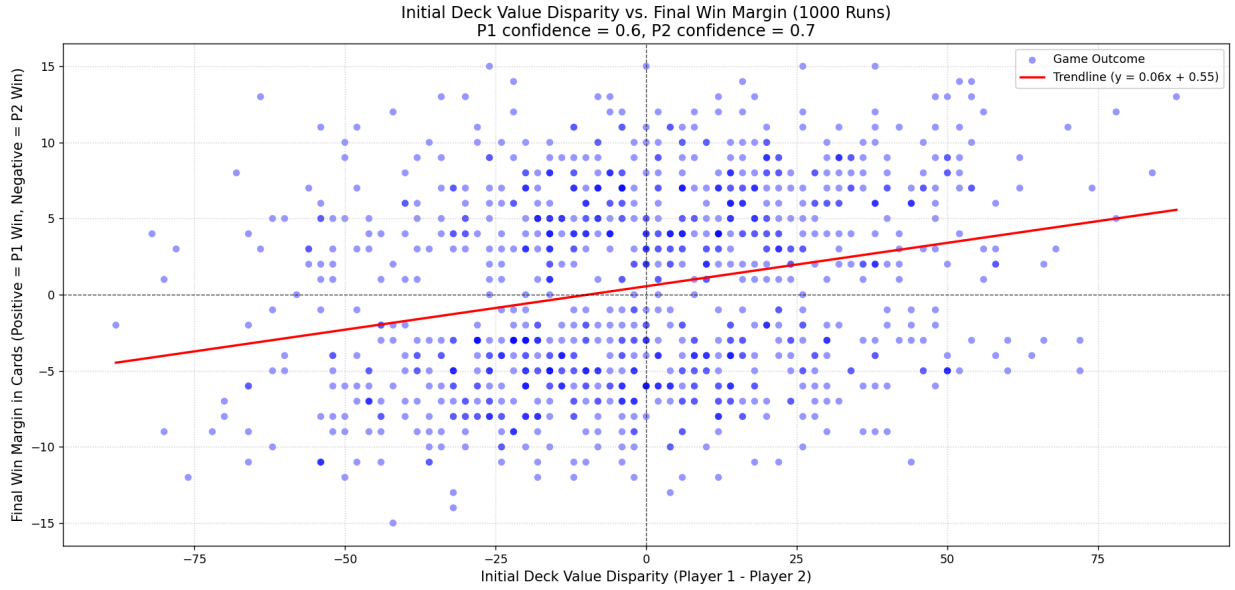
P1 confidence - 0.4, P2 confidence - 0.7



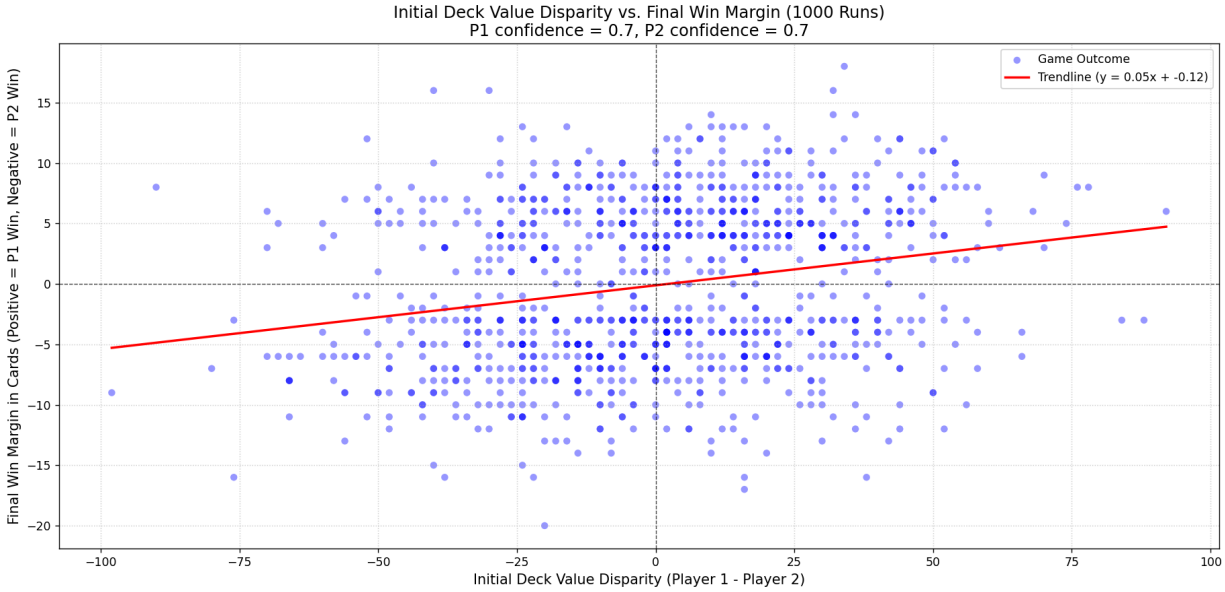
P1 confidence - 0.5, P2 confidence - 0.7



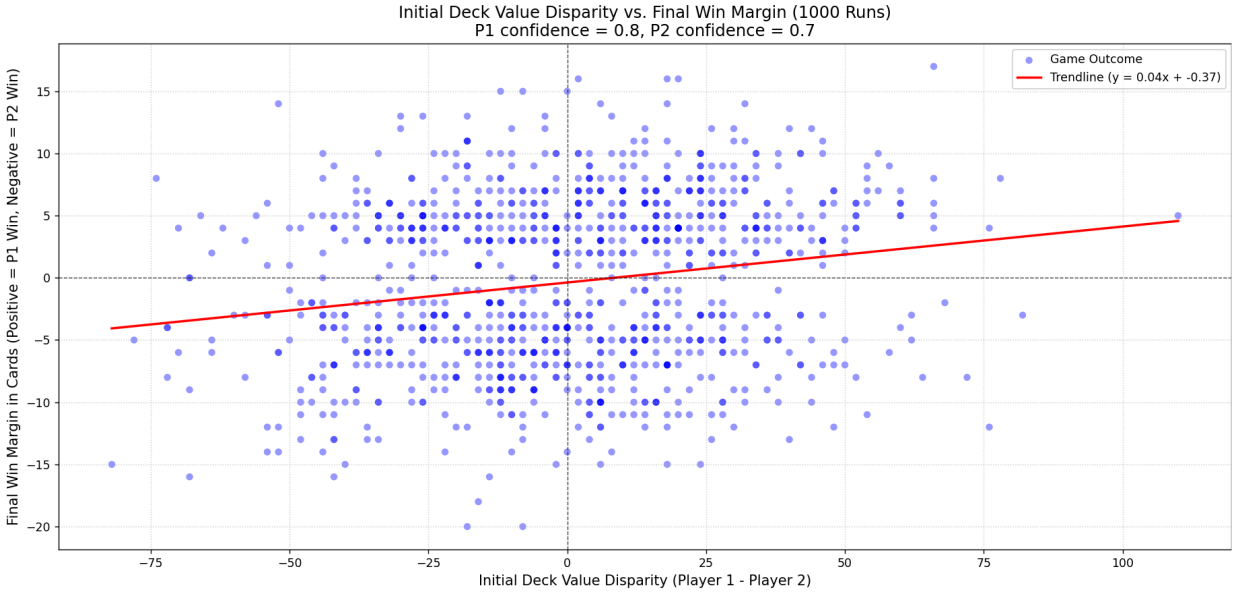
P1 confidence - 0.6, P2 confidence - 0.7



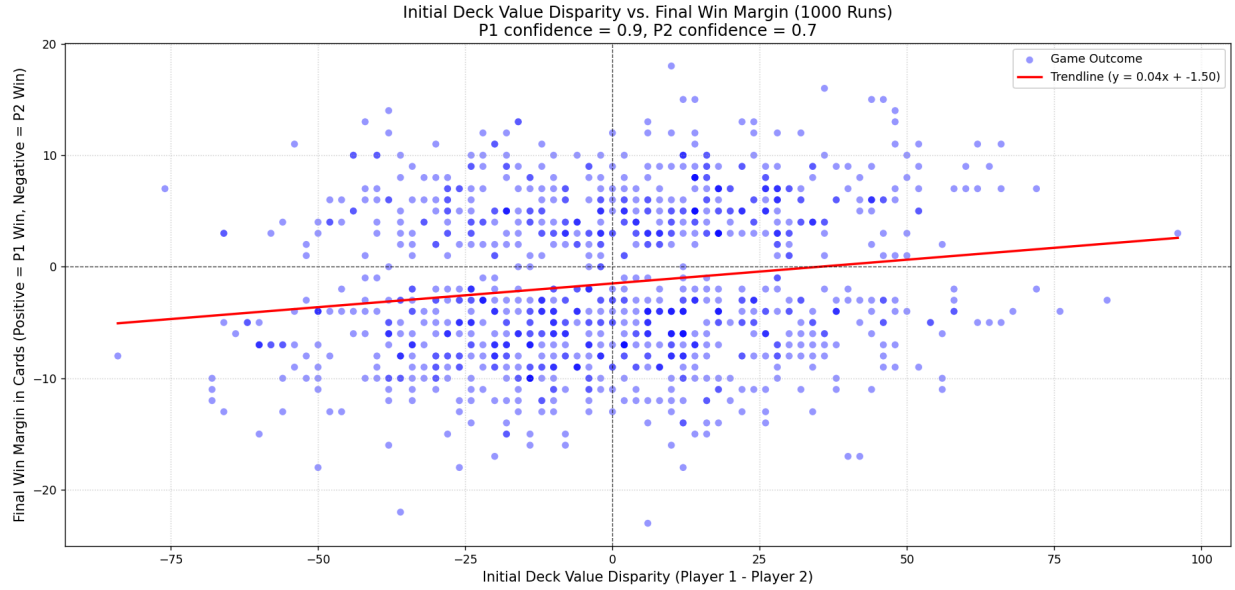
P1 confidence - 0.7, P2 confidence - 0.7



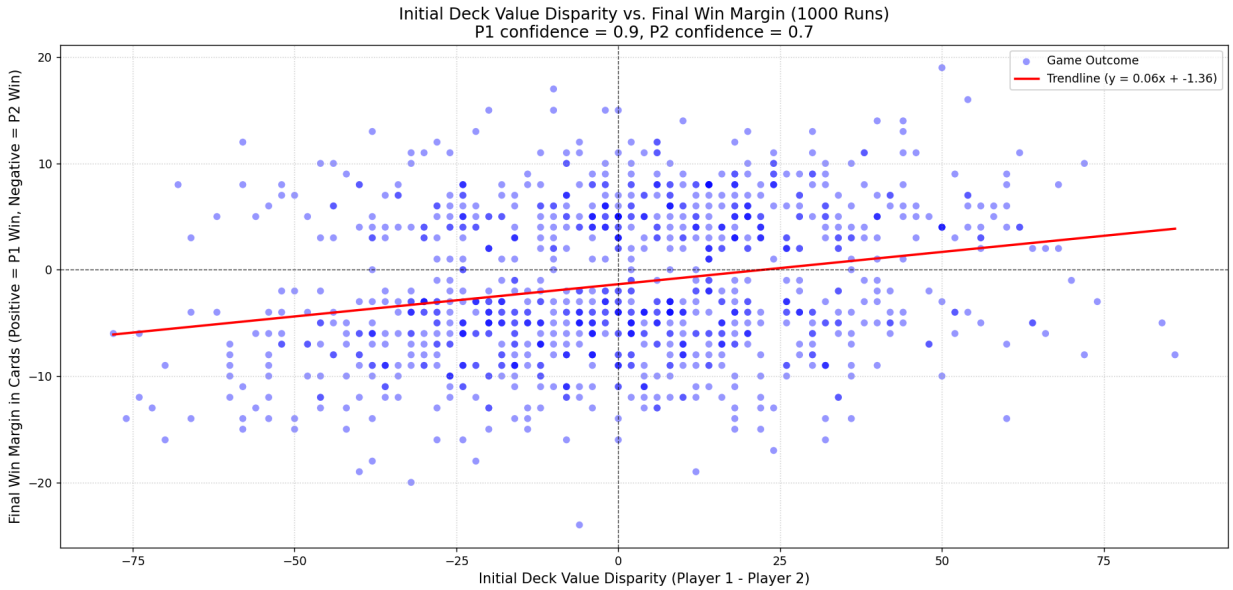
P1 confidence - 0.8, P2 confidence - 0.7



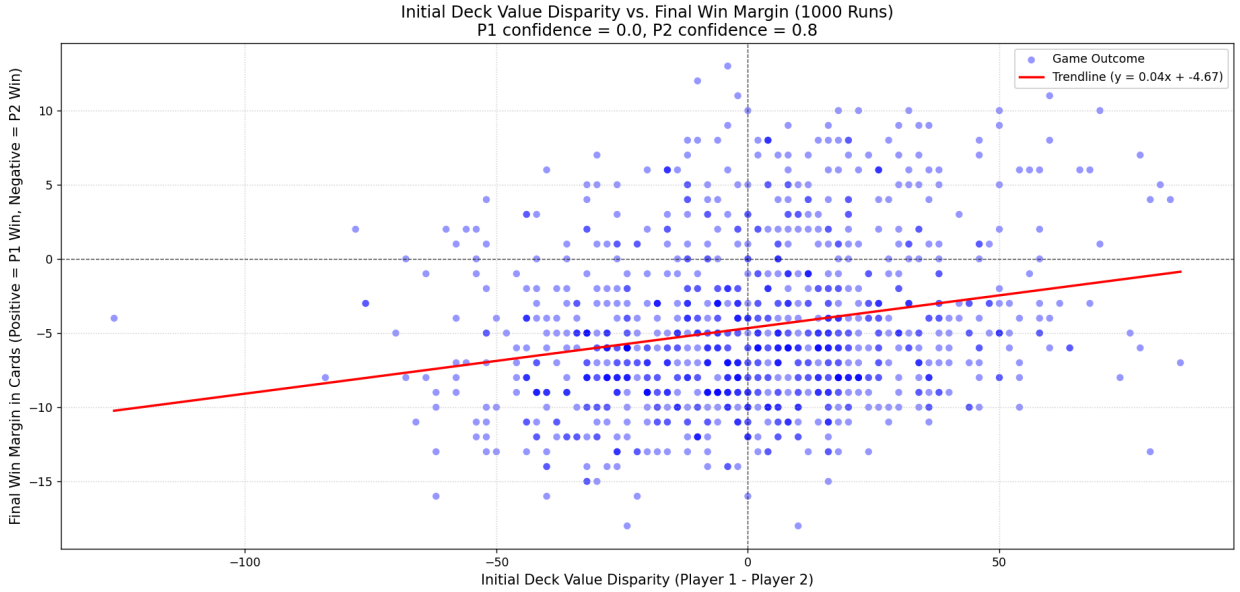
P1 confidence - 0.9, P2 confidence - 0.7



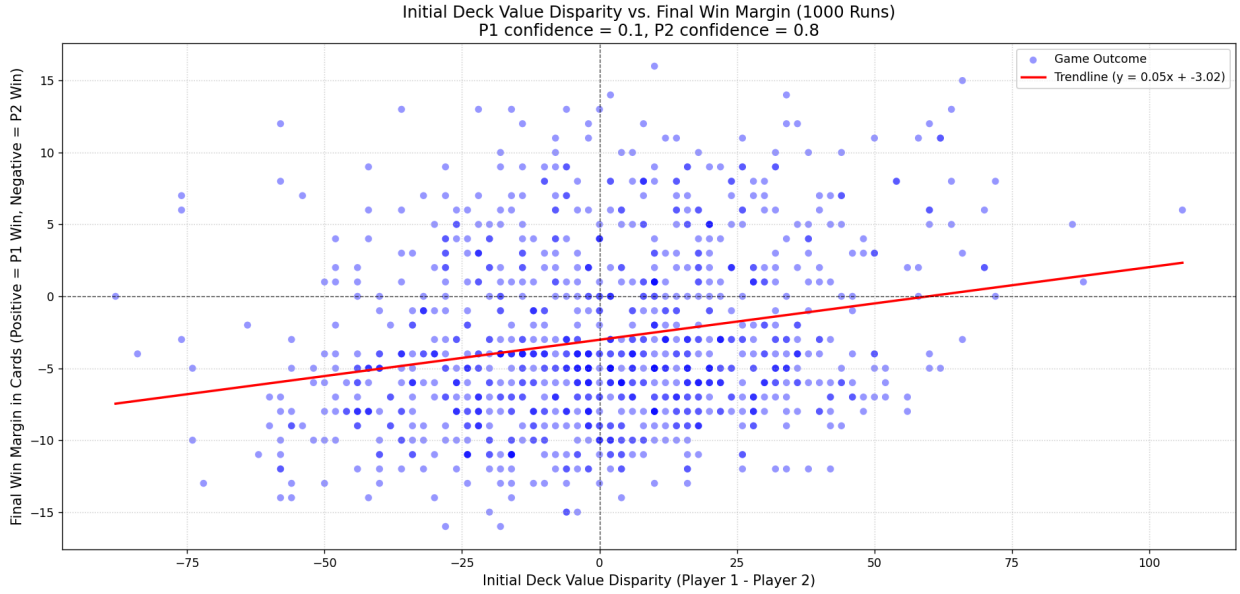
P1 confidence - 1, P2 confidence - 0.7


Initial Deck Value Disparity vs. Final Win Margin (1000 Runs)
P1 confidence = 0.9, P2 confidence = 0.7
Game Outcome
Trendline (y = 0.06x + -1.36)
Final Win Margin in Cards (Positive = P1 Win, Negative = P2 Win)
Initial Deck Value Disparity (Player 1 - Player 2)
20
10
0
−10
−20
−75
−50
−25
0
25
50
75

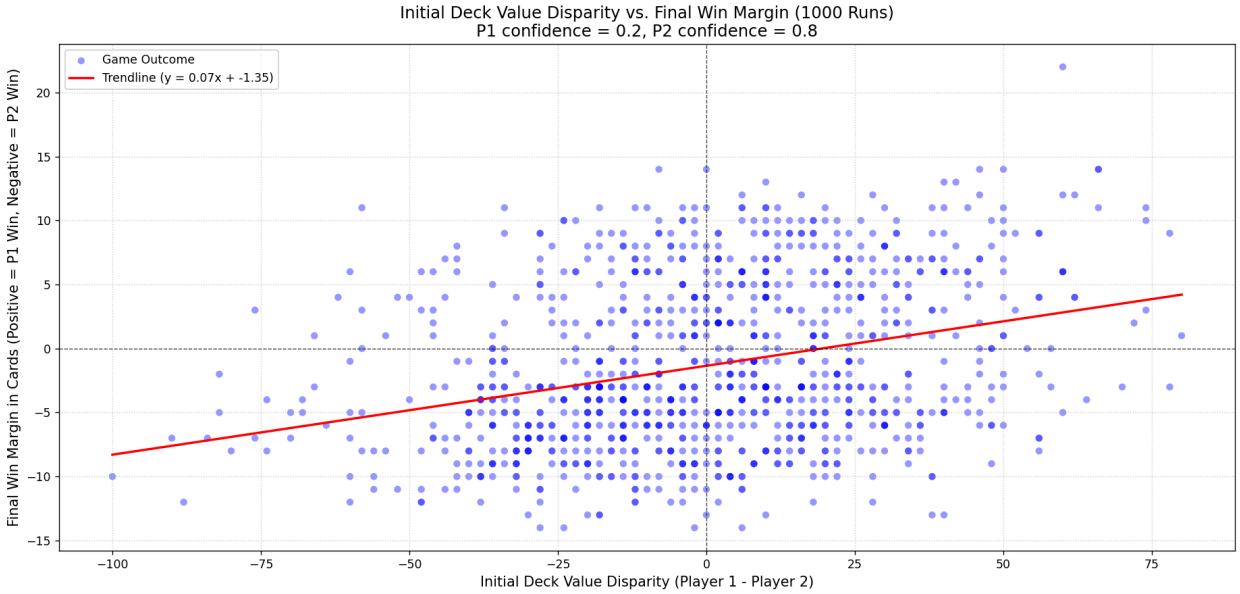
P1 confidence - 0, P2 confidence - 0.8



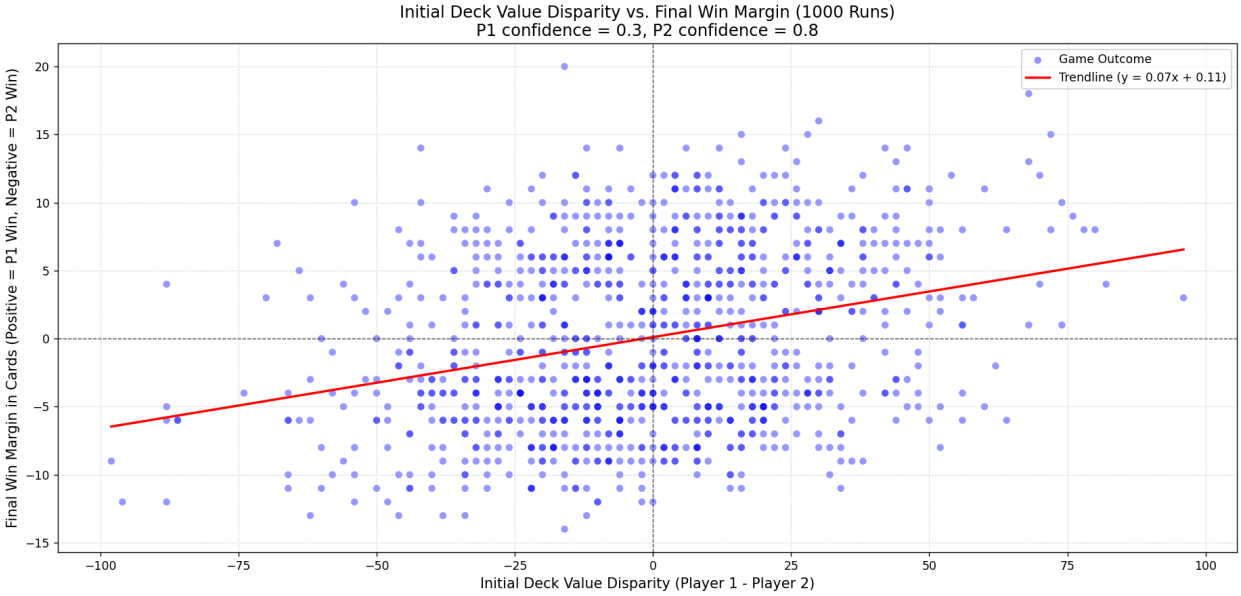
P1 confidence - 0.1, P2 confidence - 0.8



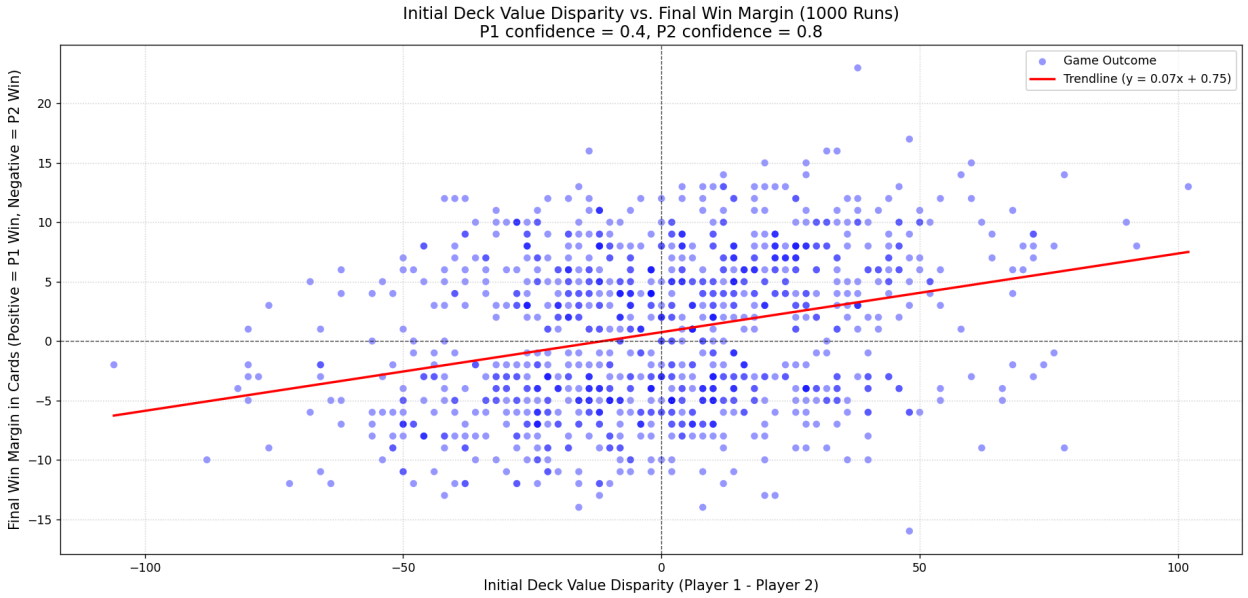
P1 confidence - 0.2, P2 confidence - 0.8



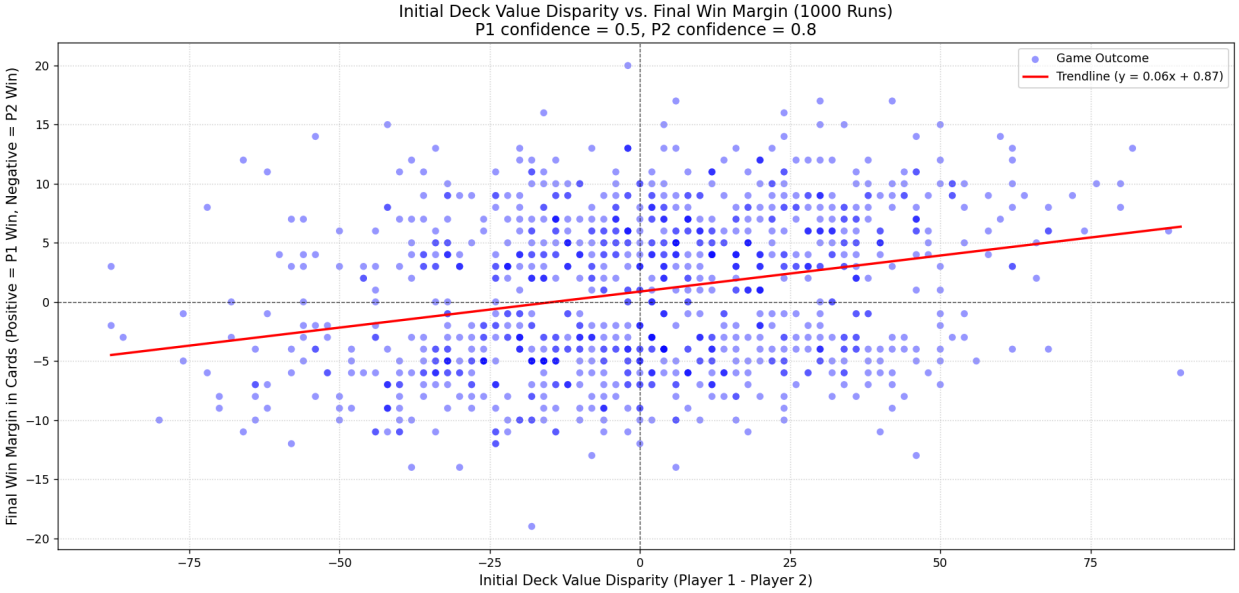
P1 confidence - 0.3, P2 confidence - 0.8



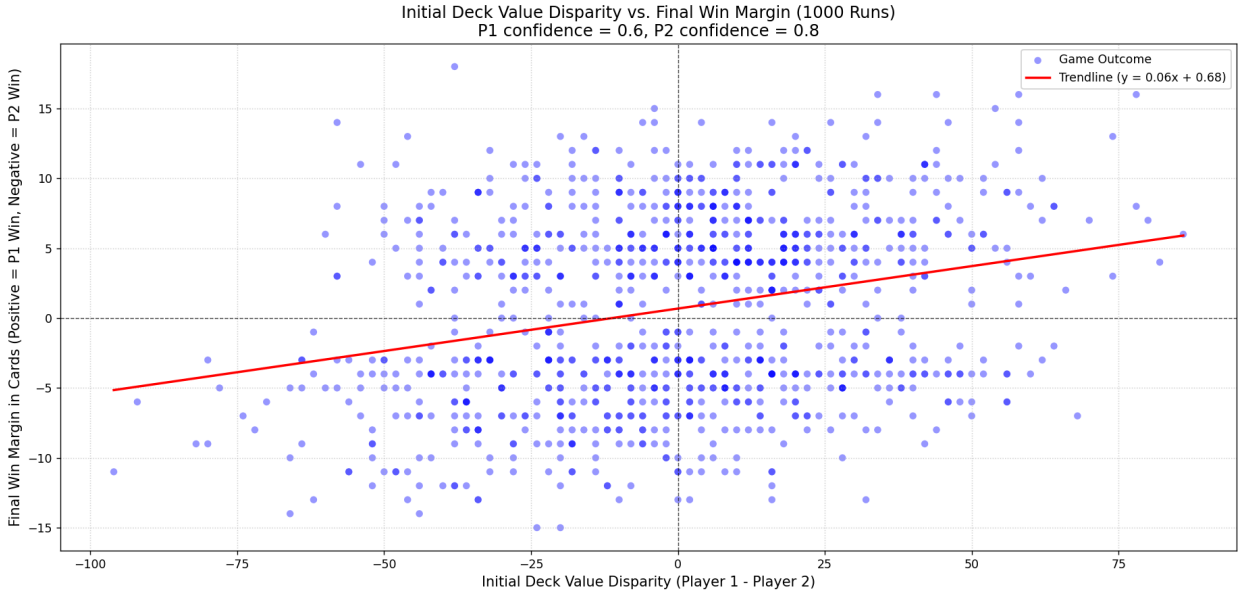
P1 confidence - 0.4, P2 confidence - 0.8



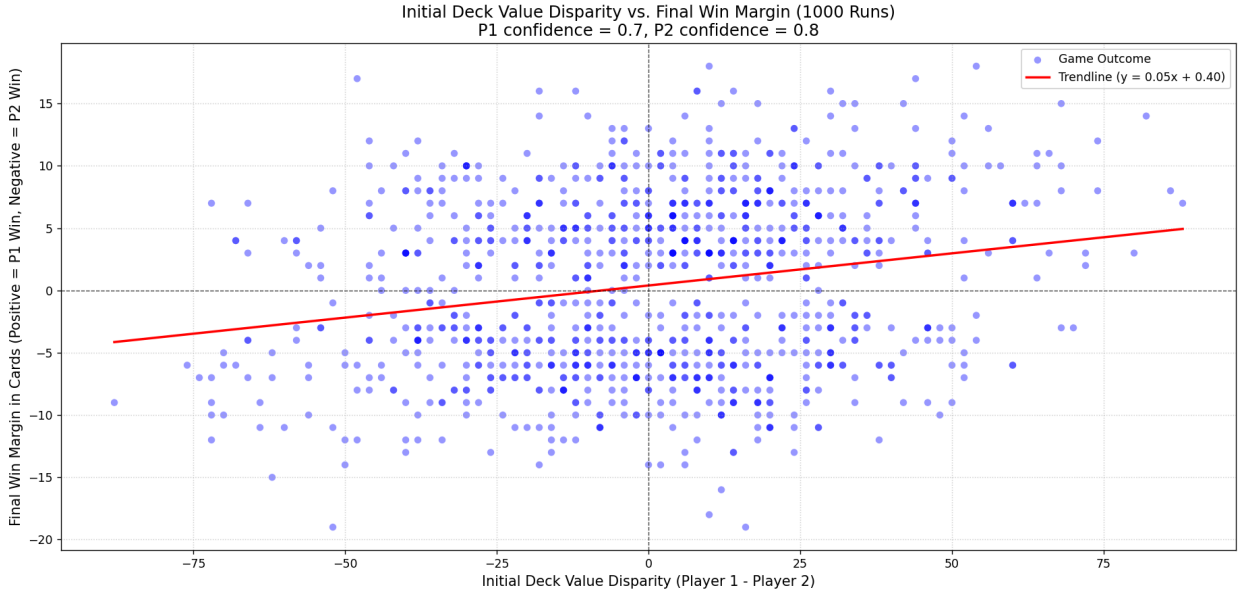
P1 confidence - 0.5, P2 confidence - 0.8



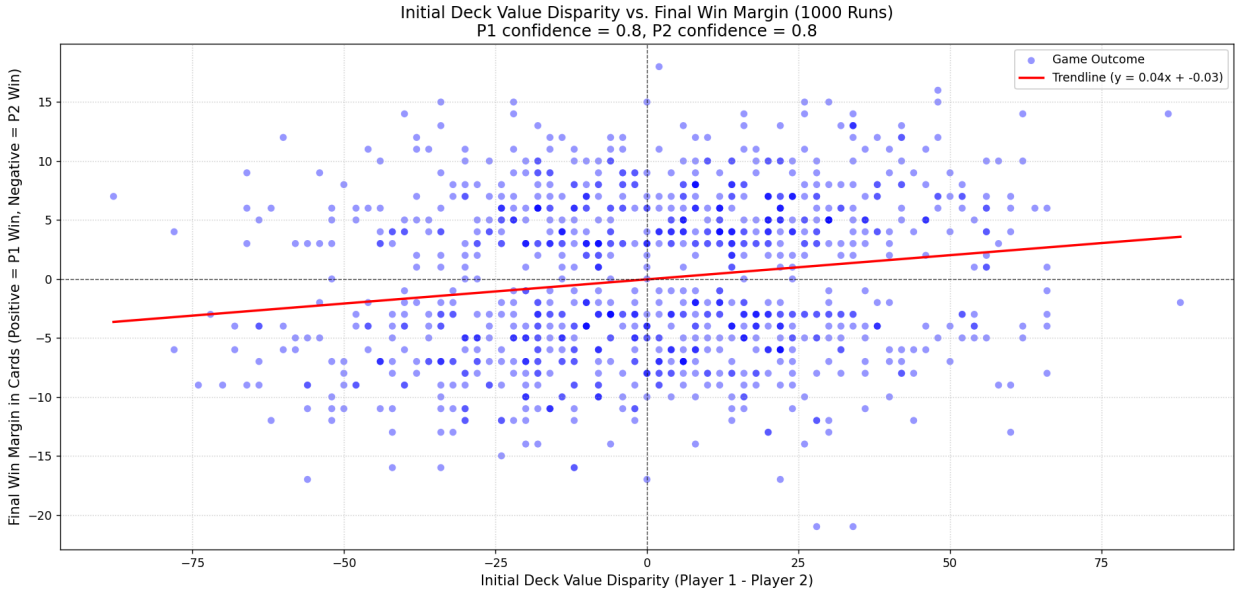
P1 confidence - 0.6, P2 confidence - 0.8



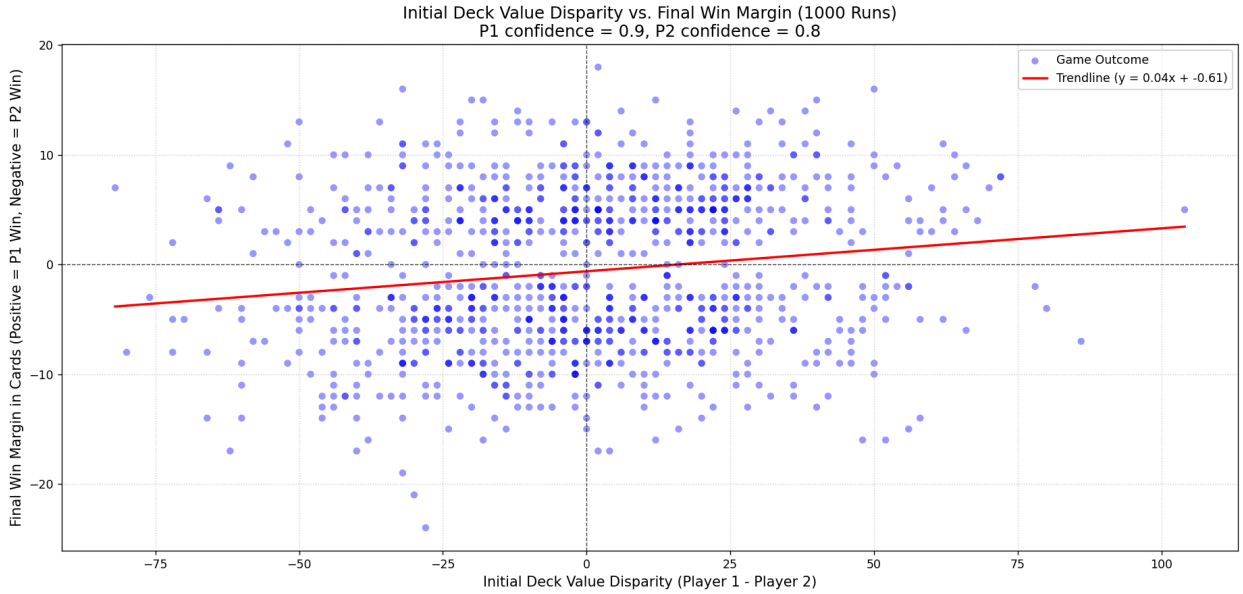
P1 confidence - 0.7, P2 confidence - 0.8



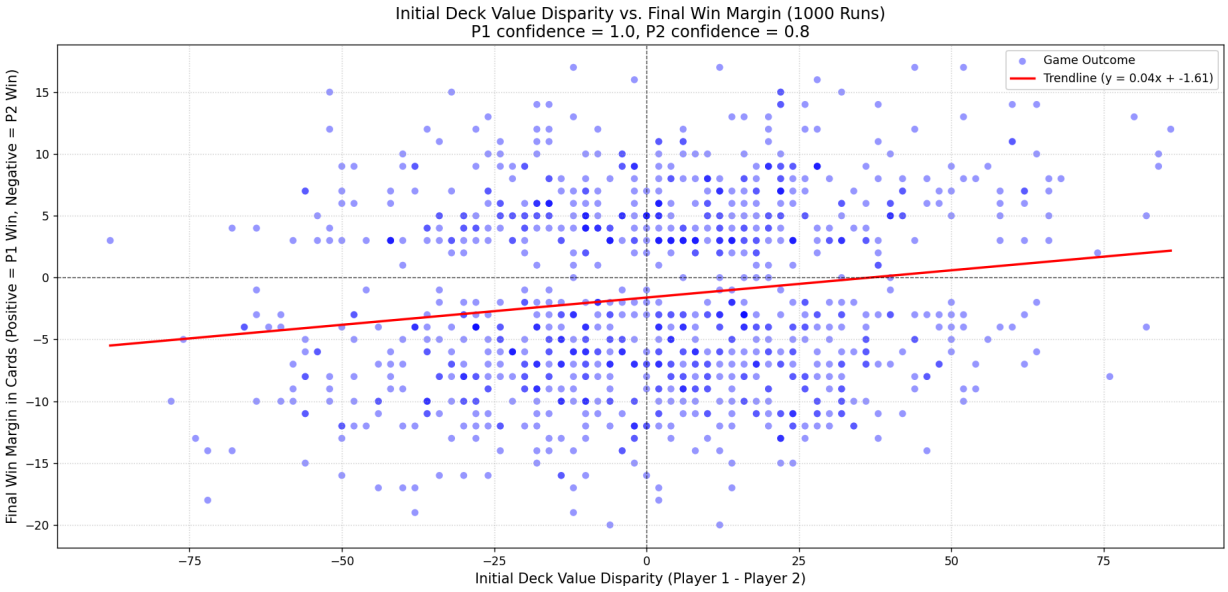
P1 confidence - 0.8, P2 confidence - 0.8



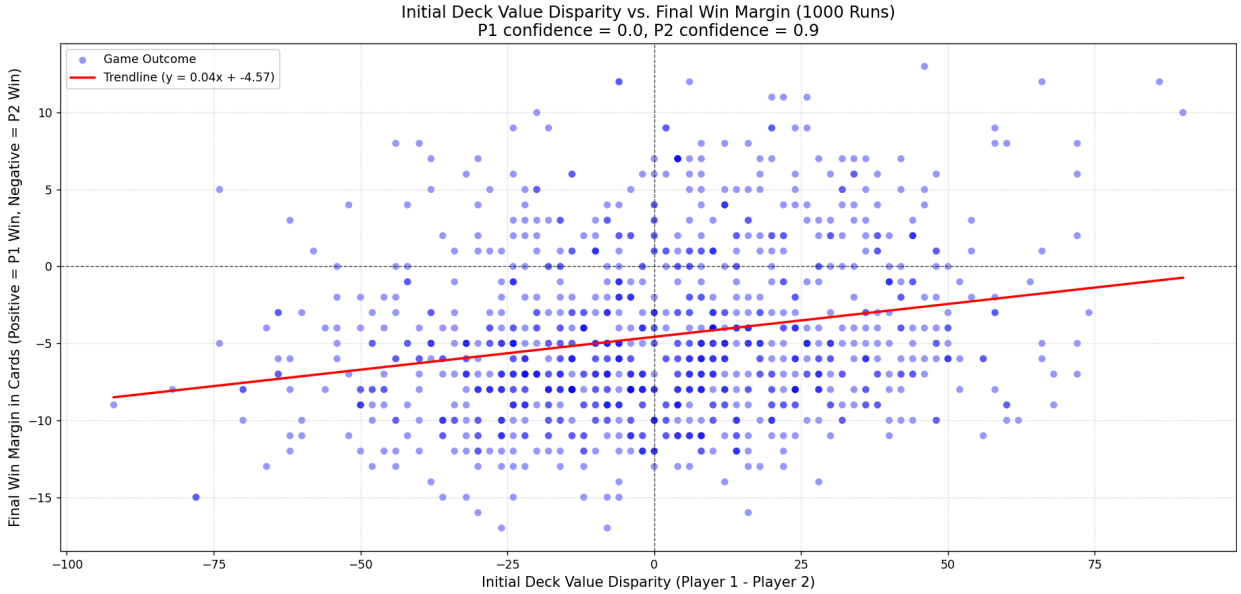
P1 confidence - 0.9, P2 confidence - 0.8



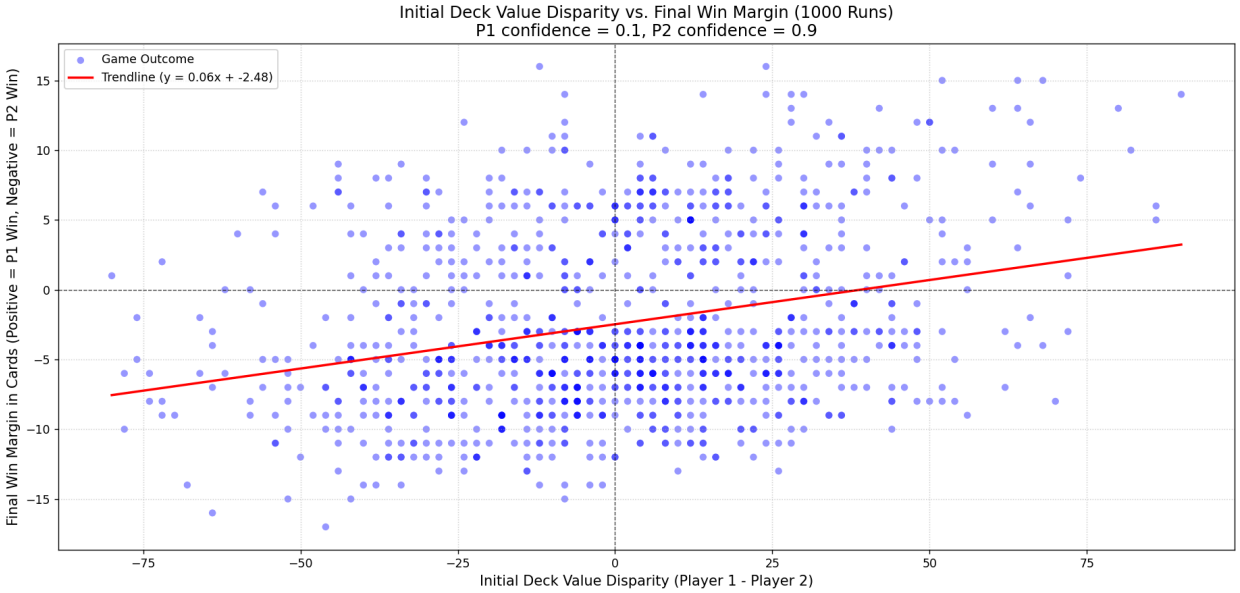
P1 confidence - 1, P2 confidence - 0.8


Initial Deck Value Disparity vs. Final Win Margin (1000 Runs)
P1 confidence = 1.0, P2 confidence = 0.8
Game Outcome
Trendline (y = 0.04x + -1.61)
Final Win Margin in Cards (Positive = P1 Win, Negative = P2 Win)
Initial Deck Value Disparity (Player 1 - Player 2)
15
10
5
0
−5
−10
−15
−20
−75
−50
−25
0
25
50
75

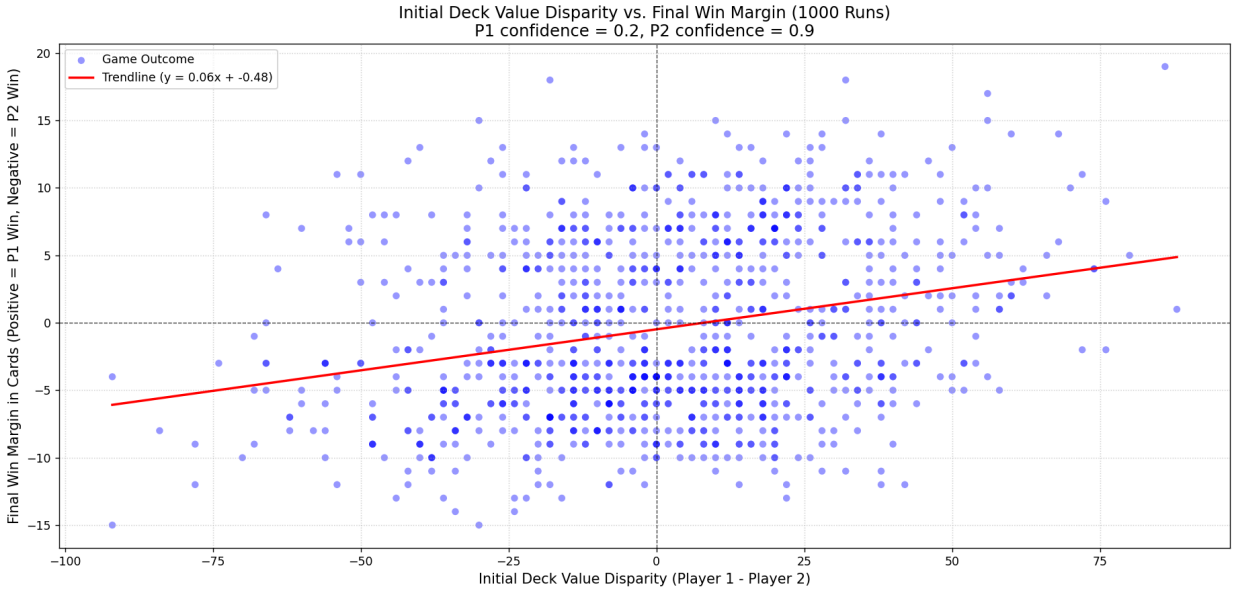
P1 confidence - 0, P2 confidence - 0.9



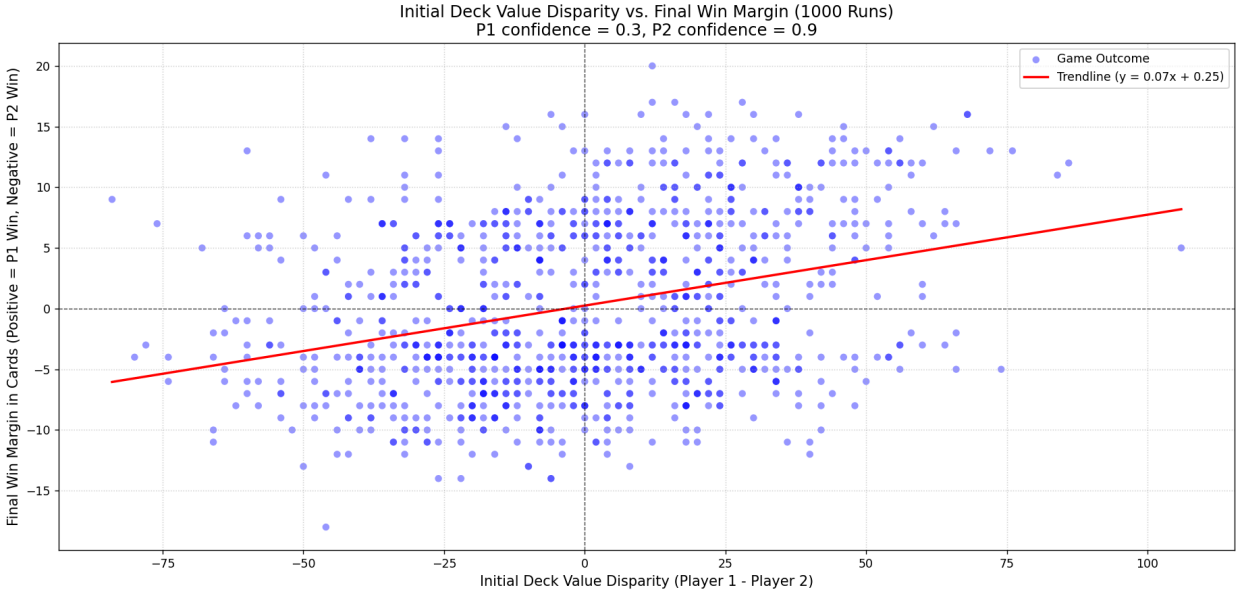
P1 confidence - 0.1, P2 confidence - 0.9



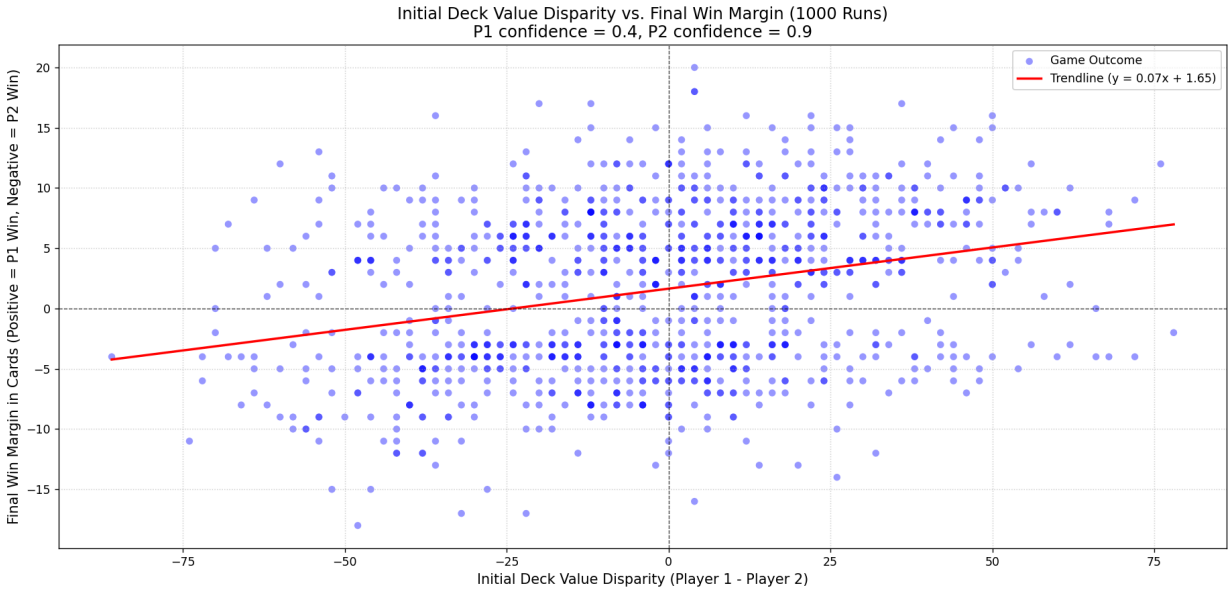
P1 confidence - 0.2, P2 confidence - 0.9



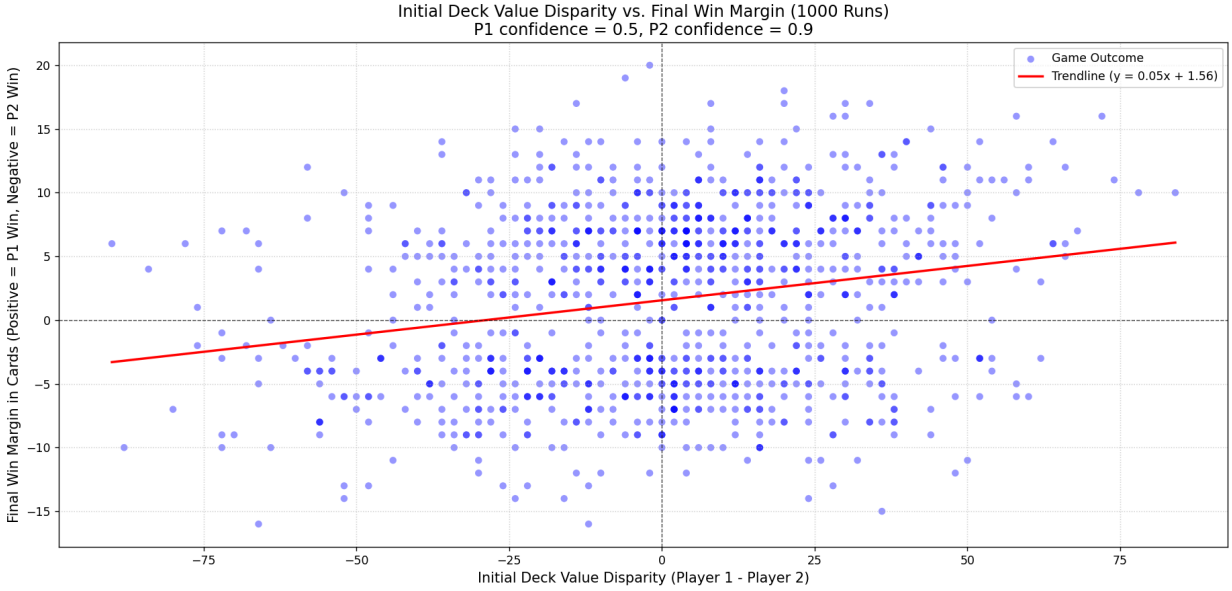
P1 confidence - 0.3, P2 confidence - 0.9



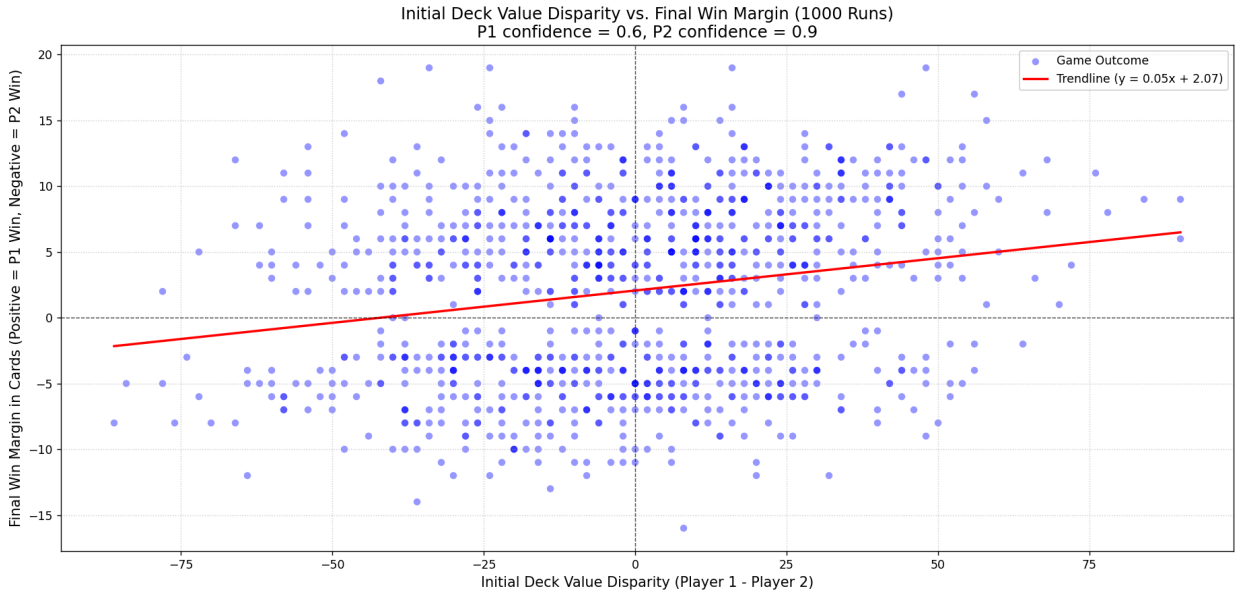
P1 confidence - 0.4, P2 confidence - 0.9



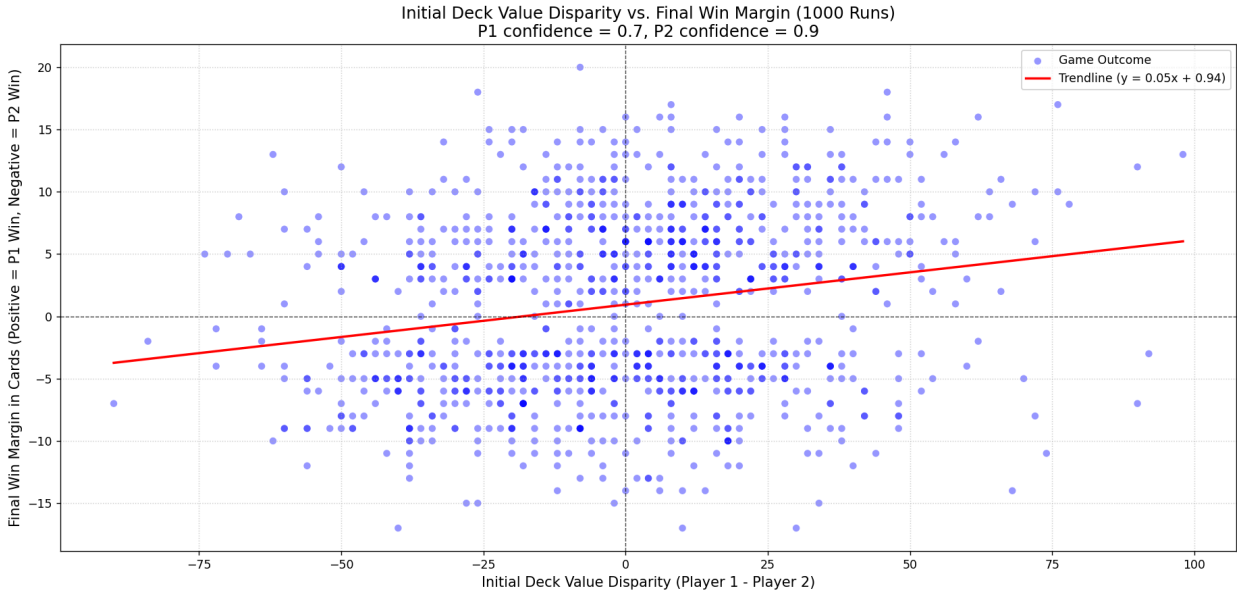
P1 confidence - 0.5, P2 confidence - 0.9



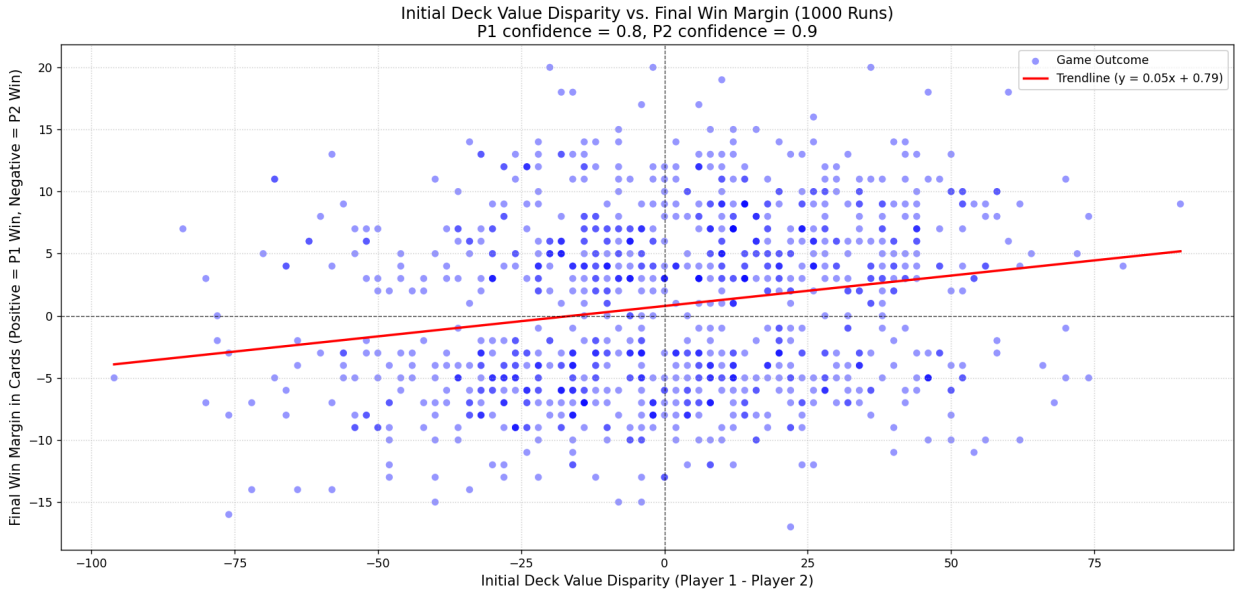
P1 confidence - 0.6, P2 confidence - 0.9



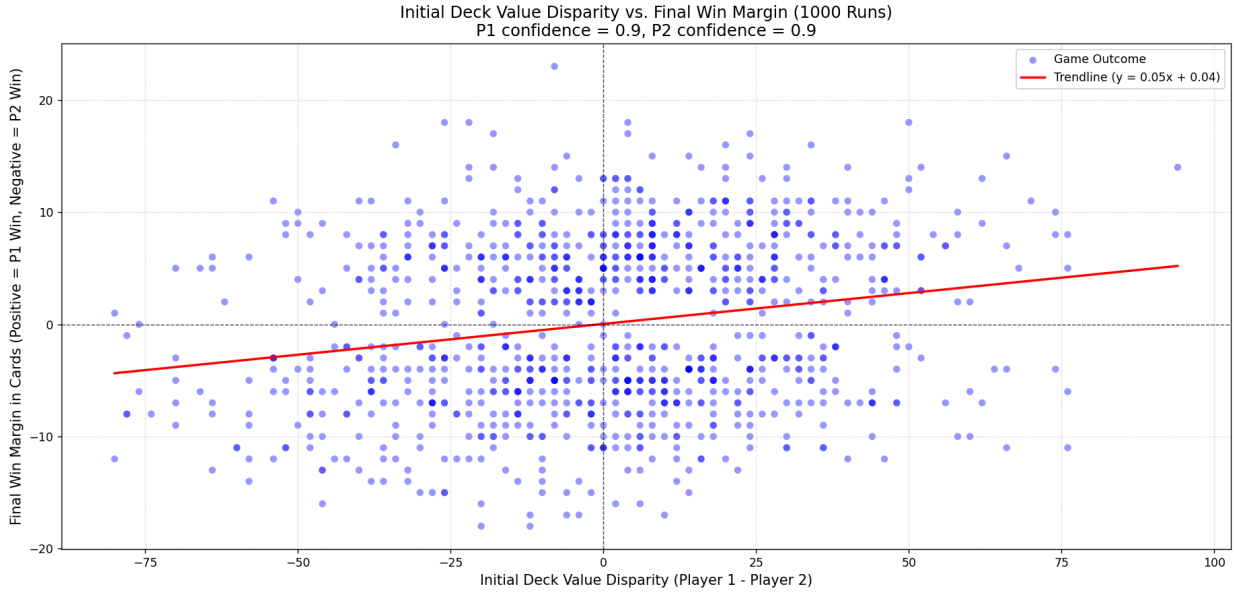
P1 confidence - 0.7, P2 confidence - 0.9



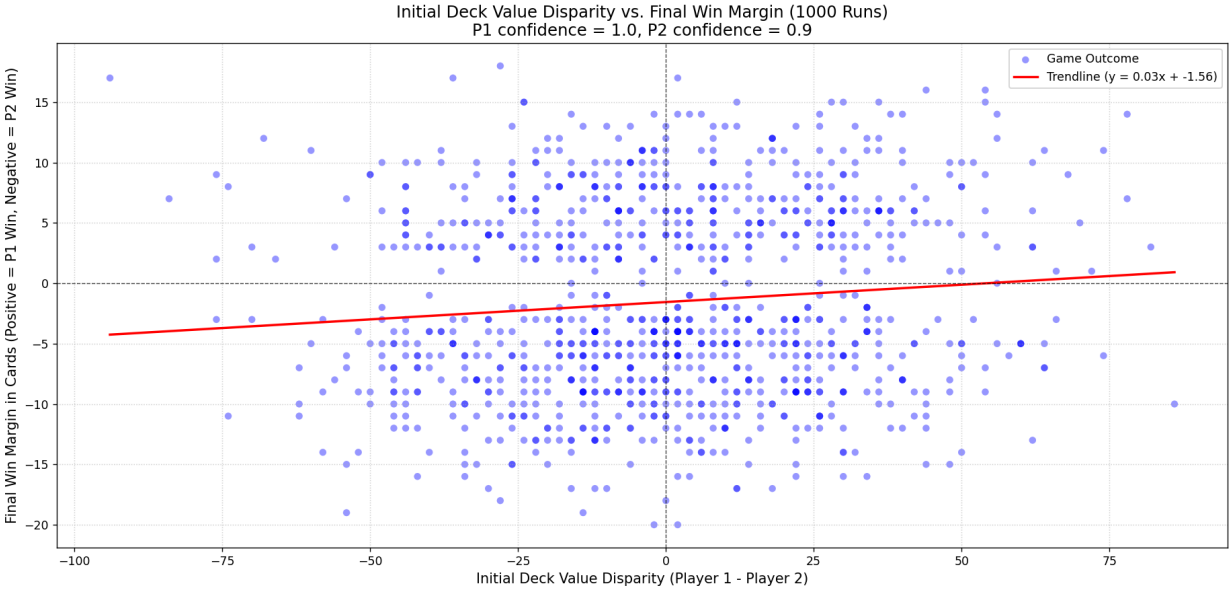
P1 confidence - 0.8, P2 confidence - 0.9



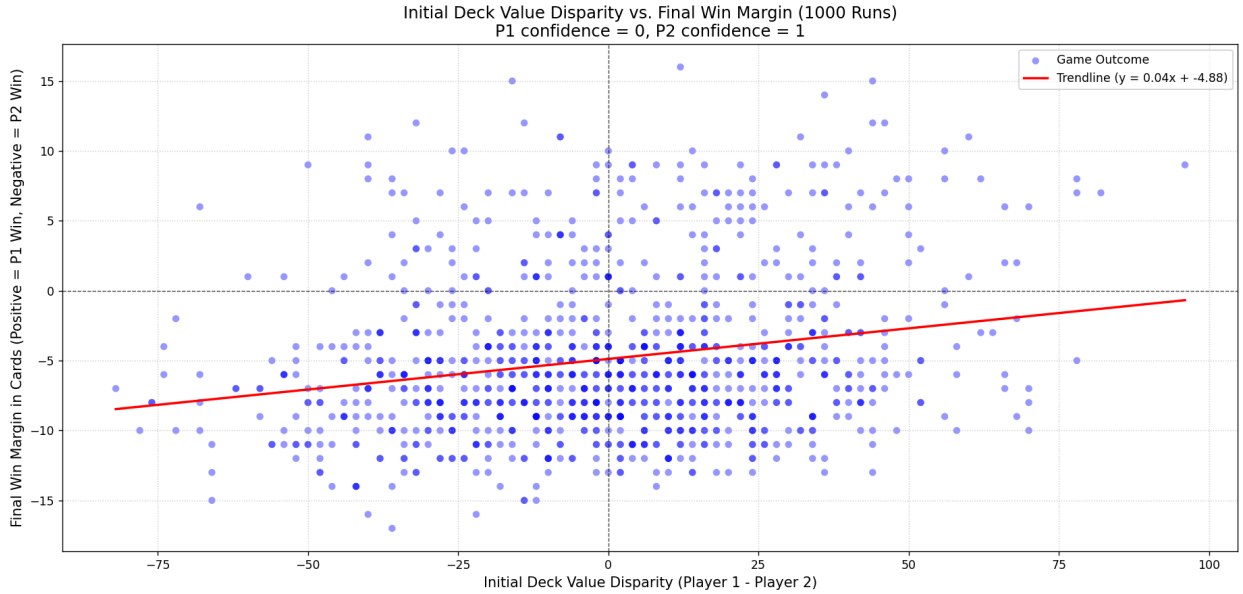
P1 confidence - 0.9, P2 confidence - 0.9



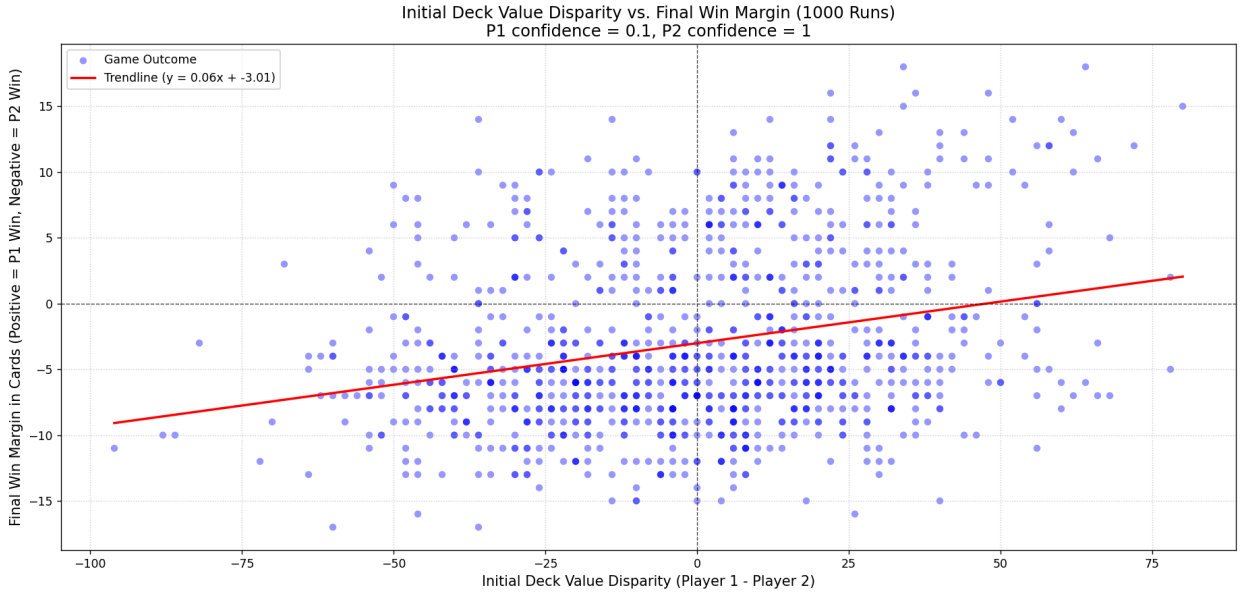
P1 confidence - 1, P2 confidence - 0.9


Initial Deck Value Disparity vs. Final Win Margin (1000 Runs)
P1 confidence = 1.0, P2 confidence = 0.9
Game Outcome
Trendline (y = 0.03x + -1.56)
Final Win Margin in Cards (Positive = P1 Win, Negative = P2 Win)
Initial Deck Value Disparity (Player 1 - Player 2)
15
10
5
0
−5
−10
−15
−20
−100
−75
−50
−25
0
25
50
75

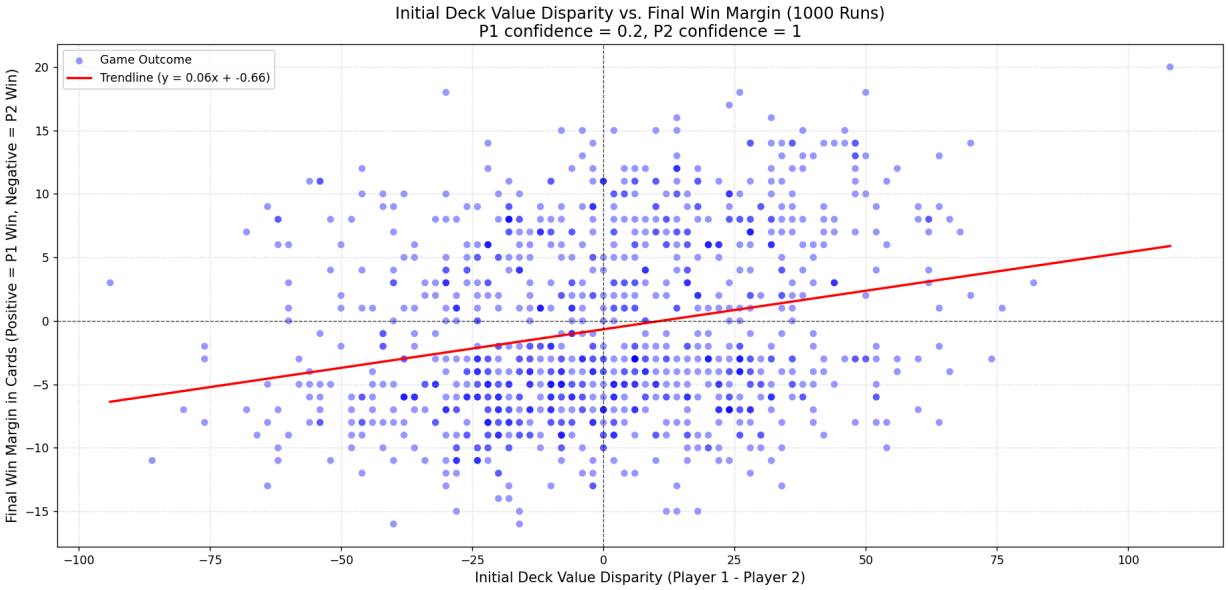
P1 confidence - 0, P2 confidence - 1



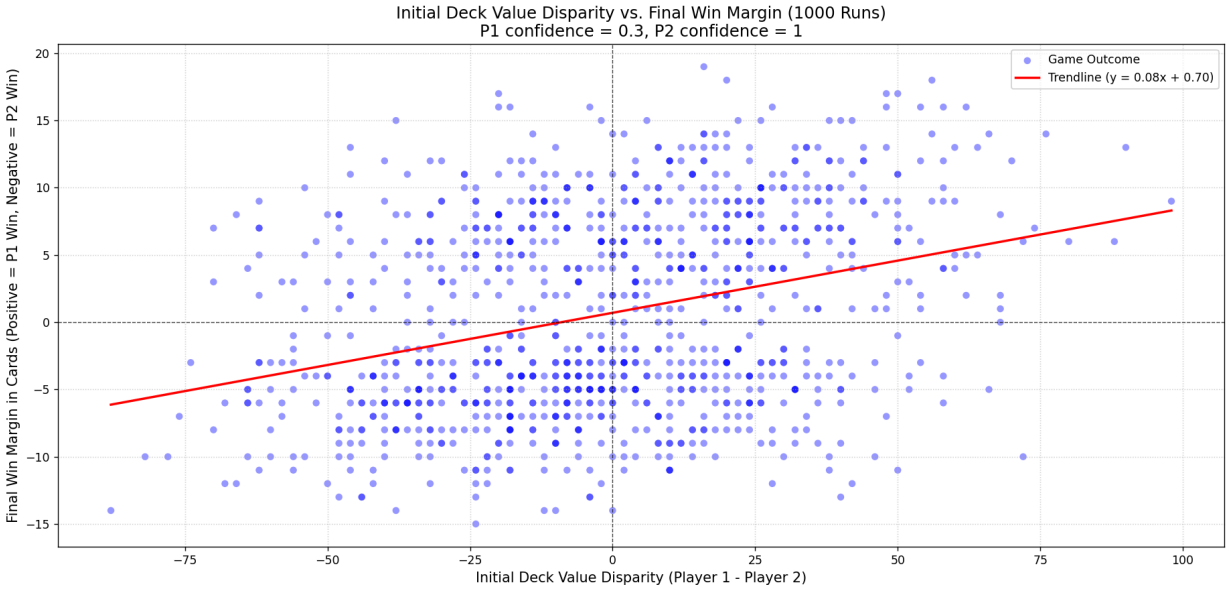
P1 confidence - 0.1, P2 confidence - 1



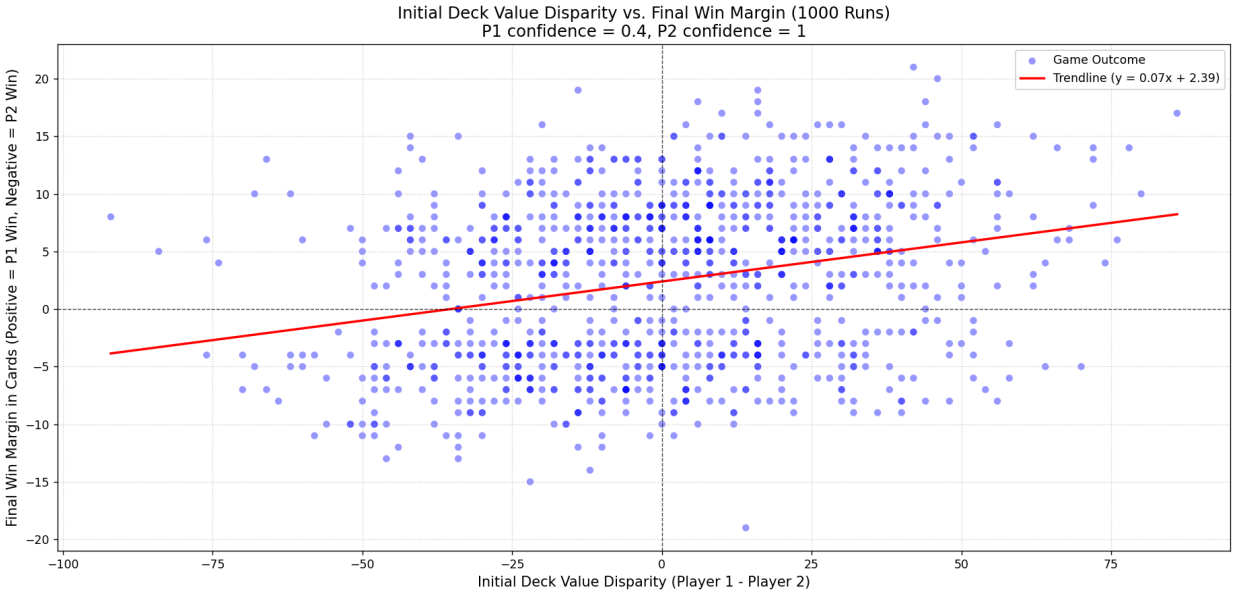
P1 confidence - 0.2, P2 confidence - 1



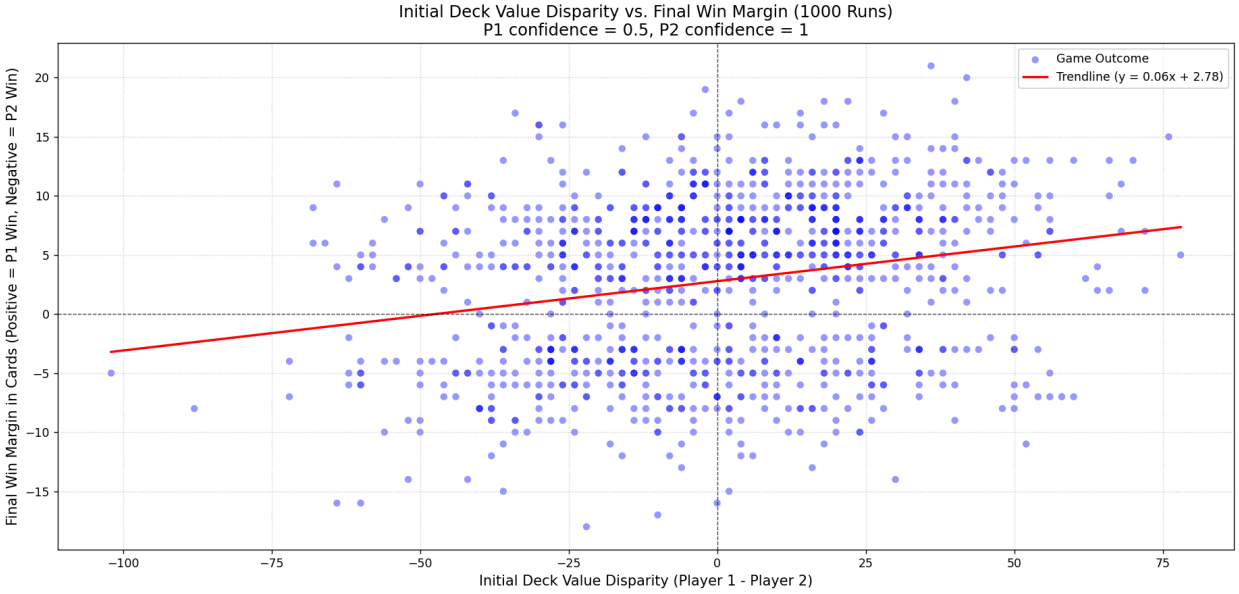
P1 confidence - 0.3, P2 confidence - 1



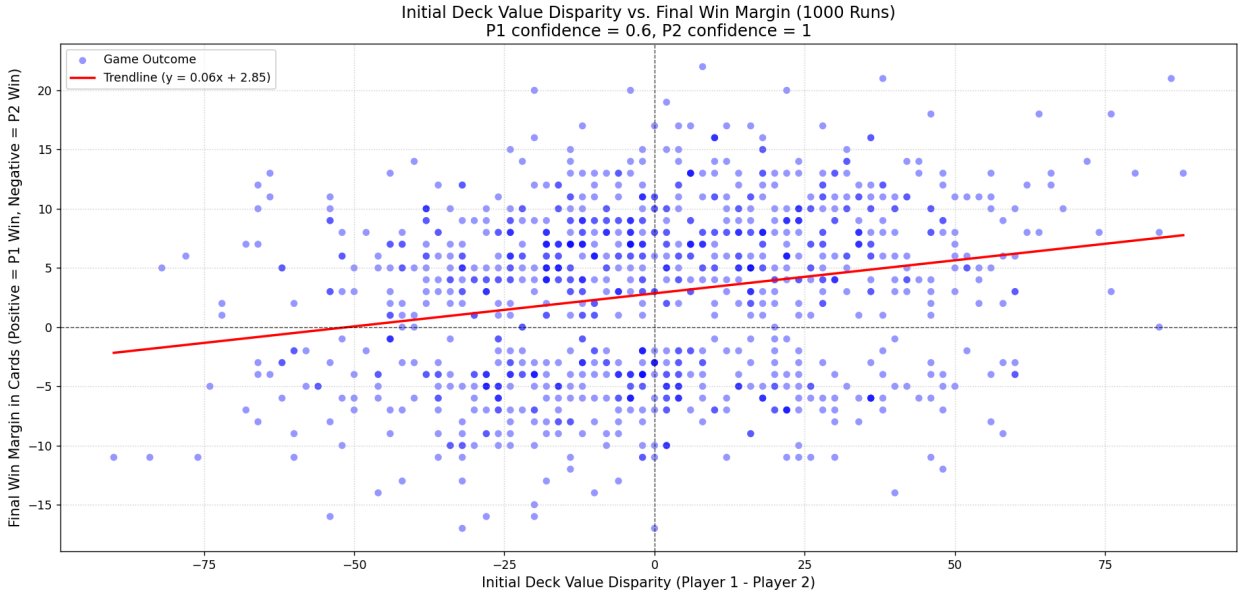
P1 confidence - 0.4, P2 confidence - 1



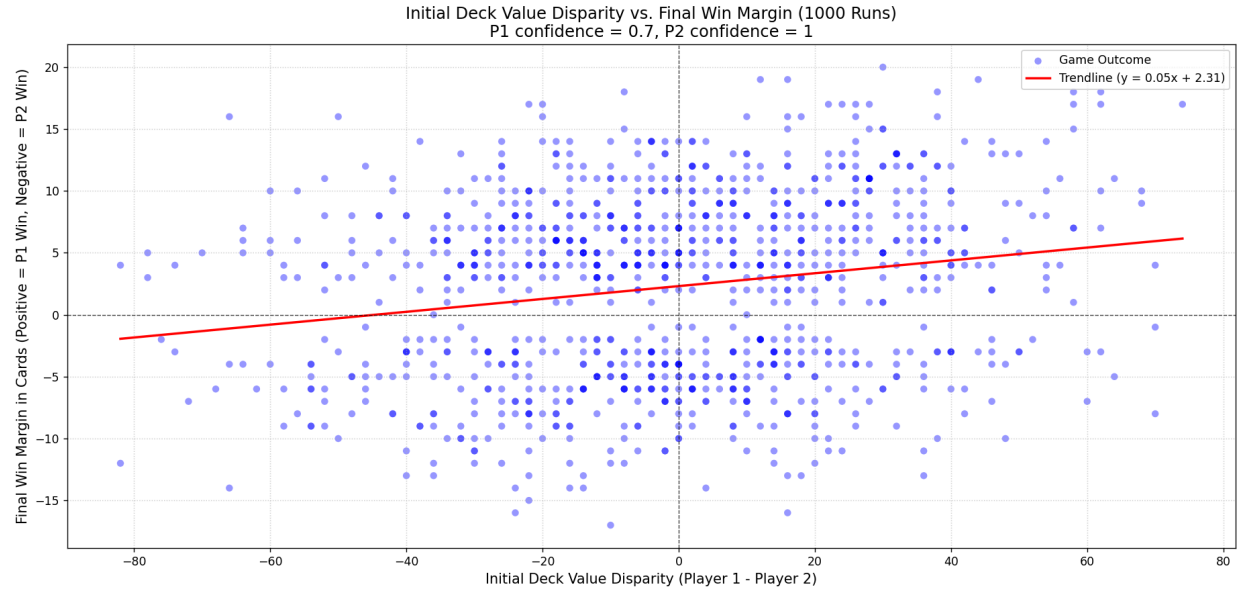
P1 confidence - 0.5, P2 confidence - 1



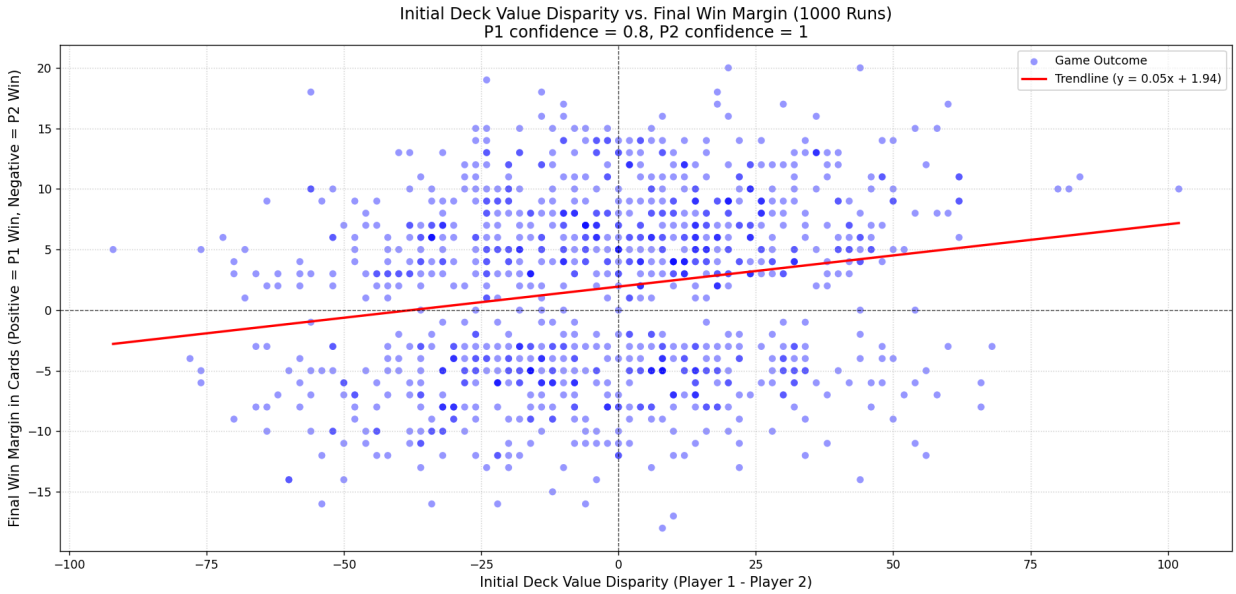
P1 confidence - 0.6, P2 confidence - 1



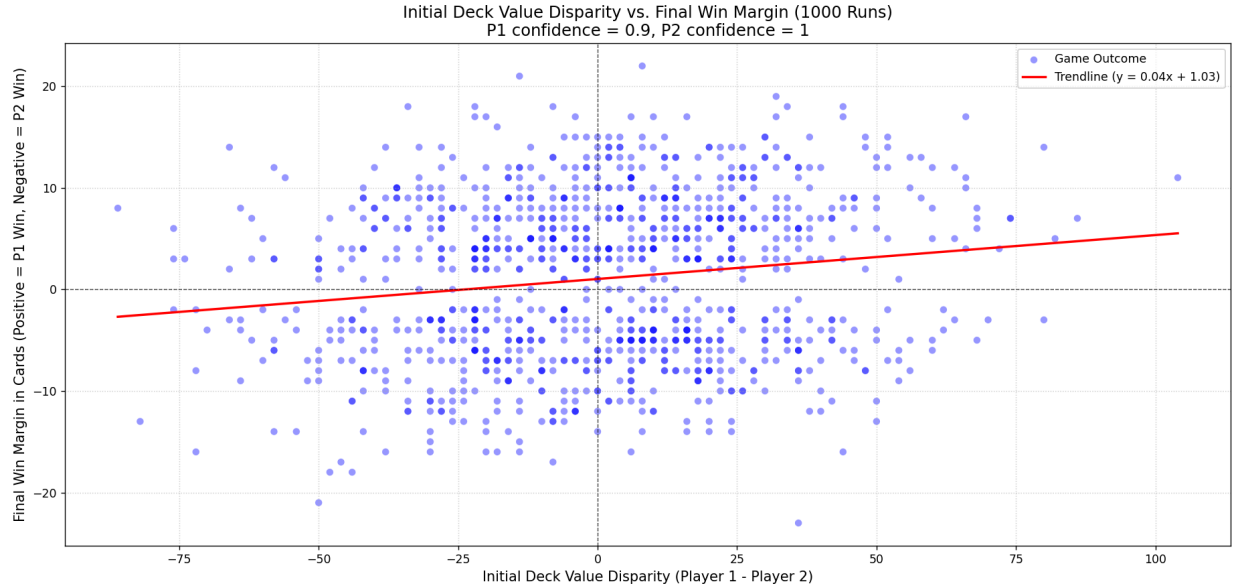
P1 confidence - 0.7, P2 confidence - 1



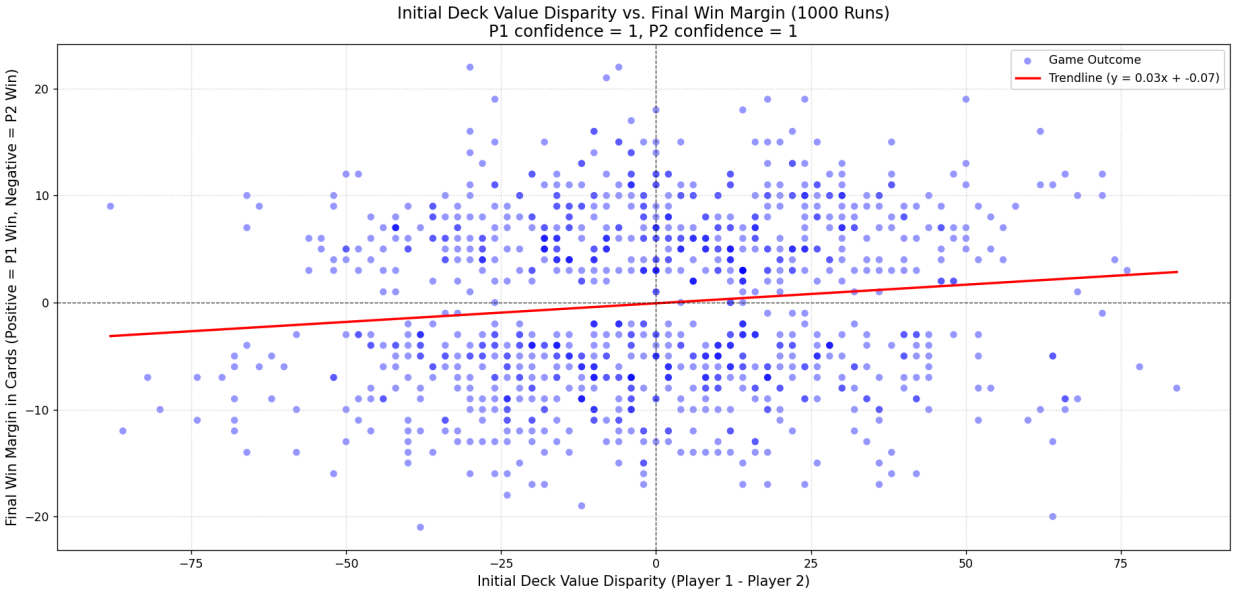
P1 confidence - 0.8, P2 confidence - 1



P1 confidence - 0.9, P2 confidence - 1



P1 confidence - 1, P2 confidence - 1



Against someone with a confidence level of 1, these are the amount of cards you will win / lose by with different confidence levels.

